

Causal Map App

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GETTING STARTED

CHAPTER CONTENTS.



Welcome to CausalMap!!

📺 If you like learning through video, there are some short Causal Map videos to get you started: [here](#). Or keep reading this Guide!

Try this: code the short example project:

- We've created a practice project for you called `example-short-uncoded-[your-username]`. This gives you your own uncoded copy of our tutorial project to experiment with freely. **You should see it** in the [Project Dropdown Menu](#) at top left of the app.
- **Select source 1** in the source selector. This will display the text from this source.
- **Practice highlighting causal claims** in the [Source Text Viewer](#) to **create links** in the [Create Links](#) tab.
- **View, edit or delete existing links** in the [Source Text Viewer](#) in the [Create Links](#) tab by clicking on the highlighted sections of text or on the corresponding links in the Map.
- **Watch** the causal relationships grow in the [Map panel](#)

Try this: view some examples of what you can do.

These examples are views of a real-life, anonymised [QuIP](#) project in the field of international development.

- Main factors [map](#).
- Main factors [table](#).
- Consequences of increased knowledge [map](#).
- Comparing groups (pivot heatmap) [table](#).

You can find them in the [Project Dropdown Menu](#).

Get help / read the documentation

The documentation you are reading now is embedded in the app via the [Help System](#) and is also available as a standalone [Guide](#).

Each section corresponds to a different part of the interface.

Within the app, you can:

- click the blue help buttons to find out more about the different parts of the app
- use the top bar buttons described in [Navbar](#) (Help, Guide, Bookmark, and Support chat).

But what even is causal mapping?

This Guide is all about how to use the app. For example it tells you how to use the different [filters](#). If you want to find out more about the ideas behind those filters, we have a new "[Ideas Garden](#)" with a more discursive look at the theory and everything that surrounds causal mapping. It's a work in progress but we are adding material every day, please bear with us.

Someone shared their work with you?

If they have to a particular map (or table), click the link and you will be automatically taken to view the corresponding map (or table), providing they have given you the correct permissions.

Alternatively if they have just asked you to log in at the app to explore a file name, say "project-x", log in and click on the dropdown list on the left-hand side and click on [project-x](#) to load it. If you can't find that file name, it means they haven't correctly shared it with the email address you logged in with.

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How CAUSALMAP WORKS

1. **Create Projects:** Create a project and upload PDF or docx documents into it as source material for analysis (we call each document a "source"). You can have multiple projects, and each project usually contains multiple sources.
 - See [Projects Panel](#) and [Upload Documents](#).
2. **Create Causal Links manually:** Identify and highlight causal claims within the source text; each highlight becomes a new causal link. After you have created multiple links you can view them as a causal map.
 - Use the [Create Links tab](#) and view results in the [Map panel](#).
3. **Create Causal Links using AI**
 - See [AI coding](#).
4. **Filter & Analyze:** Apply filters to your causal links focus on specific aspects of your data
 - Use the [Analysis Filters](#): Do qualitative causal analyseis on the selected links by filtering manipulate those links.
 - Explore [Map](#), [Factors](#), [Links](#), and [Pivot Tables](#).
5. **Collaborate:** Work together live with multiple users
 - Manage sharing in [Sharing and Permissions](#).



NEW FEATURES

The previous version of Causal Map, version 3, was already, as far as we know, the only software dedicated to causally coding causal claims within texts. Version 4 improves over Causal Map 3 in the following ways:

Speed and stability

- Fast loading and editing
- Resilient to poor internet connectivity
- Scalable to hundreds or thousands of concurrent users

Uploading and organising data

- Simplified data model to make it easier for you to import and manage your sources: we no longer break down source texts into separate statements. We treat each source text as one entire text. In cm4, there is no such thing as a statement.
- Easily create "Custom Columns" for each source (such as gender, location etc)
- Edit the data for your "Custom Columns" with a spreadsheet-like interface
- Complete management of all of your projects
- Simple upload of PDF or DOCX documents
- Tags to help you organise multiple projects
- Note that in cm3 we used to call projects "files"
- You can upload a project exported from cm3 as a new cm4 project

Managing labels

- New, more powerful and easier to use bulk editing of labels in the [links table](#) and the [factors table](#).

Filtering and analysing

- Almost all the existing links filters from cm3 are available plus
 - the option to include multiple versions of the same links filter, e.g. to narrow down a selection of links by different criteria successively
 - optional semantic filters like **cluster** and **soft recode** Require an AI subscription
- Analyse data with new pivot tables and graphs

Sharing and collaborating

- Share projects with collaborators; public projects viewable by anyone with access (login required)

- URL-based state saving for bookmarking (the same URL always takes you back to the same view)

These features require a Team subscription

- Real-time collaboration: Live updates when collaborators add links
- Interactive maps for live demonstrations

Help system

- Built-in help system
 - Help drawer with links to each section
 - Same contents used for separate Guide with links to each section

AI Coding

- Optional AI-powered state-of-the-art, paragraph by paragraph coding assistance. We call this "Human first, AI next".

Causal Map 3 features which will probably *not* be implemented:

- Deep support for standard questions across multiple sources
- Special treatment for closed questions
- Ability to view the text of multiple sources at once.



AI CODING

The **AI Coding panel** provides a streamlined AI coding workflow inside the normal left pane. It keeps the app structure familiar while reducing clutter.

You do not have to use the panel at all. With **Edit my data with AI** on, you can ask [MapCat](#) to code your sources in chat, and it works the same controls this section describes, confirming with you before anything starts. The panel is the manual route and is worth reading either way, because what MapCat sets is what these controls end up holding. The [end of this section](#) lists what you can ask MapCat for.

Quick start: If you have roughly 5–100 pages of text, you can usually ****just run everything**** and get decent results. Press ****One-click coding**** and confirm the short set-up modal, then let it run. You can then go back and adjust the coding (edit links, tweak prompts, re-run specific steps) if you want. For longer texts or high-stakes coding, work incrementally: use the ****source limit**** in Auto-code (1, 5, 20%, 50%, 100%) and the ****Links limit**** in Recode to process a sample first, check quality, then scale up. AI is switched on and off in the [Account panel](#) with the **"AI options switched on and active"** switch. If you choose an AI workflow when you sign up, it is turned on for you; otherwise you can turn it on there at any time. When AI is on, the **AI Coding panel** appears inline at the top of the **Create links** tab, ready to use.

AI usage consumes **credits** (see [Responses Panel](#)). **One credit is a fixed unit of AI spend worth about half a US cent**, so 100 credits is roughly 50 cents of usage. There is no fixed credits-per-page or credits-per-token rate: the app works out the real cost of each AI call from its token usage and the model's price, then converts that to credits at half a cent each. So the credits a task uses depend entirely on the model you choose, the length of your codebook, and how many passes you run. Credits renew monthly and do not roll over.

As a rough ballpark for a single Auto-code pass over 30 pages of text: around 30 to 40 credits on the default **Gemini 3.5 Flash**, about half that on **Gemini 3.6 Flash**, and about a sixth of it on **Gemini 3.1 Flash-Lite**, the cheapest model in the list. **Gemini 3.1 Pro Preview** costs about a third more per token than 3.5 Flash and usually writes more, so budget above that. One-click coding costs more than one Auto-code pass, because it also runs Revise codebook, Recode and the join-islands pass. Running extra prompt sections or re-runs multiplies the cost. These figures are estimates and will change as model prices change; the [Responses Panel](#) shows what your own runs actually cost.

Users with dedicated AI plans receive a larger batch of AI credits each month; other users receive 100 free AI credits per month (the free credits do not stack with paid plans). The per-plan allowances are listed under [AI credits](#).

The AI Workflow

The AI Coding panel sits at the top of the **Create links** tab, above the source text viewer. The Sources bar, the right-hand output tabs, and the **Create links** / **Filter links** / **Assess links** tabs all stay where they are; the panel does not take over the screen.

The panel is broken down into five straightforward sections (filtering is a separate step, see below):

1. **One-click coding:** Pipeline runner with a **set-up** modal first. Press **One-click coding** to choose **level of effort** (Flash vs Pro for the AI model slots), **Skip coded**, **Filter on finish**, and (if the project has links) whether to delete **every** link in the project or **only** links on the sources in scope, then confirm **Run**.
 - Pre-steps: clears Filter Links and turns the filter pipeline **on** before the modal.
 - **Scope:** One-click respects the Sources bar (empty bar = all sources, otherwise your current selection) and the sources % radio, exactly like the Auto-code button (no separate "code all sources" toggle). The modal states how many sources Auto-code will run on, including the % sample and when **Skip coded** removes already-coded sources (and **Run** is disabled if nothing is left).
 - **Auto-code prompt:** One-click coding uses the current Auto-code panel prompt and settings. The prompt is shown in the set-up modal before you run.
 - **Single source** in scope: only **Auto-code** runs (Revise codebook and Recode are skipped so labels are not merged across several AI passes). **Filter on finish** defaults off in that path but you can turn it on.
 - **Several sources** in scope: **Auto-code**, then **Revise codebook**, then **Recode**.
 - **Recode target suffix:** Choose blank (simpler — synthesised labels go straight into cause/effect) or e.g. `_recoded` (keeps raw labels, writes synthesised to temp columns so you can compare).
 - Per-step **Run** buttons still work on their own; the modal suppresses the extra confirms for the sequenced run after you confirm **Run**.
2. **Background:** Give the AI project context before coding. A status tick indicates whether enough background text is set.
3. **Auto-code:** This is where the AI reads your documents and extracts causal links.
 - You can choose to process a small sample first (e.g., **1** or **5** sources) to test your prompt, or process **100%** of them.
 - The "Skip coded" switch ensures you don't waste time and money re-processing documents that already have links.
 - Default model is **Gemini 3.5 Flash**.
4. **Revise codebook:** Once you have some causal links, the AI can review them and suggest a cleaner, more consistent list of factor labels (a "codebook"). The header tick shows whether the Recode codebook area currently contains suggestions.
 - Includes a **Target clusters** slider; see [Target clusters](#).
 - Optional **Use automatic pre-clustering** switch (default OFF).

- When pre-clustering is OFF, the AI tries to find the clusters directly from the factor list using the standard Revise codebook prompt. This prompt supports macro replacement: use **[number]** (or **[cluster_count]**) and the **effective** target cluster count is injected at run time (same as the slider logic below).
- When pre-clustering is ON, the app first groups factor labels semantically using embeddings, then sends those clustered groups to the AI with a separate labelling prompt plus a **Representatives per cluster** slider (8 to 20, default 8).
- Pre-clustering is more systematic than asking the AI to find all clusters "in its head" from a long raw list. It reduces the black-box / WEIRD-data risk a bit, and may make it easier to preserve more unusual or divergent concepts instead of collapsing them into whatever the model finds most typical.
- Default model is **Gemini 3.5 Flash**.

5. **Recode:** Apply the AI's suggested, cleaned-up labels back to your existing causal links. Paste the codebook (from Revise codebook or your own), add a recode instruction, and run.

- In **Semantic** mode (formerly "AI factors") the AI returns index mappings (row → codebook item) rather than full label text, reducing tokens and improving reliability; in **Causal** mode (formerly "AI links") it returns a recoded cause and effect for each link.
- **Semantic** and **Causal** each have their **own recode-instruction box, default and history** (separate prompt channels), and only the box for the selected mode is shown. Semantic default: *"For each raw label give me the number of the best-matching codebook item by meaning. Use 0 when no codebook item fits. Never invent labels."* Causal default: *"Recode each link to the codebook: pick the best-matching label for its cause and for its effect, using the link quote for context."*
- **Skip recoded:** When on, only processes links that have at least one unrecoded label (cause or effect). Use this when recoding again to focus on remaining work.
- **Links limit** (1, 5, 20%, 50%, 100%): When not 100%, a random sample of links is recoded. Non-sampled links keep their existing recoded values (or stay blank on first run).
- The header progress bar is segmented: grey = empty recoded fields, orange = recoded equals original cause/effect, green = recoded non-empty and different.
- Default model is **Gemini 3.5 Flash**. After coding, **filtering** happens in the adjacent **Filter links** tab (the normal Filter Links pipeline). When **Filter on finish** is **on** in the One-click set-up, completing the run applies these analysis filters to the pipeline: **Factor Frequency** (top 12, counted by **citations**) → **Link Frequency** (top 30, counted by **citations**). The global **Label set** controls which **cause/effect** columns Recode writes to (no separate "recode suffix" in this panel).

What the coding prompt decides

The conventions the AI follows when it turns text into links are defined by the **Auto-code prompt**, which you can read and edit in the Auto-code section (it is also shown in the One-click set-up modal before a run). That prompt is the single source of truth, so to see exactly how any case is handled, or to change it, read or edit the prompt rather than relying on a summary. In outline it covers:

- **Hierarchical labels:** factors can be labelled **General**; **specific** (two or more free levels), most general first, so the map can later be zoomed out by truncating at a semicolon. Whether to use them is a choice: it suits a large corpus with a multi-level scheme, and is left off for a small project or a fixed codebook. When it is on, the coder is also told what the top (most general) level should read like, for example a social-science concept, an actor name, or a codebook's general categories. You can ask MapCat to turn hierarchical labels on or off.
- **Actor focus** (default on): factors name who or what the cause and effect are about, with the actor as the subject and no passive voice, which suits maps that track who affects what. You can ask MapCat to turn this off for a concept or thematic map, and the coder then labels in the source's own words instead.
- **Opposites:** a leading ~ marks the reverse pole of a factor (for example ~**Employment** for "less employment"), so both polarities share one node.
- **Failed influences:** a link aimed at an outcome that did not follow is still coded, with the **failure** column set to 1 and the arrow pointing at what the influence was for.
- **Every segment accounted for** (default on): the coder splits each source into short segments and must account for each one, finding the causal claims in it or recording that it has none, rather than stopping once it has found a few links. This raises recall on long chunks. To keep it reliable, chunks are processed at 16,000 characters even when you set a larger chunk size.

The prompt states the full rules and their interactions; this list is only an orientation.

One-click coding (AI)

- Sequencer for the AI pipeline, with one **set-up** modal (see step 1 under **The AI Workflow**).
- **Run** starts **Auto-code**; with **more than one** source in scope it continues with **Revise codebook** then **Recode**, stopping on the first non-successful stage. With **exactly one** source in scope, it **stops after Auto-code**.
- One-click uses the same panel prompts/settings as the separate step buttons, while skipping the extra per-step run confirmations after you confirm the one-click set-up modal.
- Existing links on the sources in scope are always replaced: they are deleted just before the run so the new coding does not duplicate them. Links on other sources are kept. The set-up modal shows the counts.
- **Recode target:** Use the global **Label set** below the Sources bar. Create a new suffix there first if you want Recode to fill **cause_suffix** / **effect_suffix** instead of only the default columns.
- **Join islands (final step, default on):** after coding, one-click runs a join pass so the map comes out connected rather than left in many small islands. It merges near-duplicate factor labels and adds cross-island links, each backed by a verbatim quote from the same source. These added links are the least certain part of the map, so review them if precision matters. The join stays within a single source; connecting across sources is Recode's job. If it fails it is skipped and the rest of the run still completes. The pass decides for itself when to stop: when the map has come out in one piece, when a round adds nothing, or when the island count stops falling (it allows one grace round, because a merge can enable a bridge in the round after it). It reports which of those ended it, so a run that

converged early does not read as one that ran out of budget. Through MapCat you can set the join to **off** or to **merge only** (tidy and consolidate labels without adding any new links); leave the number of rounds alone, because a number you name overrides the stopping rule above. **merge only** suits reproducible coding or very long sources where no invented links are wanted. You can also name the model the join pass should use, separately from the coding model, for example "code with Flash then join islands with Pro".

Background (AI)

- Sets shared project context used by AI coding prompts.
- The status tick indicates whether enough background text is present.

Auto-code (AI)

- Runs AI coding across selected/all sources using your prompt and model.
- **Layout (top → bottom):** **Model**, **Skip coded**, **Add source prompt**, and **source limit** row; then the **Prompt sections** editor; then **Advanced** (chunk size, concurrency, temperature, thinking, etc.).
- Use source limit + skip coded options to test quickly and avoid rework.
- **Add source prompt** (switch): when ON, each source's optional *Source Prompt* (edit above the source text when viewing a source) is prepended to your main Auto-code prompt for that source. Saved per project in the browser. Use when sources need different context; skip when one background prompt in **Background** is enough.
- **Status line** under the settings shows progress, per-chunk detail, and stop — same behaviour as the former "Code with AI" card.
- **Prompt sections:** use **Add section** / **Remove** in the UI to split one saved prompt into reproducible iterations. Internally this is still stored as one prompt with **====** separator lines. Later sections see prior user/assistant turns. Only the last iteration's result is written to links; all iterations appear in Responses. This is best for workflows where coding is genuinely better in stages, such as first building the network, then adding columns like Time or Certainty, then running a checking pass.
- **Standard prompt builder:** the button above the sections opens a panel of tested prompt parts — the core coding rules plus optional extensions (hierarchical labels; in-vivo labels, which is actor focus turned off; opposites; despite; packages; a whole-network framing; and the sentiment, type and emotion columns), a prune pass, and your project codebook. The ticks are built from the prompt manifest rather than a fixed list, so a part added later appears here on its own. Tick what you want and **Replace prompt:** the composed prompt lands in the section cards, fully editable, and saves to history as normal. Parts with prerequisites tick them automatically (despite needs opposites).
 - **Project codebook:** one of the ticks is your own codebook, the factor labels saved under Project Details > **Edit codebook**. Ticking it puts the whole list into the prompt, so the coding run works to your labels instead of inventing its own. The tick is greyed out until the project has a codebook, and it shows how many labels it will use. Two ways to use it:

- **Guided** (the default): the coder uses one of your labels wherever one fits, and writes a new label in the same style where none does. Your list grows.
- **Strict**: the list is closed. The coder reproduces a label as written, and does not code a claim it cannot express in your labels. Your list stays exactly as it is, and some claims in the text go uncoded, which is the point of choosing it.
- On a project using hierarchical **General**; **specific** labels, the codebook is read as the list of general concepts, and the specific part after the semicolon still comes from the claim.
- MapCat reaches the same thing from chat: it can read the codebook, add labels to it, and put it into the coding prompt in either mode. Asked that way it appends the codebook section to the prompt you already have, rather than replacing your prompt with a composed one.
- This changes the prompt for the NEXT run. To bring links you have already coded onto the codebook, use **Recode** instead.
- **Per-section model**: each section after the first has its own model dropdown, defaulting to **Main model**. Use it to code with a cheap high-recall model and prune with a stronger one (for example a Flash model coding and Gemini 3.1 Pro Preview pruning), which testing showed removes most wrong links while keeping the good ones. All sections in one run must be from the same provider.
- **Rerun from here**: each prompt section has a small rerun button. Use it to continue a stable multi-section prompt without paying again for earlier successful stages, not as an open-ended chat workflow for coding maps. Section 1 reruns normally. Later sections reuse the latest successful earlier iteration history only when the earlier source text, prompt sections and chunk bounds are unchanged; otherwise the run fails loudly. You can add new sections under a successful run and rerun from the first new section.
- **Confirm** before a run shows model, chunking, word count, and cost estimate. **Stop** cancels after current chunk tasks finish.
- **Chunk size** (Advanced) defaults to **16K** characters. Larger chunks give the model more context per call and fewer islands to join afterwards; lower it only if a model struggles with long inputs. The every-segment-accounted-for pass caps the effective size at 16,000 characters even when you set a larger value.
- Timeouts scale by model and iteration count (cap ~540s total). **Concurrency** (1–5) is in Advanced; raise for speed, lower if you see 429/timeouts.
- **Which prompt a run used**: every confirmed run saves the prompt it sent and stamps that prompt's id on the run, so a map can be traced back to the wording that made it. Ask MapCat to list this project's coding prompts with the runs each one produced, and to put one back in the panel. It restores by id rather than by "the most recent", because the panel always shows the newest saved prompt and a later run moves it, which is what used to make a prompt look as though it had reverted on its own.
- **The prompt a run sends is not quite the prompt you see**: the run appends the per-segment accounting instructions (see [What the coding prompt decides](#)) and, when it is on, each source's own Source Prompt. Ask MapCat to read out the effective coding prompt if you want the exact string that goes to the model.

- [Tips on using the prompt history](#) (same chrome as other prompt fields).
- Default model is **Gemini 3.5 Flash** (precise and EU-resident; in testing it produced very few wrong links at moderate volume). One-click coding inherits this default.

While a coding run is going

- Before a run starts, it shows how many sources are in scope, coloured by how much of the project that is: red for every source, amber for a large slice, grey for a small one, with the chunk count, chunk size and model underneath. A five-second countdown then runs, so you can press **Stop** to abort before any AI work starts if the scope looks wrong. One-click runs skip the countdown, because they confirm the scope earlier, but still show the header.
- While the run works, a progress panel shows the current chunk, an estimated finish time, and a **Stop run** button. The browser tab title also tracks progress, so you can watch it from another tab.
- Press **Continue in the background** to hide the panel and carry on working. The run keeps going on the server and tells you when it finishes; reopen the panel any time from the AI Coding panel. Server runs are safe to close the tab on; the panel says so once the run is server-side.
- Only one coding run happens at a time. While a run is live the **Run** buttons become a stop control and the other AI actions are greyed out, so you cannot start a second run over the same sources by accident.

Revise codebook (AI)

- Suggests a cleaner consolidated codebook from existing links.
- Use this after you have enough coded links for a representative sample.
- Header tick indicates whether the Recode codebook area currently has content.
- **Target clusters:** see [Target clusters](#).
- Optional **Use automatic pre-clustering** switch (default OFF).
- With pre-clustering OFF, the AI clusters the factor list directly from the Revise codebook prompt. That prompt supports `[number] / [cluster_count]`.
- With pre-clustering ON, embeddings are used first to group labels semantically, then the AI only has to label those grouped clusters. This is a bit more systematic, less dependent on the AI doing all clustering internally as a black box, and may help preserve unusual or divergent concepts.
- Pre-clustering also adds a **Representatives per cluster** slider (8-20, default 8) and uses a separate labelling prompt.
- Default model is **Gemini 3.5 Flash**.

Target clusters (Revise codebook)

- The **Target clusters** control is a slider with **50 positions**. The **far left** is **Default**; moving right sets an explicit target of **2** through **50** clusters (one step per cluster count).
- **Default** (far left): the app derives a target count K from the number of **unique factor labels** n in the **current filtered pipeline** (same scope as Revise codebook): $K = \min(\lfloor n/3 \rfloor, 25)$ — at most **25**, or roughly **one label in three** as clusters, whichever is smaller.

- **Explicit** positions (not Default): the requested K is the number shown by the slider (**2–50**). If K is **greater than** n , the run uses n instead (you cannot have more clusters than distinct labels); the app may show a short notice when that cap applies.
- Pre-clustering, embedding clustering, and `[number]` / `[cluster_count]` in prompts all use this **effective** K .

Recode (AI)

- Applies your codebook back onto existing links, turning raw factor labels into cleaner synthesised ones.
- The **Recoding** radio buttons map onto the kinds of recoding (see [Different kinds of coding and recoding](#)). Overall the four run in order of **increasing cost and increasing quality**, left to right: Magnetic, then Semantic (formerly “AI factors”), then Causal (formerly “AI links”), then Hard. Pick the cheapest one that is good enough for your data. The names say what the model reads. Magnetic and Semantic both judge a label by what it means, one by vector and one by reading it. Causal judges the claim the link makes, which is why it needs the quote.
 - **Magnetic** (soft recoding): embedding similarity to codebook lines, the same magnet machinery as the pipeline's soft-recode path, with a similarity threshold. No AI call, and only as good as the embedding space.
 - **Semantic** (factors recoding, formerly “AI factors”): the model relabels each unique factor label to its best-matching codebook line, reading the label and nothing else. Cheap, because it costs one row per distinct label however many links carry it, and blind to what any of those links actually claimed. You can bring other factor columns into play (citation count, source count) when you write the instruction.
 - **Causal** (links recoding, formerly “AI links”): the model relabels each link from its cause, effect and the actual quote, so it judges by the causal claim rather than by the wording, almost like recoding from scratch. It can also send the same label two ways in two links where the quotes show it doing different work, which Semantic cannot do at all. Dearer, because cost follows the number of links. This is the default.
 - **Hard** (hard recoding): re-codes the source text from scratch against the codebook rather than relabelling existing links. Pick the sources (**Coded sources**, the ones already coded, or **All sources**), confirm, and it re-runs AI coding with the codebook as a closed list (the default instruction forbids inventing labels; your own instruction may relax that), replacing those sources' links. It uses the Auto-code model and chunk settings.
 - The names refer to the mapping method, not "soft recoding" in the filter sense.
 - **Watch the batch size**: with a large set of links and long quotes, a recode run is split into multiple calls, and each call can settle on slightly different labels. If you want a small fixed set of top-level labels, develop them first with Answers mode or the Cluster part of the Soft Recode filter, then recode against that codebook.
 - **Recode target**: the global [Label set](#). The default set writes the standard cause/effect labels; a named set writes that set's own cause and effect columns instead, leaving the default pair

untouched.

- Supports sampled recoding and skip-recoded behavior (skip-recoded only applies when using a non-default label set).
- Header bar shows recode coverage mix across all cause/effect recoded fields.
- Default model is **Gemini 3.5 Flash**.

Filter links (AI)

- This is the same Filter Links workflow, run from the adjacent **Filter links** tab (not embedded in the AI Coding panel).
- Use it to refine/select links before reviewing outputs on the right.
- When **One-click coding** finishes with **Filter on finish** enabled, the app applies top-12 factor frequency (citations), then top-30 link frequency (citations) (no longer injects the deprecated Temporary Factor Labels filter).

Advanced Settings

Each section header is clickable and opens/collapses its settings panel. Section headers also include contextual **Help** buttons. The advanced sections are inline (not flyouts), and only one section is expanded at a time.

Inside advanced panels you can:

- Edit the exact **Prompt** the AI uses.
- View your prompt history and load previous prompts.
- Change the **AI Model** (e.g., switch to a "Pro" model for complex reasoning, or a "Flash" model for speed).
- Tweak technical settings like chunk size, concurrency, and temperature.

Coding by chat instead

Everything above can be asked for in words. With **Edit my data with AI** on and an AI plan, [MapCat](#) drives this panel for you.

- **Code my sources.** MapCat asks a couple of plain questions about how you want them coded, sets the controls, and starts the run. Ask for the plain coding pass and it runs **Auto-code** alone; ask for the whole pipeline and it runs one-click coding, with revise, recode, join islands and the finishing filters behind it.
- **Set up before running.** Level of effort or a named model, chunk size, skip coded, filter on finish, which sources are in scope, hierarchical labels on or off, actor focus on or off, the recode mode, the recode instruction, and the codebook.
- **See what it chose.** Before a run, MapCat shows a default-collapsed **coding decisions** card listing the conventions it settled on, and the same decisions go into the run's logs, so a map can be

explained afterwards rather than guessed at.

- **Afterwards.** Ask it to join up a fragmented map, suggest a revised codebook, recode the links to a codebook you agree in chat (which is how you code top-down to your own themes or theory of change), or delete the coding for a set of sources and start again.
- **Trace and restore prompts.** Ask which prompt a run used, what you have tried on this project, and to put a particular one back in the panel.

Every coding run confirms in the chat before it starts. That card names the sources in scope, so you see what is about to be coded before you agree to it.



NAVBAR

 **What you can do here:** Use the buttons at the top of the app to open Help and the Guide, bookmark views, contact support, and manage your account.

About menu (version + What's new)

- Use the About dropdown at top-left to see the **current version** and open the **What's new** changelog.
- If you need to check your version quickly, see: [Do I have the latest version?](#)

Bookmark button

- Click the yellow bookmark icon to save the **current view** (map/table state) as a bookmark.
- A small popover opens (**Add / update Bookmark**) where you can add an optional Description.
- Open **Existing bookmarks** to see current-project bookmarks in one compact list (number, description, and quick actions like Load/Copy).
- To update one, click the row's **Update** icon, then click **Overwrite**. **That bookmark is fully replaced by your current screen state.**
- Bookmarks are managed in: [Bookmarks](#) and [Bookmarks and Reports](#).

Guide button

- Opens the standalone Guide (same content as the in-app help, but as a full page).
- See: [Help System](#).

Help button

- Opens the in-app Help drawer where you can **search** and follow links; internal links also highlight the corresponding UI element.
- See: [Help System](#).

Chat to the support team

- Click **Chat to the support team** to send us a message.
 - We get notified and will reply in the same chat window.
 - If you have a new reply waiting, the chat button turns yellow until you open it.

Account / Login

- If you are not logged in, click **Login**.

- When logged in, the user menu lets you open the **Account** panel and **Logout**.
- **Subscription badges** in the navbar show your plan and upgrade nudges; see [Subscriptions](#). An **AI** ✓ or **AI** ✗ badge shows AI status (tooltip: "Manage AI in Account tab"); click it to open the Account tab.



THE LEFT AND RIGHT PANELS

The app uses a two-pane layout with a draggable border between them (default split 30:70).

The left hand side of the app is all about selecting sources then creating and filtering links.

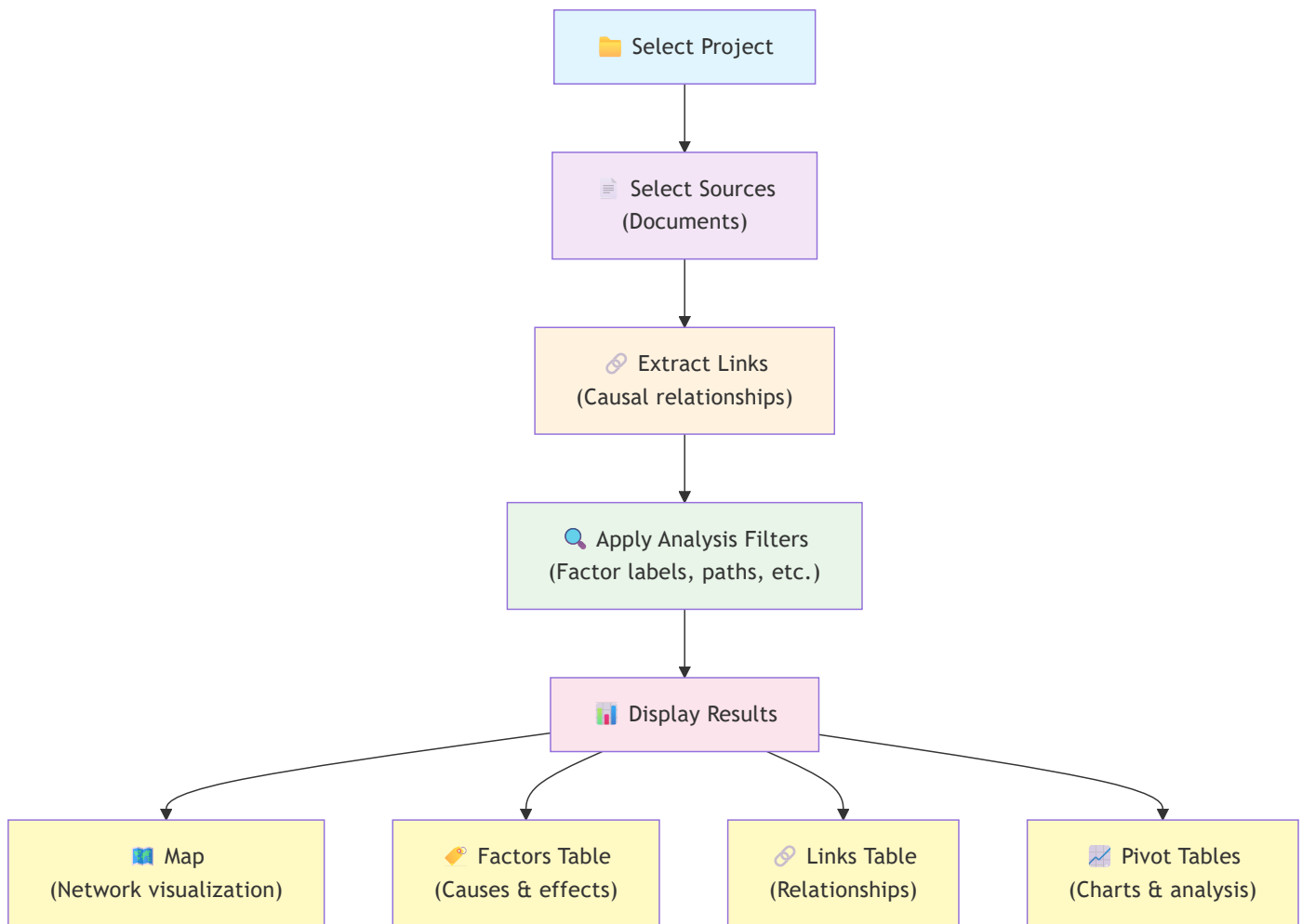
The right hand side (the pink tabs) is all about presenting the results.

Left-hand side

- **Project Dropdown Menu**: select a project including its links and documents
- **Sources Dropdown Menu**: choose one or more sources. **(Leaving it empty includes *all* sources)**.
- **Create Links tab**: Read and code source text. In the **Source text** header you get current source ID, source info toggle, source Prev/Next, highlight navigation, and Help. If **two or more** sources are selected, a small **pen-to-square** icon on a pill in the Sources dropdown marks which source is open in the viewer.
- **Filter Links tab**: Do qualitative causal analyseis on the selected links by filtering and manipulating them.

The Links Pipeline

The diagram shows the Links Pipeline: The top four boxes here correspond to the left side of the app and are called the "Links Pipeline": each step selects and filters links. The resulting links are then displayed in the pink output tabs on the right side.



Right-hand side (pink tabs)

- **Outputs:** these all show the same filtered links from the Links Pipeline but in different formats
 - **Map:** visual network of links
 - **Factors:** editable factor list (toggle available to bypass analysis filters)
 - **Links:** editable links table (toggle available to bypass analysis filters)
 - **Pivot Tables:** additional analysis and charts

Right-hand side (other tabs)

The right-hand side also contains other tabs not influenced by the pipeline:

- **Help:** help drawer and docs
- **Projects:** manage projects
- **Sources:** manage sources
- **Settings:** application preferences
- **Account:** your account
- **Logs:** application logs
- **Bookmarks:** saved views
- **Responses:** AI logs and usage



TIPS FOR USING THE APP

Tips for Using the Dropdown Menus

There are many dropdown menus throughout the app.

- Most dropdowns allow multiple selections: you can select more than one thing at once
- Most dropdowns allow you to type and create new entries which are not already in the list.
 - Type part of a word and click "Create new..." to add new items
 - Press Enter to complete selections
- Pressing Tab always moves you to the next field (doesn't complete selection). See also [Search/replace](#) for bulk editing patterns.

Edit an item

- **Backspace editing:** Position cursor after an existing selection and press backspace to edit it

Click to the right of the item, then press backspace to edit the label, and press tab to complete.



Dropdown editing tip (backspace)

Delete the first and subsequent items using the keyboard

Click after the last item, use the left arrow on your keyboard to go back to previous items, then press Delete on your keyboard to delete items after the cursor.



Dropdown deletion tip (arrow keys + delete)

Tips for Using Tables

Most tables include:

- **Checkboxes** ☒ for selecting multiple rows
- **Bulk action buttons** when you have selected one or more rows (edit, delete)
- **Action buttons** within individual rows to apply actions (edit, delete etc) just to that row
- **Filtering** using the filter row below headers
- **Pagination** with 10/25/50/100 items per page
- **Re-ordering columns** by dragging the column headers
- **Sorting** by clicking column headers; use **Shift+click** to sort by multiple columns. The column you Shift+click becomes the primary sort, and the previous column(s) break ties. Shift+click the same column again to reverse its order.

If you want to sort by source count and then by citation count, click first on citation count (click again to sort in the opposite direction) and then shift+click source_count. Shift+click again to reverse the order. Now, we have sorted by source_count but citation_count is used to break any ties.

Tips for Using Prompts and other text windows

Require an AI subscription

When you use text windows, your texts are automatically saved so you can reuse them later.

Text history is available in:

- **AI Coding** (Process Sources tab)
- **AI Answers** (Answers tab)
- **Map Vignettes** (Vignettes tab)
- **Soft Recode filters** (both label prompts and magnet lists)
- **Auto Recode filter**

How to use text history:

- **Dropdown menu:** Lists saved prompts (newest first). The first real entry is always the **most recently saved** text for that channel.
- **< and > buttons:** Navigate between newer and older prompts (tooltips note that running also auto-saves the current text).
- **Save (disk) button** (where shown): Saves the current text to project history; tooltips note that **running the action also auto-saves** the same way.
- **Text area:** Shows the selected prompt and lets you edit it.
- **Expand button:** Optionally edit your text in a larger editor (multiple cursors, search/replace, etc.).
- **Trigger button:** Runs the AI with your current prompt and saves it to history.

What loads when you open a panel

- If you **already have saved prompts** for that feature, the box loads the **newest** one and the dropdown points at it — not whatever happened to be in the field from a bookmark/URL default.
- If you have **no** saved prompts yet, you see the **built-in default** for that feature (from app defaults / `prompts.json` where applicable) until you save or run.
- The dropdown can include a row — **Not in history** —: that means you are **not** tied to a saved row (e.g. you picked that option after editing). Normal use does not require it.

How prompts are organized:

- Your current project's prompts appear first (most recent at top)
- Then prompts from other projects you can access
- Each prompt shows when it was last used

- Duplicate prompts are automatically removed

Using the controls:

- Select any prompt from the dropdown to load it
- Use < and > buttons to move through your text history
- Edit the prompt in the text area as needed
- Click the action button (Process, Ask, Generate, etc.) to run it

Saving: Manual save and run both persist the current text. If you try to save text that is **already identical** to the **newest** saved prompt (same wording), the app skips writing a duplicate row.

Tip: Any lines beginning with // in your prompt will be recorded in the history etc but not actually sent to the AI. You can use this to make notes e.g. at the top of your prompt: "//Sarah's version with tweaked summary"



PROJECTS BAR

 **What you can do here:** Choose which Project (project) you want to work on. Use the File menu for quick actions like creating new Projects, uploading documents, or sharing your work with others.

- A small locked indicator at the top-right of **#project-selector-header** shows only when a project is read-only.
- An archived icon appears only when the project is archived. Archived projects are automatically read-only and only visible to owners, collaborators, and admins (not visible to the public even if marked public).
- **Versions shortcut** : if the current project has any **named versions**, a small history button appears just left of the pencil. It opens the same Versions window as the File menu, one click instead of three. Projects with no named versions do not show it.
- **New Project or Upload Sources** : Opens a modal to create a new project or add sources to the current project. Choose files and/or **Or paste text** (plain text becomes a source file named **pasted-text.txt**), then Continue. A **Show Advanced** switch (default off) skips the **Confirm Upload** screen and uses defaults (keep filenames as IDs, no custom columns, no split). When on, the Confirm Upload modal appears for adjusting filenames, adding custom columns, or splitting sources (including delimiter split for pasted text).

File menu

Quick access to common actions:

Manage the Current Project

- **Edit** : Modify settings and sharing
- **Upload sources** : Add documents (PDF, DOCX, RTF, TXT, MD) or XLSX with a **statements** tab to the current Project
- **Clone** : Create a complete copy of the Project under a new name
- **Merge** : Copy sources and links from selected projects into this one
- **Clone filtered** : Create a copy of the current Project but only containing the sources and links as currently filtered
- **Archive** : Hide from main list
- **Delete all links** : Remove every causal link in this Project (sources remain)
- **Download** : Export as XLSX
- **Versions** : Restore and create backups of this Project
- **Update sources (Excel)** : Upload an XLSX to update sources in-place ("round-tripping").
 - **Tabs**: Must contain a **sources** tab (lowercase). Other tabs are ignored.

- **Required (case-insensitive headers):** Must include an `id` column. IDs must exist in the current project.
- **Columns:** You can include only the columns you want to change. Any missing columns are left unchanged.
- **Updates:** `title`, `filename`, `content` (if those columns exist). Any other columns are merged into `metadata.custom_columns`.
- **Manage projects** : Opens the `Projects` tab as a shortcut

Create new project

- **Quick upload** : Create a new project or add sources to the current project (same as the green `+` button in the Files bar)
- **New Project** : Create an empty Project which you can then import sources into
- **Import XLSX** : Import a complete new Project from Excel ("round-tripping").
 - **Tabs:** If the XLSX has 1 tab, it is treated as the sources tab (whatever its name). If it has multiple tabs, it must contain `sources` (lowercase); `links` (lowercase) is optional.
 - **Long source text:** If the XLSX contains a `source_content_chunks` tab (exported by this app when source content exceeds spreadsheet per-cell limits), imports will rehydrate `sources.content` from it.
 - **Format note:** CausalMap can import many document/file formats elsewhere in the app, but a BathSDR-style QuIP Excel workbook is not supported by this XLSX importer.
 - **`sources` columns (case-insensitive headers):** source id comes from `id` (preferred) or `source_id`; source text comes from `content` (preferred) or `text`. Any other (non-empty) columns are imported as custom columns (into `metadata.custom_columns`).
 - **`links` columns (optional):** uses `cause`, `effect`, `sentiment`, `tags`, `source_id`, `selected_text` (and remaps `source_id` based on the old ids from the sources sheet).
 - Any other non-empty `links` columns are imported into `links.metadata.custom_columns`.
 - For round-trip consistency, incoming `custom_<name>` headers are normalized to `<name>` before writing `links.metadata.custom_columns`.
- **Import cm3** : Import a complete new Project downloaded from CausalMap3.
 - **Tabs:** Requires `links` and `statements` (lowercase). Optional: `sources`, `questions`.
 - **`statements` columns (case-insensitive headers):** requires `source_id` and `text` (optional `statement_id/id`, optional `question_id`).
 - **`links` columns used:** `from_label`, `to_label`, `quote`, `hashtags`, `sentiment`, `statement_id` (used to map links to sources via the statements sheet).

Project Dropdown

-  **Project** (dropdown): lists all the projects you have created or been invited to; selecting one opens it.
-  **On change:** the rest of the app resets to defaults (sources selection + analysis filters pipeline).

- On startup, the app auto-selects the most recent **viewable** project (owned by you, shared with you, or public).
- Admins can see all projects in the Projects table, but the dropdown never auto-selects or loads a non-viewable project; admins may open the Edit Project modal for non-viewable projects from the table only.

Project Details button

- A small pencil button sits to the right of the [Project Dropdown](#).
- Clicking it opens the Project Details screen, the same as you get by clicking the first item in the [File menu](#), which we describe next:

Project Details screen

Manage every aspect of the current project. You can reach this management pane by clicking Edit from the [File menu](#). Manage other projects by clicking the edit button in the corresponding row of the [Projects Panel](#).

 **What you can do here:** - **Open it:** File → Edit, or the edit icon in the Projects table. - **When it appears:** Also opens on project load/change unless you turn it off for this project. - **Save vs instant changes:** - Use the *Save* button to apply changes to **Name**, **Tags**, and **Description**. - **Archived, Locked, Public**, and **Collaborators** update instantly.

Details

- **Name:** Rename the project. Click *Rename* to save.
- **Tags:** Comma-separated tags for quick grouping/searching.
- **Description:** Free-text notes about the project.
- **Edit codebook:** Toggle to reveal a text area where you can list factor labels (one per line). These are added to the cause/effect dropdowns in the link editor; existing options are kept.
 - **AI coding does not read this list on its own.** The codebook here is for coding by hand. To make the AI code to these labels, tick **Project codebook** in the AI Coding panel's Standard prompt builder, or ask MapCat to add the codebook to the coding prompt. You choose how binding it is: **guided**, where the AI uses one of your labels wherever one fits and writes a new label in the same style where none does, or **strict**, where the list is closed and the AI drops any claim it cannot express in your labels. To bring links you have ALREADY coded onto the list instead, use Recode (see AI Coding) with the codebook pasted in.
 - MapCat can also read the codebook back to you, and add labels to it or replace it, if you would rather not edit the box by hand.
- **AI Processing Region:** Choose where AI processing occurs for GDPR/data residency compliance:
 - **EU (Belgium - europe-west1)** - Default. Recommended for EU data residency requirements.
 - **UK (London - europe-west2)** - UK has GDPR adequacy decision, suitable for EU/UK compliance.

- **US (Virginia - us-east5)** - US East region.
- Setting is saved per-project and auto-saves on change (with confirmation warning).
- All subsequent AI coding for this project uses the selected region.
- **Archived:** Hide the project from the main list and make it read-only. Archived projects are only visible to owners, collaborators, and admins (hidden from public view even if marked public). Applied immediately. Unarchiving restores normal visibility and editability.
- **Info line:** Created / Modified / Owner, plus counts for links/sources/words, and quick actions:
 - *Versions:* Open the versions manager.
 - *Delete embeddings:* Remove factor embeddings for this project (advanced).
- **Show on open:** Toggle "Show this screen when opening this project" at the top to auto-open or suppress this screen for this project.

Sharing

- **Locked:** Make the project read-only. Editing is disabled until unlocked. Applied immediately.
- **Public:** Allow all signed-in users to view the project (read-only). Applied immediately.
- **Collaborators:** See current collaborators, add by email, and choose permission:
 - *Viewer:* Read-only
 - *Editor:* Read & write

Bookmarks

- If available, view saved “bookmarked” views of your data and open the Bookmark Manager.

This screen also shows when a project loads or is changed, except:

- for new users (the help drawer is still being opened to welcome them),
- if you have already clicked "don't show" for this project.



SOURCES BAR

 **What you can do here:** Choose which source documents (e.g. interviews or reports) from your current project you want to focus on. You can select one or more sources. Use this to narrow your analysis to specific interviews, reports, or other source materials. - The text of the selected source is shown below in the Create Links panel. - Selecting these sources also fetches only their links and no others, starting off the [Links Pipeline](../links-pipeline/): only the links from the currently selected sources are available for further filtering, and are finally shown in the output tabs.

Factor label set (global)

Treat **label sets** like **different versions of the same document**: you keep a **default** “main” version (the usual factor labels on each link) and you can spin up **extra named versions** to experiment—try a different coding scheme, run [AI Recode](#) on a copy, or compare schemes—without changing the main story until you choose to. **Most people** finish by **copying one experimental version back into the default** so the map and tables show that version as the “real” one, while the other versions stay in the project if you need them later.

- **Which version am I using?** The **Label set** button sits under the Sources bar and shows the version you are on. Open it for the list of every version in this project; the one you are using has a tick. Pick another to view and edit **that** version’s labels in the [Links Pipeline](#), map, links table, and link editor. Switching refreshes those outputs so they match.
- **New experimental version:** choose **Duplicate “”** at the bottom of the menu, give it a name, and confirm. The app fills a new column-pair for **every link** with a copy of the version you were on, then switches you to it. Handy before Recode, so the AI writes into the experiment rather than the default. There is no way to make an empty version, and that is on purpose: a version with no labels shows blank for every link.
- **Delete a version:** the bin icon on its row in the menu. The default has none, because it cannot be deleted. Anything stored only in that version goes with it, and a saved view or bookmark that used it will show nothing.
- **Done experimenting — make it “official”:** choose **Promote to default**, which appears only when you are on a non-default version. That copies the version you are on into the **default** columns for every link. Your other named versions are **not** deleted. What the default held before is not kept anywhere, so duplicate it first if you might want it back.
- **What each version is:** the menu shows the name you gave it with a line beside it saying when it was made, which version it was copied from, what wrote it and how many links it covered. A version made before the app recorded that shows its stored name and nothing else.

- **Renaming** a version is not possible yet, though the name is now stored separately from the columns, so it is a small change rather than a rewrite of every link. Duplicate it under the name you want and delete the old one.
- **Temporary Factor Labels** (old filter in the pipeline): **deprecated** for new work—use this control instead; old bookmarks may still reference the filter.

Sources Dropdown

- 🖱️ **Sources** (multi-select dropdown): pick one or more sources from the current project (typing searches the list).
- 🖱️ **Empty selection**: means “all sources”.
- Selected sources appear as pills; when **two or more** are selected, a **pen-to-square** icon appears to the right of the pill for whichever source is currently open in the Source Text Viewer (not shown when only one source is selected).

Default behavior when switching projects:

- When you load a project via the **Project Dropdown** or **Projects Panel**, and nothing else specifies which sources to load, the app auto-selects the **first source** so the Source Text Viewer shows something immediately.
- When loading from a **URL/bookmark**, we do **not** change the sources selection. An empty Sources selection means **all sources** (by design).
- Source loading/viewing behavior is the same for **coded and uncoded** sources (having 0 links must not block selector state, text loading, or source navigation).

Source Groups sub-panel

🔗 **What this does:** Filter your analysis by participant demographics or document characteristics, using the [custom columns](../custom-columns/) you have defined for your project. For example, show only responses from "women aged 25-35" or interviews from "urban areas." Perfect for comparing how different groups see causal relationships.

Controls:

- 🖱️ **Field** (dropdown): choose which source metadata/custom column to group by (includes built-in fields like title/projectname).
- 🖱️ **Value** (multi-select dropdown): choose one or more values (list is filtered by Field).
- 🖱️ **Random N** (buttons): load a deterministic random sample of sources from the whole project (e.g. Random 5 / 10 / 20).
- 🖱️ **Random N / Group** (buttons): after choosing Field, load up to N deterministic random sources per value of that Field.
- 🖱️ **Clear** (button): reset the Source Groups selector.

The sampling buttons make a random selection, but deterministically, so the same sources are chosen again if you click the same button again.

The second, "Value" dropdown is filtered to show only valid values for the selected field. Previous/next buttons cycle through values of the selected group.

The effect is to retain only links where the selected custom column has the selected values.

This dropdown is NOT a filter and it does NOT get saved/restored in URL. It is a loader: when we click it, the app automatically loads corresponding sources into the sources selector. These sources then DO form part of the links pipeline and ARE restored from the URL.

There is a similar filter in the [Analysis Filters](#).



CREATE LINKS TAB

Qualitative causal mapping involves taking passages of text, e.g. from interviews or documents, and identifying sections which make causal claims. We highlight each of these sections and specify a causal factor at each end of each link (for example Lost job or Went hungry). This means creating a new factor or reusing an existing one. Usually we create these factors inductively as we code, and revise and review and consolidate them as part of the process, as with any other kind of qualitative content analysis.

To code a causal link,

- With your mouse, highlight a piece of text within the statement which makes a causal claim. Your selection must remain within one statement and must not cross into another statement.
- Watch how that passage is copied for you into the "Quote" window. (Usually, you don't need to think about this window: you can edit the text if you really need to but it **has to remain an exact quote of one part of the text.**)
- Start to type the name of the influence factors at the **start** of the link(s) which you are going to make, in the first drop-down menu.
- If there is an existing factor which matches what you want, you can select it.
- Otherwise, you will create a new factor with the contents of what you have typed; finish what you have typed with a comma or a tab character if you want to continue to select or create another factor.
- If you want to create more than one link, you can select or create additional factors in the same box (as shown in the video below).
- When you have finished, press Enter.
- Repeat the process in the other box to specify the factors at the **end** of the link (or ends of the links).
- Press the green Save button which is now active.
- The link is created in the Map window.
- When you have finished coding one source, click the right arrow in the **Source text** header to code the next source.

Source Text Viewer

✨ **What you can do here:** Read your source documents and create causal links by highlighting text.

When you highlight a passage that claims or implies that one thing influences or causes another, a popup lets you identify the cause and effect. This is where you do the core work of mapping out causal relationships from your source material, a process which we call **coding**

The text viewer shows full text from one source at a time. The **Source text** header contains source ID, source info toggle, source Prev/Next, highlight navigation, and Help. When you select several sources, it initially shows the first in the selection; with **two or more** selected, a **pen-to-square** icon on a pill in the Sources dropdown marks which source is on screen. You can:

- Highlight sections to identify causal claims. Highlighting opens the causal link editor.
- Examine and edit existing highlights by clicking on them.

Full text search (selected sources)

Below the **Sources** bar, **Full text search...** opens a small panel. Type **at least three characters** to search **source text** in the current scope (same rule as the Sources dropdown: **no selection = all sources** in project order; **more than one** source selected = only those; **exactly one** selected still searches **all** project sources for this tool—only the strict multi-select narrows the list). Matches are highlighted in the viewer in a **separate colour** from causal link highlights; they do **not** change the links table or filters.

Use **First / Previous / Next / Last** to jump between hits. The counter shows **hits** and how many **sources** actually contain a match (out of how many are in scope). When the next hit is in another source, the viewer **loads that source** without changing what is selected in the Sources dropdown (so **empty = all** stays empty). **×** clears the query and closes the panel; **clicking Full text search... again** while the panel is open does the same.

Inside the header, there is an info  icon which toggles open/shut a panel beneath it which shows the values of the custom columns for the current source e.g. gender etc.

Navigation Controls

Navigate sources (buttons in the **Source text** header):

- **< Previous source**
- **> Next source**

Navigate highlights within the current source:

- **⏮ First highlight in source**
- **⏪ Previous highlight**
- **⏩ Next highlight**
- **⏭ Last highlight in source** (useful if you haven't finished coding the whole text yet and want to see the last highlight)

Source selection is controlled by the Sources dropdown. When several sources are selected, the viewer starts on the first; with **two or more** selected, a **pen-to-square** icon on a pill shows which source is open (hidden when only one source is selected).

- **Multiple sources selected:** Main Previous/Next pages between the selected sources and only changes which source is displayed in the text viewer. It does **not** change the actual selected-sources set.
- **Exactly one source selected:** Main Previous/Next pages through **all project sources** (project order). In this mode it **does** change the selected source, so both source selection and text viewer

content update together.

- **No source selected (empty = all):** Main Previous/Next picks and then pages in project order like the one-source case, so selection changes as you page.
- These navigation rules apply equally when selected sources currently have zero links.

Dealing with long documents in the source text viewer

For documents longer than ~30-40 pages, the text viewer automatically splits content into manageable chunks for better performance.

- 🖱️ **Next chunk** (button at end of chunk): appears at the end of each chunk (except the last).

Visual Highlighting

Each section of coded text, each causal claim, is shown with a highlight.

For overlapping or identical highlights with multiple links, overlaps are shown with varying color opacity. Clicking on multiple highlights shows a link selector for each section.

- Multiple highlights shown with varying color opacity
- Click on overlapping highlights to select specific links
- When highlights overlap, the link chooser lists each link with its own cause to effect, since overlapping links need not share the same bundle.

Each link's quote is found in the source text even when small things differ, such as straight versus curly quotes, spacing, accents or the kind of character mismatches that come from PDFs.

When the AI codes your text, it sometimes returns a quote that does not quite match the source. Every AI quote is checked against the source before the link is saved:

- If the quote is found, the link is saved and lined up with the exact source wording.
- If it matches only roughly, the link is still saved but given the **unverified_quote** tag so you can review it.
- If the quote cannot be found in the source at all, the link is not saved, because the AI may have invented it. The summary at the end of a run tells you how many links were skipped or tagged this way.

To check the rough matches, filter the links by the **unverified_quote** tag.

Link Editor screen

Opens when you highlight text or click on existing links.

Fields:

- 🖱️ **Project** (read-only): current project name.

- 🖱️ **Source** (text): source document id for this link; editable and prefilled when you code from the text viewer. Save requires a non-empty value; if you change it from what it was when the overlay opened, the app asks you to confirm.
- 🖱️ **Cause selector** (multi-select dropdown + free text): choose one or more causes (project-wide list, sorted by frequency).
- 🖱️ **Effect selector** (multi-select dropdown + free text): choose one or more effects (same project-wide list). After you select Causes, the Effect dropdown boosts the most common effects of those causes to the top.
- 🖱️ **Quote** (text area): editable evidence text; supports ellipses like *Actual quote [this text is ignored] quote continues...*
- 🖱️ **Chain** (toggle): if on, saving keeps the editor open and uses the previous Effect as the next Cause.
- 🖱️ **Plain coding** (toggle): if on, saves a self-loop (Cause = Effect) tagged `#plain_coding` so it counts as “theme present” rather than a causal claim. Plain codings can be hidden with the `Exclude self-loops` filter.
- 🖱️ **Package** (toggle): if on, the Causes are recorded as claimed to work *together* on the Effect (a causal package, e.g. "you need both oxygen and a spark to make fire"). Each created link gets the same auto-generated code in its `package` column, and a small field appears for the **package type**: pick `AND`, `OR` or `DESPITE`, or type your own. With several Effects, the causes into each Effect form their own package (one code per effect). Filter packaged links later with the Everything filter on `package` (one package, or any non-empty code) or `package_type`; the links table also has bulk package/unpackage buttons for selected rows.
- 🖱️ **Tags** (multi-select input): add tags like `#hypothetical` or `check`.
- 🖱️ **Favourites** (3 toggle buttons): heart / exclamation / star.
- 🖱️ **Custom fields** (collapsed panel): choose a small subset of link custom columns to edit in-place for this overlay. You can also type a new field name here to create it for the project. The selection is remembered per project; untouched custom columns are preserved.

Actions:

- 🖱️ **Save** (button): create/update link(s).
- 🖱️ **Delete** (button): remove an existing link.
- 🖱️ **Cancel** (button): close without saving.

Update behavior:

- If you save with exactly one Cause and one Effect, the editor updates the existing link row in place.
- If you save with multiple Causes and/or Effects, the editor applies causes × effects and recreates rows for that cross-product.

Links in Causal Map only have one cause and one effect. You can add multiple causes and/or effects to the boxes, and the system creates all combinations when saving. So if you put `unemployment` and `violence` in the Cause box, and `stress` and `worry` in the Effect box, the system will create four links.

About the factor label dropdown menus

By creating links, you also create the names of your factors.

In Causal Map, a factor is its label. Once you create a label, there is nothing else to add.

Factor names which contain semicolons ; get special treatment as they separate the different parts of [Hierarchical factors](#) .

After beginning to create links between factors, already-coded factors will appear in the dropdown menus in the to and from factor boxes. For added convenience. The most frequently coded factors will appear at the top of this list

#doubtful? #hypothetical? Adding link tags

Link tags

Link tags are available as a special kind of memo when coding a link: you can use them to provide any kind of additional information.

There is no need to actually use a hash # at the start of a link tag, though you can if you want. Just use any unique single word which is easy to search and filter on, like #nutrition or nutrition# or nutrition–.

As usual in Causal Map, you can apply one or more tags, and you can either select existing tags or create new ones on the fly.

Later, you can filter the map (see [🌟 Transforms Filters: Include or exclude tags](#)) to show only links containing or beginning or ending with specific hashtags (or parts of hashtags), and also for links which *do not* contain specific hashtags or parts of hashtags.

You can also use tags to narrow down your searches in [🔗 The Manage Links tab](#).

You can display [tags on your map](#).

Conceptually, there are two kinds of tag.

Ordinary link tags

You can use any tag which does not begin with a ? to record any other information about the link, e.g.:

- respondent doesn't like this connection
- respondent feels good about the outcome
- for you, the analyst, e.g.
 - respondent is answering a different question
 - to tag links you want to come back and review.

Weak tags

Weak tags are a special kind of tag. They are *caveats*. If you use weak tags, you should make sure that by default your maps do not include any link with a weak tag.

This is just a convention, it makes no difference to the Causal Map app.

They begin with ? and are used to mark any link which you are not sure is always valid across the global context for the whole global map, for example:

- **the causal connection** is only valid for a specific context, e.g.
 - the respondent says this is only true for their village, not for other villages e.g. ?village X
 - a link is only projected for the future e.g. ?future
- you are unsure about **the claim about the causal connection**
 - a link is only a hypothesis e.g. ?hypothetical
 - you as the analyst are not confident in the claim e.g. ?doubtful
 - the source themselves are not sure e.g. ?source seems unsure
 - to add other qualifying information e.g. ?probably hearsay
 - to mark the fact that a connection is **weak or non-existent**, e.g.
 - Respondent makes a substantive claim that X does *not* influence Y, e.g. ?zero influence
 - Respondent makes a substantive claim that X only insignificantly influences Y, e.g. ?weak



FILTER LINKS TAB

Do qualitative causal analyses on the selected links by filtering or manipulating them.

 **What you can do here:** Apply filters to focus your analysis on specific aspects of your data. You can trace causal pathways, group similar concepts, filter by themes or demographics, and much more. Think of this as your analysis toolkit - combine different filters to explore your data from different angles.

The Filter System: overview

Use filters to narrow down and/or transform the links you want to study. Filters are applied in order, from top to bottom. You can drag and drop them to reorder them.

Position in the pipeline is part of what a filter does. Each filter sees only the links handed down to it, so moving one up or down can change the final result even when the set of filters is the same. The clearest case is any filter that counts: **Link Frequency** and **Factor Frequency** apply their threshold to the links they receive, so anything above them that removes links (Exclude self-loops, Exclude Tags, a Factor Label filter) changes the counts and therefore changes which links and factors pass. Transform filters (Zoom, Collapse, Combine Opposites and the recoding family) change the labels themselves, so anything below them works on the new labels. Only assume order does not matter when you have checked that nothing downstream counts or relabels.

Filter ordering rules

Any order of filters means something, so these are not restrictions on what you can build. They are the two orderings that usually give an answer to a question nobody asked. The app asks about them: it colours the filter pane and puts the question at the top, you can dismiss it, and it appears again the next time you change the chain. MapCat follows the same rules when it builds a chain for you, and will explain either one if you ask.

The order to build in: filters that choose which sources or links you are looking at, then tracing, then Combine Opposites, then the filters that rewrite labels, then the frequency filters, then Exclude self-loops.

Rule 1. Put tracing at the top of the chain. **Path tracing**, source tracing (path tracing with threads turned on), the **Pathways** filter and a **Factor Label** filter set to more than one step all exist to find causal paths that are really in the data. Put a filter that rewrites factor labels above them (**Zoom**, **Collapse**, **Combine Opposites**, **Remove Brackets**, **Soft Relabel**, **Soft Recode Plus**, **Cluster**, **Auto Recode**) and it merges factors that were two different things in the coding, so you trace a route through a junction nobody described. Put a frequency filter above them and it removes links the paths are made of, so you

find no route where the data has one. Either way you are tracing a map that is no longer the coded data. Trace first, and simplify afterwards for presentation.

Rule 2. Put the frequency filters at the bottom of the chain. Link Frequency and Factor Frequency count the links they are given, so they mean something different at every position. Put them after the filters that decide what you are looking at. Take an ego network: filter to the neighbourhood first and then to the most frequent material within it, and you learn what people said most about that factor. The other way round you keep the most popular material in the whole corpus and then cut it down to the ego, which is a different map and usually not the one you wanted. You can put an **Exclude self-loops** filter after them.

Rule 3. Put **Combine Opposites above the filters that replace labels.** Combine Opposites finds each pair by reading the labels it is given: a numeric tag such as **[~3]** against **[3]**, or a label starting with **~** against the same label without it. **Remove Brackets** deletes the square-bracket tags, and **Collapse**, **Soft Relabel**, **Soft Recode Plus**, **Cluster** and **Auto Recode** replace a label with a new one that no longer carries the marker. Put one of those above Combine Opposites and it finds no pairs at all, so the negative and the positive stay on the map as two separate factors and nothing tells you the pairing was lost. Combine the opposites first, then replace labels for presentation. **Zoom** is the exception: it truncates the hierarchy and keeps the marker on whatever it keeps.

More rules will follow. Maintainers: the rules, their wording and the checker are single-sourced in **js/filter-type-flags.js** (**FILTER_ORDER_RULES**, **checkFilterOrder**, **suggestFilterPosition**), which feeds the filter-pane warning, MapCat's prompt and MapCat's default insertion position. Change them there and update this section to match.

- **Default filter:** starts with a Factor Label Filter.
-  **Add Filter +** (button): insert a filter at the start or between existing ones.
-  **Enable/Disable** (toggle on each filter): turn an individual filter on/off.
-  **Remove X** (button): delete a filter.
-  **Collapse** (button on each filter): hide/show that filter's controls.
-  **Clear All X** (button): reset to the default single Factor Label Filter.

This row of controls stays at the top of the pane while the filters below it scroll, so Add Filter and the enabled switch are within reach however long the chain is.

The funnel icon on the **Filter links** tab itself is a second switch for the same **enabled** state: click it to bypass every filter, click again to turn them back on. The icon shows a line through it whenever the filters change nothing, either because the pipeline is off or because every filter in it is off.

Transform filters vs permanent relabelling

Most filters leave factor labels untouched, but these 'Transform filters' filters temporarily relabel factors:

- **Zoom**

- Collapse
- Remove Brackets
- Soft Recode Plus
- Auto Recode
- Soft Relabel
- Temporary Factor Labels
- Cluster

No filters actually change your original coding.

💡 **Tip:** If you want to permanently rename your factors (write new labels to storage), there are several ways to do that:

- Search and replace factors
- Search and replace links

For example, after clustering (which may give labels like C11), click a factor on the map and rename it (e.g., "Wellbeing") to save the new name permanently.

Link Frequency Filter

 **What this does:** Focus on the most important causal relationships by filtering out rare ones. Choose "Top 10" to see only the most frequently mentioned connections, or set a minimum threshold like "at least 3 sources" to ensure reliability.

- 🖱️ **Threshold** (slider 1–100): choose N (for Top) or k (for Minimum).
- 🖱️ **Type** (radio buttons): **Top** vs **Minimum**.
- 🖱️ **Count by** (radio buttons): **Sources** vs **Citations**.

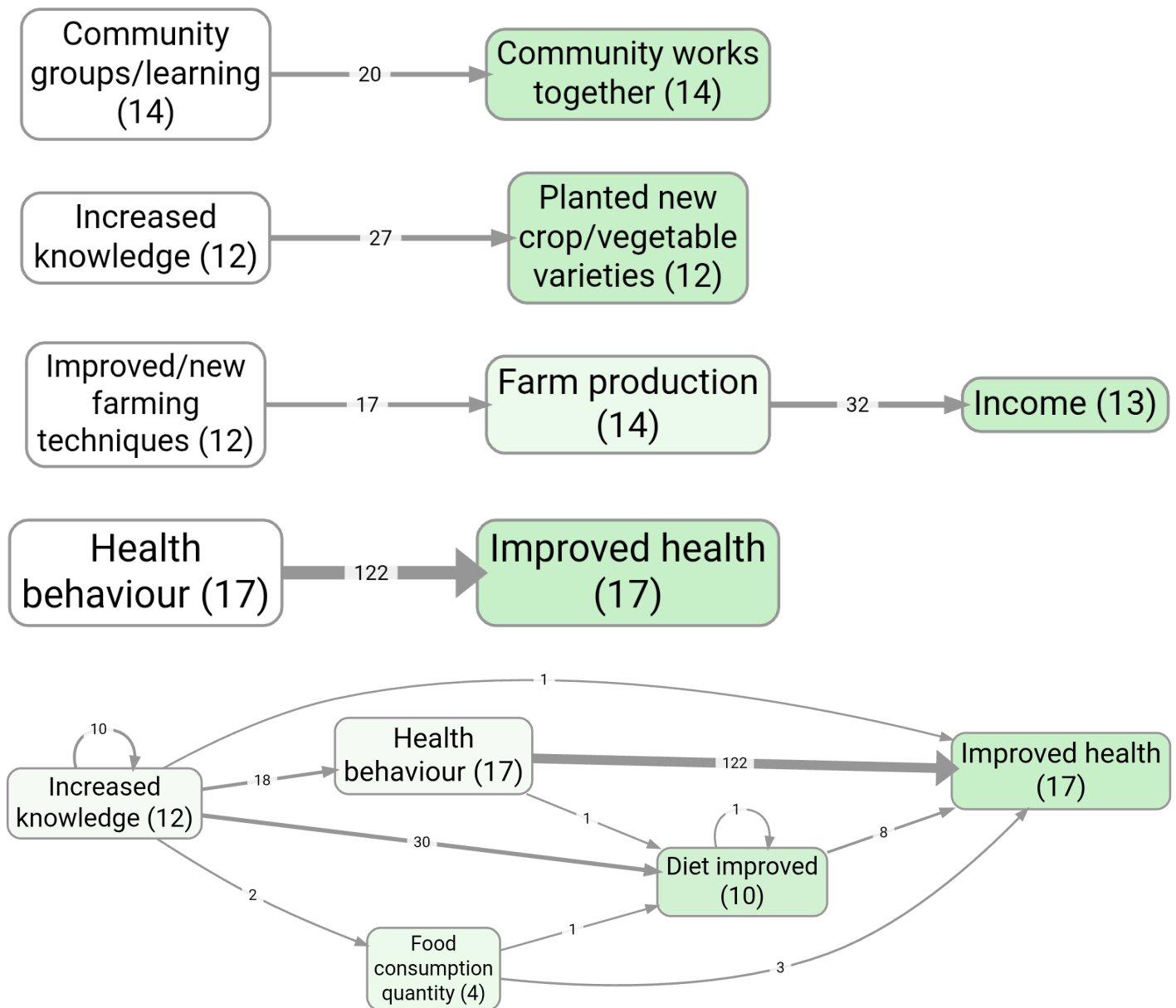
Put this filter at the bottom of the chain (rule 2). It counts the links it is given, so higher up it ranks the whole corpus and the filters below it only trim what survived.

Examples:

- **Minimum 6 Sources:** Only links mentioned by 6+ sources
- **Top 6:** Only the 6 most frequent link bundles

Example bookmarks (contrast):

- Link frequency (top links): #1124
- Factor frequency (top factors): #266



By default, setting the slider to 6 means we are selecting only links with at least 6 citations.

If you switch to “Sources”, we are selecting only links with at least 6 sources.

If you switch to “Top” we are selecting only the top 6 links by citation count, etc. The selection respects ties, so that if there are several links with the same count, either all of them or none of them will be selected.

Factor Frequency Filter

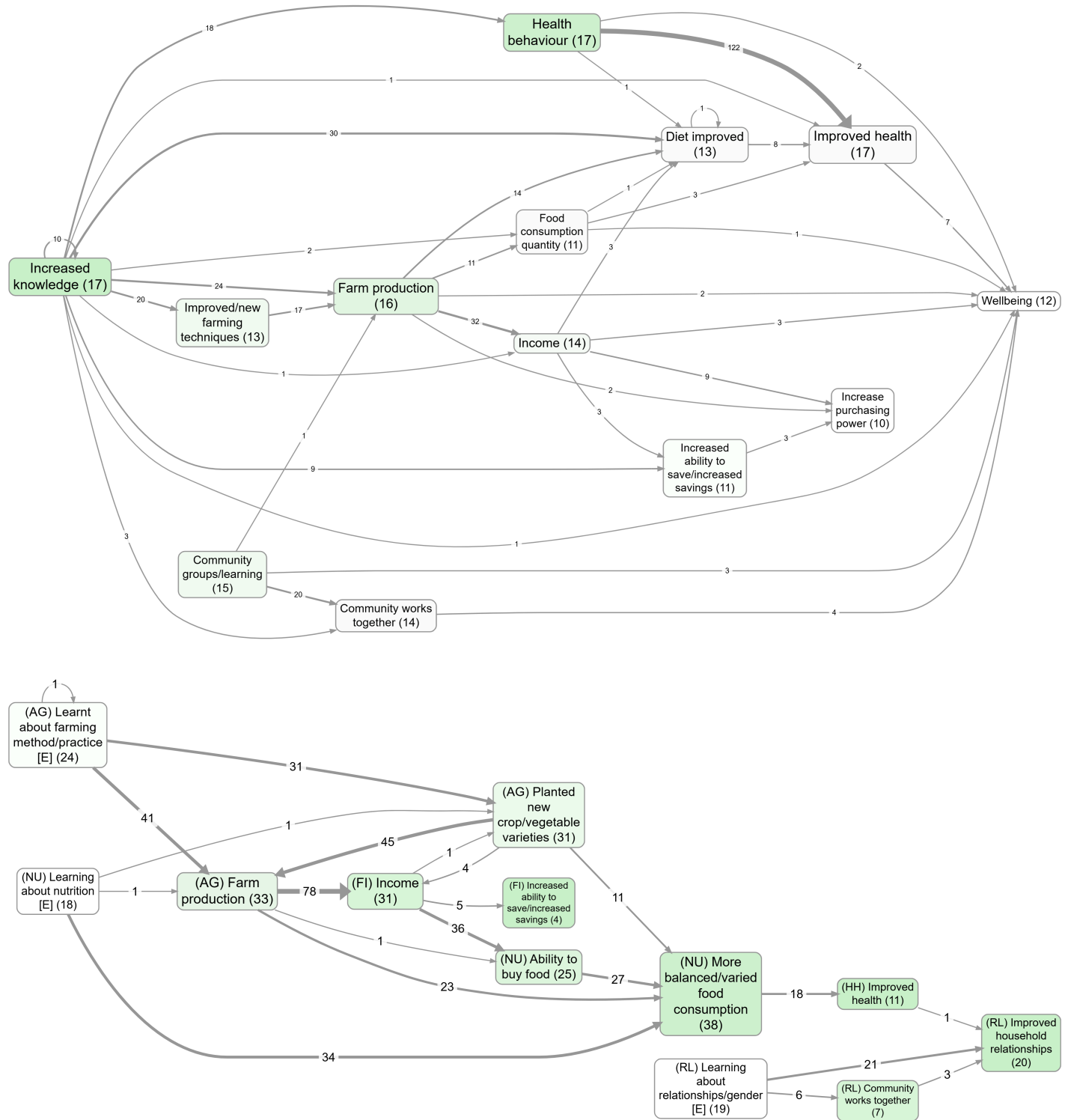
Ideas Garden: [Factor and link frequency](#)

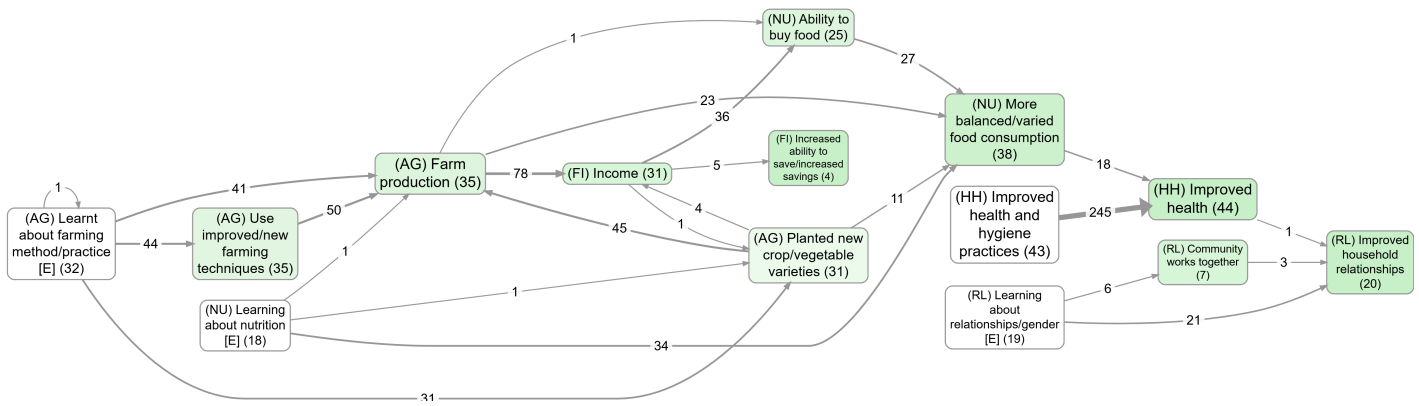
 **What this does:** Similar to Link Frequency, but focuses on the most important factors (causes and effects). Show only the most frequently mentioned themes or concepts to identify the key issues in your data.

Same controls as **Link Frequency** but applies to factors instead of links, and the same rule: it goes at the bottom of the chain (**rule 2**).

Example bookmarks:

- Factor importance colouring (top factors)
- Top factors (no zoom): #983 vs with zoom: #984





Source Groups filter

 **What this does:** Filter your analysis by participant demographics or document characteristics. For example, show only responses from "women aged 25-35" or interviews from "urban areas." Perfect for comparing how different groups see causal relationships. This is very similar to the [Source Groups widget](../source-groups-sub-panel/) in the Sources sub-panel, but having it here too means you can add multiple source filters to the pipeline.

- provides
 - a prepopulated dropdown called Field with all the metadata fields plus title and projectname
 - another multi-select called Value. Multiple values work as OR: either/any count as a match
 - a previous/next button pair to cycle through values of the selected group
 - Example: Add two Source Groups filters in the pipeline to combine criteria (e.g., first filter Field = gender → Value = women, then another filter Field = region → Value = X) so you see links from women AND from region X.

Example bookmarks:

- Village 1 — splitting by group
- Village 2 — splitting by group

Everything Filter

 **What this does:** Filter your analysis by any characteristic of your links and their sources. Useful for anything not covered by the other filters, for example, - show only links with negative sentiment - Show only links from one source Also shows source separators and their values, often used for common sections within multiple sources texts.

- **Field dropdown** with all fields in the links table
- **Value selector** filtered by selected field
- **Navigation buttons** to cycle through values
- **Clear button** to reset

Animate Filter

► **What this does:** Picks one link field (for example **year**) and drives map animation frames from that field, respecting pipeline order.

- **Field dropdown:** choose the link field used for frame values.
- **Cumulative toggle** (default off): when on, each frame includes all values up to and including the current one.
- **Refresh button:** reload available fields.

Notes:

- The filter does not directly remove links during normal pipeline runs; it tags frame values at its exact pipeline position.
- Map play controls appear when an Animate filter has a selected field.
- Filter order matters: moving Animate changes which links are visible in each frame.

Factor Label Filter

Ideas Garden: **Focus or exclude factors**

 **What this does:** Show links connected to factors you care about (e.g. "Education"). Choose how many steps to look upstream (causes) and downstream (effects).

Widgets:

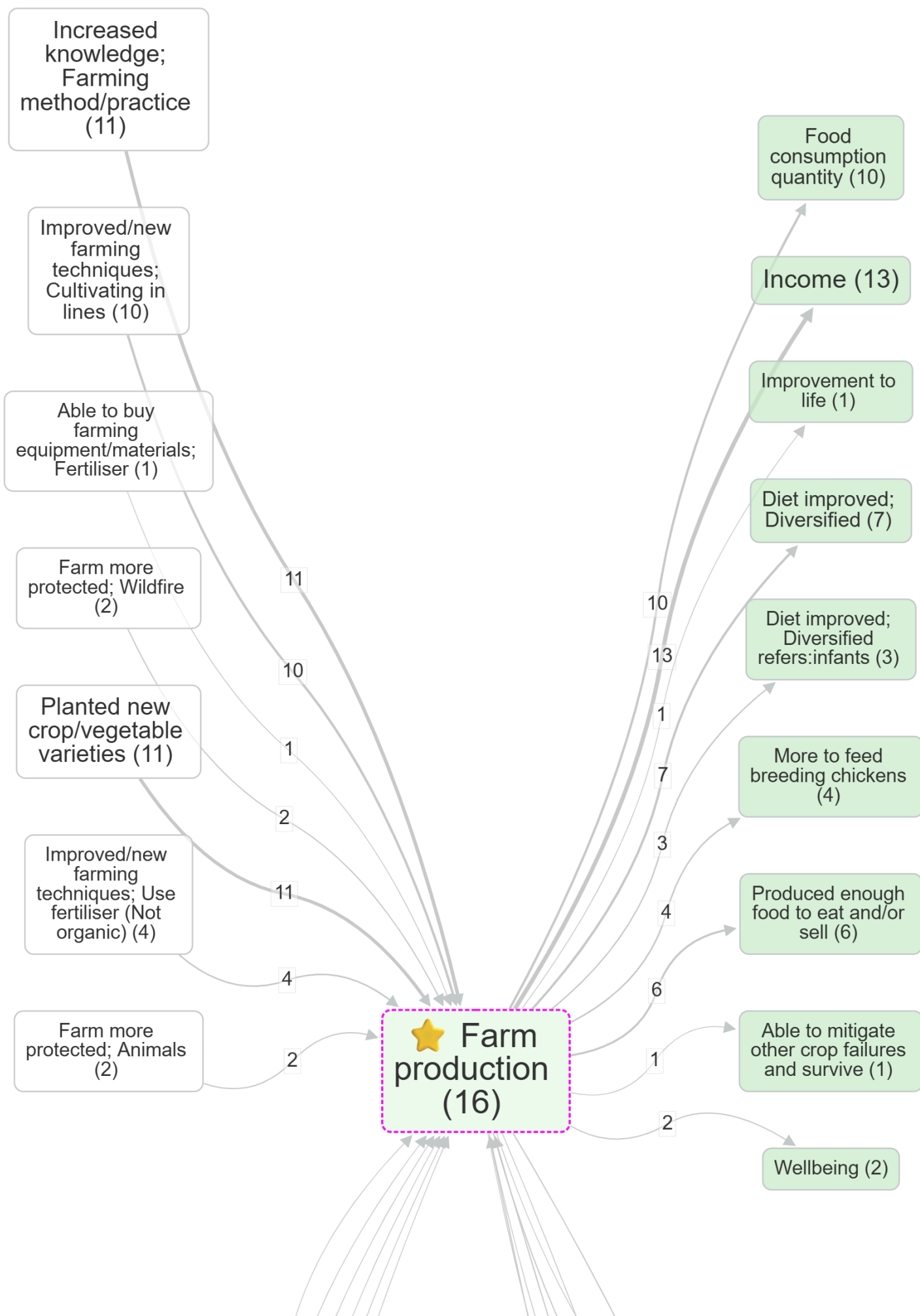
-  **Factor selector** (multi-select dropdown): choose one or more target factors.
-  **Show All** (toggle): show labels from the whole project (otherwise only labels visible at this pipeline stage).
-  **Steps Up** (radio button group 0–5): how many steps upstream (causes) to include.
-  **Steps Down** (radio button group 0–5): how many steps downstream (effects) to include.
-  **Source tracing** (toggle): keep only links that lie on a within-source path (single-source narrative constraint).
-  **Highlight** (toggle, default on): show/hide the dashed magenta border around matching factors.
-  **Matching** (radio buttons): Start / Anywhere / Exact (case-insensitive).

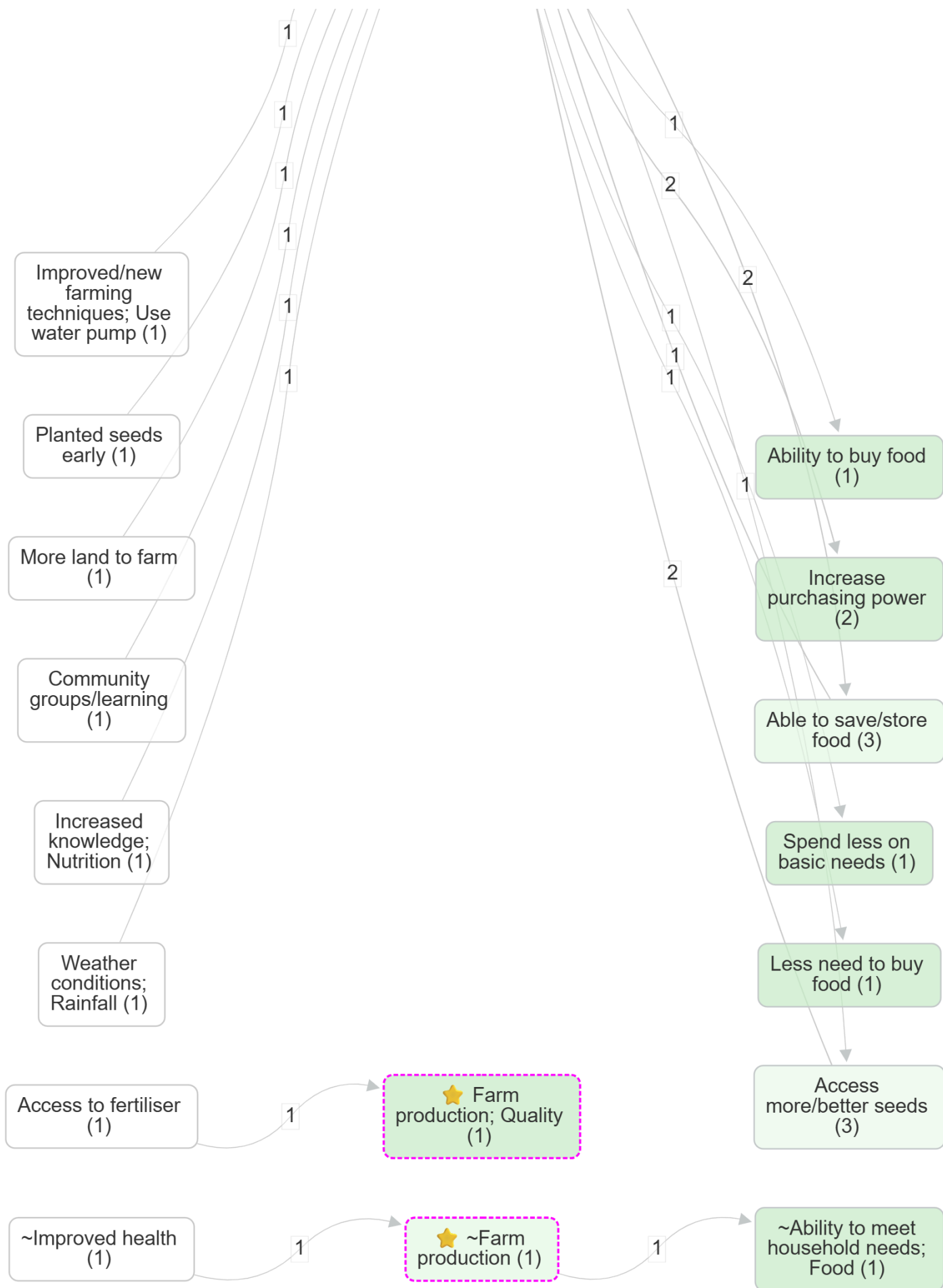
How to use:

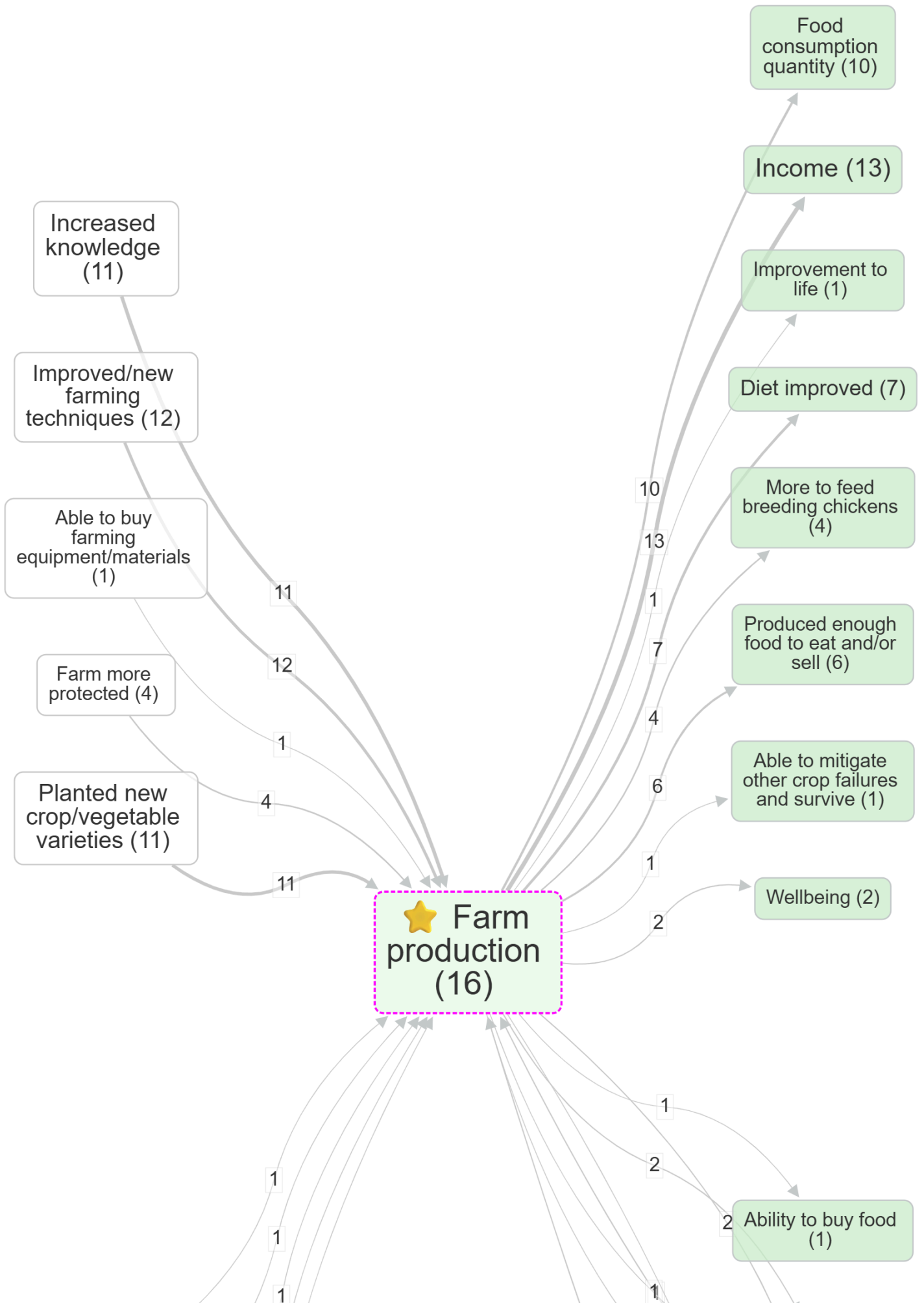
1. Select one or more factors.
2. Set Steps Up/Down to widen or narrow the neighbourhood (for interviews, chains longer than ~4 steps are uncommon).
3. (Optional) Turn on Source tracing to require paths from a single source.
4. (Optional) Turn off Highlight to hide the custom highlighting.
5. The map and tables update to show only links on those paths.

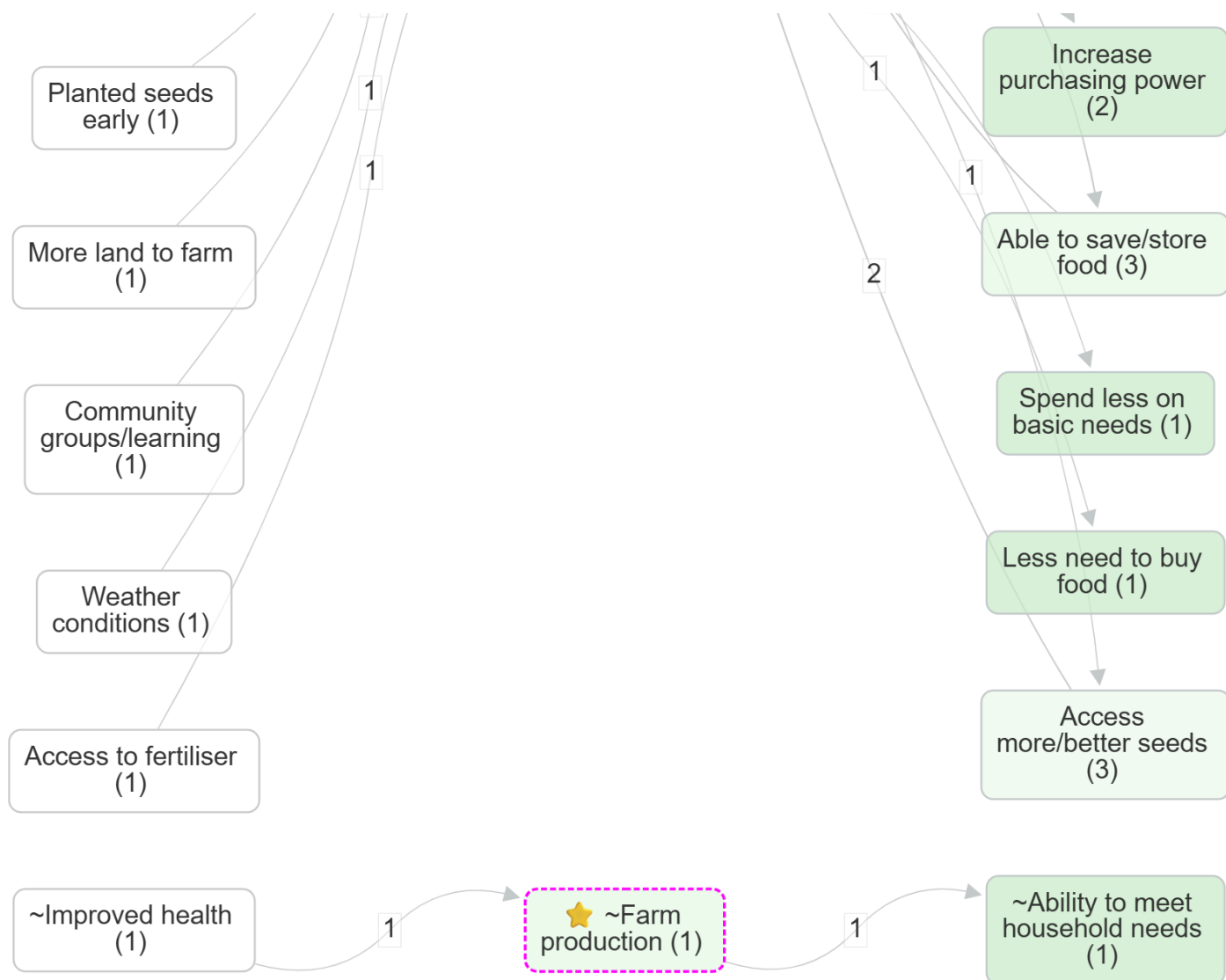
Example bookmarks:

- Upstream influences on wellbeing
- Focus on a factor, then compare the effect of zooming: Farm production (no zoom) vs Farm production (Zoom level 1)
- Health behaviour, without zooming
- Upstream focus (contrast): without source tracing #270 vs with source tracing #534









Tips:

- **Order matters:** This filter runs wherever you place it in the pipeline. If you have **Zoom** above it, then your “focused” factors may be zoomed-out labels (and the dropdown options will reflect that).
- **Simplify first, unless you are using steps:** at one step up and down you are picking a set of factors, so applying **Zoom**, **Collapse** or **Remove Brackets** first often makes the labels easier to choose from. At more than one step you are tracing a neighbourhood, so **rule 1** applies and this filter belongs above anything that rewrites labels.
- **Collapse alternative:** If you want to merge several factors into one *without* changing the neighbours, consider using **Collapse** instead of focusing.

All label/tag filters have three radio buttons below the selectize input called Match: Start (default), Anywhere or Exact to control how search terms match against labels/tags:

- **Start:** Match only at the beginning of text (default)
- **Anywhere:** Match anywhere within the text
- **Exact:** Match the entire text exactly

For include filters, multiple search terms are treated as OR (any match).

For exclude filters, default behavior is AND (all terms must match), unless you turn on **Exclude Any**.

Focused factors show with a dashed magenta border in the map (when Highlight toggle is on).

Exclude Factor Label filter

 **What this does:** Remove unwanted factors from your analysis. Type factors like "Unclear" or "Other" to hide them from your map and tables. Useful for cleaning up your data by removing vague or irrelevant categories.

- **Factor selector** for factors to exclude. By default shows only labels from links currently visible at this stage of the filter pipeline. Use the **Show All** toggle to display all factor labels from the entire project instead.
- **Matching options:** Start / Anywhere / Exact
- **Exclude Any** toggle (default OFF):
 - OFF = exclude only when **all** selected texts match (AND)
 - ON = exclude when **any** selected text matches (OR)

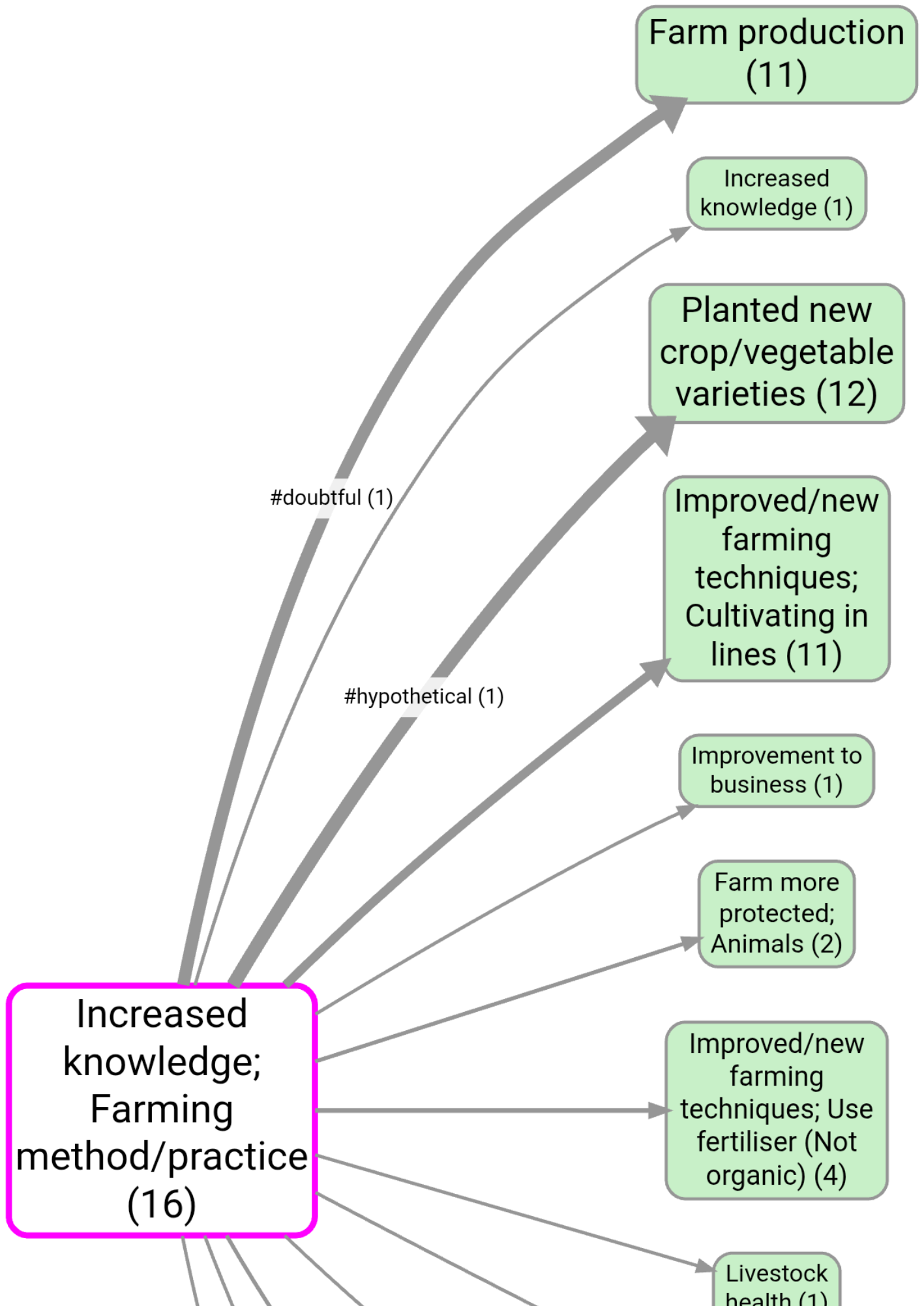
Tags Filter

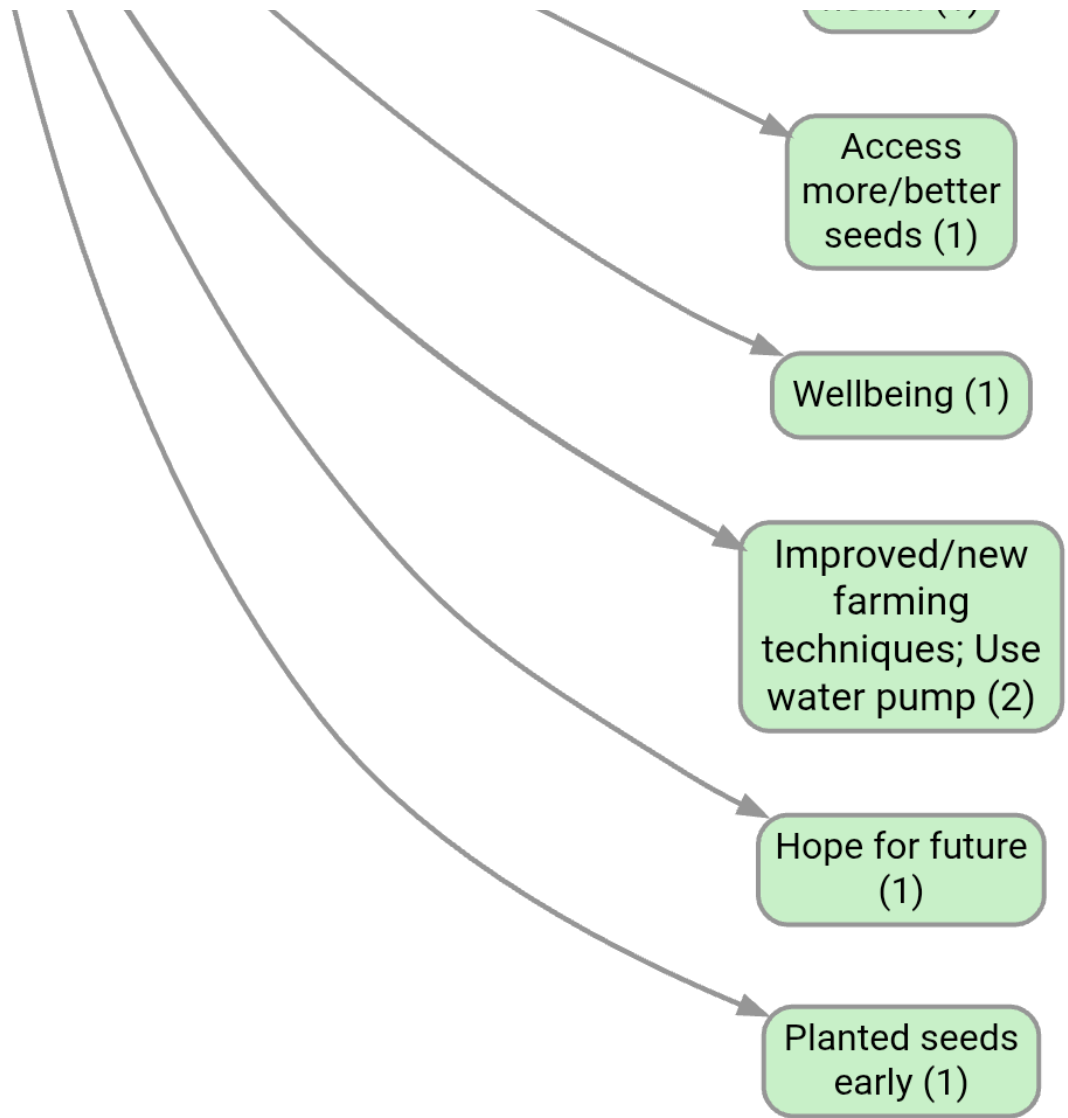
 **What this does:** Filter your analysis by the tags you've added to links. Show only links tagged as "#important" or "#policy" to focus on specific themes or types of relationships you've identified.

- **Tag selector** with existing link tags from current project
- **Matching options:** Start / Anywhere / Exact

Example bookmark:

- Displaying **#hypothetical** / **#doubtful** tags on links: **#1126**





Exclude Tags filter ✕

🚫 **What this does:** Remove specific types of links from your analysis. Exclude links tagged as "#uncertain" or "#duplicate" to focus on higher-quality data. Helpful for filtering out questionable or irrelevant causal claims.

- Same as Link Tag filter except *exclude* links containing these tags.
- **Exclude Any** toggle (default OFF):
 - OFF = exclude only when **all** selected texts match (AND)
 - ON = exclude when **any** selected text matches (OR)

Exclude self-loops Filter ↺

You can exclude self-loops from the maps, but that is more of a visual change. This is a real filter as part of the filter pipeline. For example, if you are using a filter like **Link Frequency** that might be retaining link bundles which are actually self-loops, so you might get unexpected results if you use the map setting to remove the self-loops. So this filter is a better way. It simply removes all links which are self-loops from the links table.

Where you put it in the pipeline changes the result. Like every filter, this one only sees the links passed down from the filters above it, and only the filters below it see what it leaves behind. That matters most for the two frequency filters, because they count the links they are given. A self-loop counts **twice** towards its factor (once going out, once coming in), and it is one more link in a factor's or a bundle's tally. So putting Exclude self-loops **above** a [Link Frequency](#) or [Factor Frequency](#) filter lowers those counts before the threshold is applied, and factors or links that would have survived on a "top 20" or "at least 3" rule can now drop out. Putting it **below** lets the self-loops count towards the threshold first and then removes them from what you see. Both are reasonable, they answer different questions, so decide which you want rather than treating the position as tidying up.

Path Tracing Filter

Ideas Garden: [Path tracing and source tracing](#)

 **What this does:** Find causal pathways between two specific points. Set a starting factor (like "Poverty") and an ending factor (like "Poor Health") to see all the causal chains that connect them. Great for understanding how problems and solutions are linked.

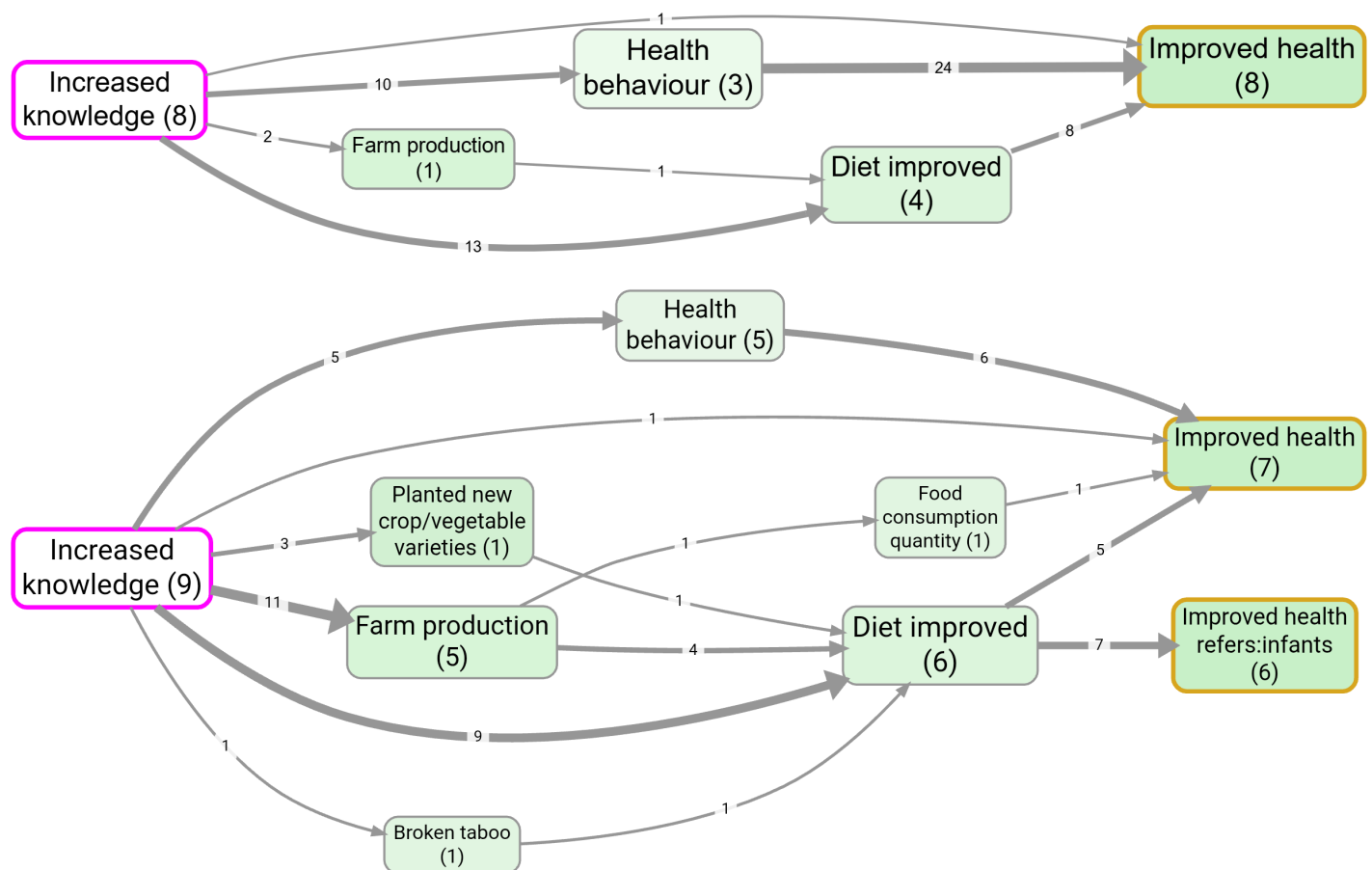
- **From selector** (multi-select dropdown): starting factors (uses the same label source rules as other filters; see **Show All**).
- **To selector** (multi-select dropdown): ending factors.
- **Show All** (toggle): show labels from the whole project (otherwise only labels visible at this pipeline stage).
- **Matching** (radio buttons): Start / Anywhere / Exact.
- **Max Steps** (radio button group 1–5): maximum path length.
- **Source Tracing** (toggle): keep only paths where every link in the path comes from the same source.
- **Highlight** (toggle, default on): show/hide special highlighting (From factors = dashed magenta; To factors = dashed dark-yellow).
- **Only Indirect** (toggle, default off): remove direct From→To links so you only see mediated paths (only applies when both From and To are non-empty).
- **Open From × To matrix:** opens a modal table with From factors in rows and To factors in columns, using this filter's From, To, Max Steps, and Source Tracing settings. This is useful when you have several From and/or To factors and need to compare the individual From→To counts rather than just inspect the combined map. In the modal, choose **Source counts** or **Citation counts**, set heatmap formatting (global / by rows / by columns / off), exclude empty rows/columns, then copy a screenshot or export XLSX. The matrix is computed from links before this filter, so preceding filters are respected.
-  **Check sources tell a continuous story** → (link, only shown when **Source Tracing** is on): jumps to the [Links Panel](#) in [Print View](#) with quotes grouped by **Source** then by **Bundle**, sorted by Source. Use it to read the quotes per source and verify that each source really tells one continuous causal story from the From factor to the To factor.

Motivation:

- Use this when you want to explain *how* an outcome is reached (“what are the chains that lead to X?”) or explore “routes” between two concepts (e.g. **training** → **income**).
- Turn on **Source Tracing** when you want coherent within-source narratives rather than paths stitched together across different sources.
- **Source Tracing** only guarantees that *every link in every bundle in the map is part of a complete pathway of links from From to To told by one specific source* — but does not really guarantee that the source actually narrates the *whole* path as one connected story. The **Check sources tell a continuous story** → link is a quick way to read the underlying quotes per source and judge that yourself, instead of trusting the structural filter alone.

Example bookmark:

- Source tracing from Increased knowledge to Improved health
- Upstream influences on wellbeing; source tracing
- Contrast (order matters): source tracing + then zooming out: [#1125](#)



Notes:

- **Order matters, and this filter belongs at the top (rule 1):** the app applies it at its position in the pipeline, so a transform filter above it (e.g. **Zoom**, **Collapse**, **Combine Opposites**) changes which labels you can select and which paths exist, and a frequency filter above it removes links the paths are made of. Trace first, then simplify for presentation.

- **Empty selectors:**

- If put nothing in **From** then all paths from the factors matched by the To box of up to specified length are returned.
- If put nothing in **To** then all paths from the factors matched by the From box of up to specified length are returned.

- **Non-matching selectors:**

- If you **From** or **To** matches nothing, no paths are returned.

Pathways Filter

 **What this does:** Surfaces typical causal stories. Unlike Path Tracing, you do not pick a start or an end. The filter reads the quotes inside each source and finds short passages where one source tells a whole chain of factors close together in its own text, then ranks those chains by how many sources tell them and shows you one at a time.

This is an exploratory filter for finding typical pathways. It does not answer a definitive high-stakes evaluation question; it surfaces common stories so you can read and judge them.

How it works: within each source it looks for a **passage**, a set of quotes, one for each step of a chain $f_0 \rightarrow f_1 \rightarrow \dots \rightarrow f_N$, that all sit within a window of **proximity** characters (from the start of the first quote to the end of the last). That grounds each chain in a piece of more or less contiguous narrative, so "A→B at the start of a long document and B→C near the end" is not counted as one story. It then lists every qualifying chain for every source and ranks the chains by how many sources tell them.

- **Length** (radio buttons 1–6): the number of links N in the chain. Only chains of exactly this length are shown; shorter, more direct routes are not. Self-loops may form a step.
- **Rank** (< / >): which ranked chain to show. Rank 1 is the most frequent across sources; step right to browse rarer ones. When you step past the last chain the map goes empty, which tells you there are no more.
- **Proximity** (characters): how close together the quotes of a chain must be in a source's text to count as one passage. Smaller means tighter, more clearly contiguous stories; larger is more permissive.
- **Successive** (toggle, default off): also require the quotes to appear in chain order in the text (non-decreasing start positions along $f_0 \rightarrow \dots \rightarrow f_N$), so the source narrates the steps in the right order rather than merely nearby.
- **Tightest fit only** (toggle, default off): by default the filter counts *every* separate (non-overlapping) passage that tells a chain in each source, and the map keeps the links of all of them, so the chain shows its full citation weight. This is also what gives a useful ranking when you have only one source: every chain is then told by one source, so the chain narrated most often inside that source wins. Turn this on to revert to the older behaviour, which counts and shows only the single tightest passage per source (so each step shows just one citation).

Ranking order: number of sources that tell the chain (most first), then the total number of separate passages that tell it across all sources, then a stable tiebreaker. That passage tally is what ranks the chains on a single-source project, where the source count is always one. The map shows the links of all of the chosen chain's passages (or just the tightest one when **Tightest fit only** is on); sister links inside the same bundles are dropped, because the output is always one chain.

Once you have a chain you like on the map, use **Vignettes** to turn it into a short written story: with the chain showing, run a **Typical-source vignette** to have the AI write the narrative from the quotes behind that chain, or a **Whole-map vignette** to describe the chain as a whole. So Pathways finds the typical story and Vignettes writes it up.

Notes:

- **Quote offsets required:** a link can only join a passage if its quote has a recorded position in the source. Links without offsets do not participate.
- **Order matters, and this filter belongs at the top (rule 1):** the app applies it at its position in the pipeline, so a transform filter above it (e.g. **Zoom**, **Collapse**) changes which chains exist. Put it below the filters that choose which sources or links you are looking at, and above anything that rewrites labels or counts.
- **Heuristic:** on very large or dense sources the search is capped, so the ranking is indicative rather than exhaustive.

Recalculate quote positions

Found in **Project** → **Edit project** → **Quote positions**.

Each coded link has a quote from a source. To know *where* in the source that quote is, the app records a character position (offset) for it. Source-text highlighting uses these positions, and so does the **Pathways filter**, to judge whether the steps of a story are close together in the text.

Positions are normally filled in as you read sources, so older projects or ones never opened in text view can have many links without them. **Recalculate quote positions** does the whole project in one pass: it matches each link's quote against its source text and stores the position.

- Use it if the Pathways filter finds little or nothing, or after re-importing or editing source text.
- A quote that no longer matches its source text is reported as "unmatched" and left unchanged; fix the quote or the source and run it again.
- Needs edit access to the project.
- The Pathways filter also runs this automatically, once per project, when it notices many links without positions.

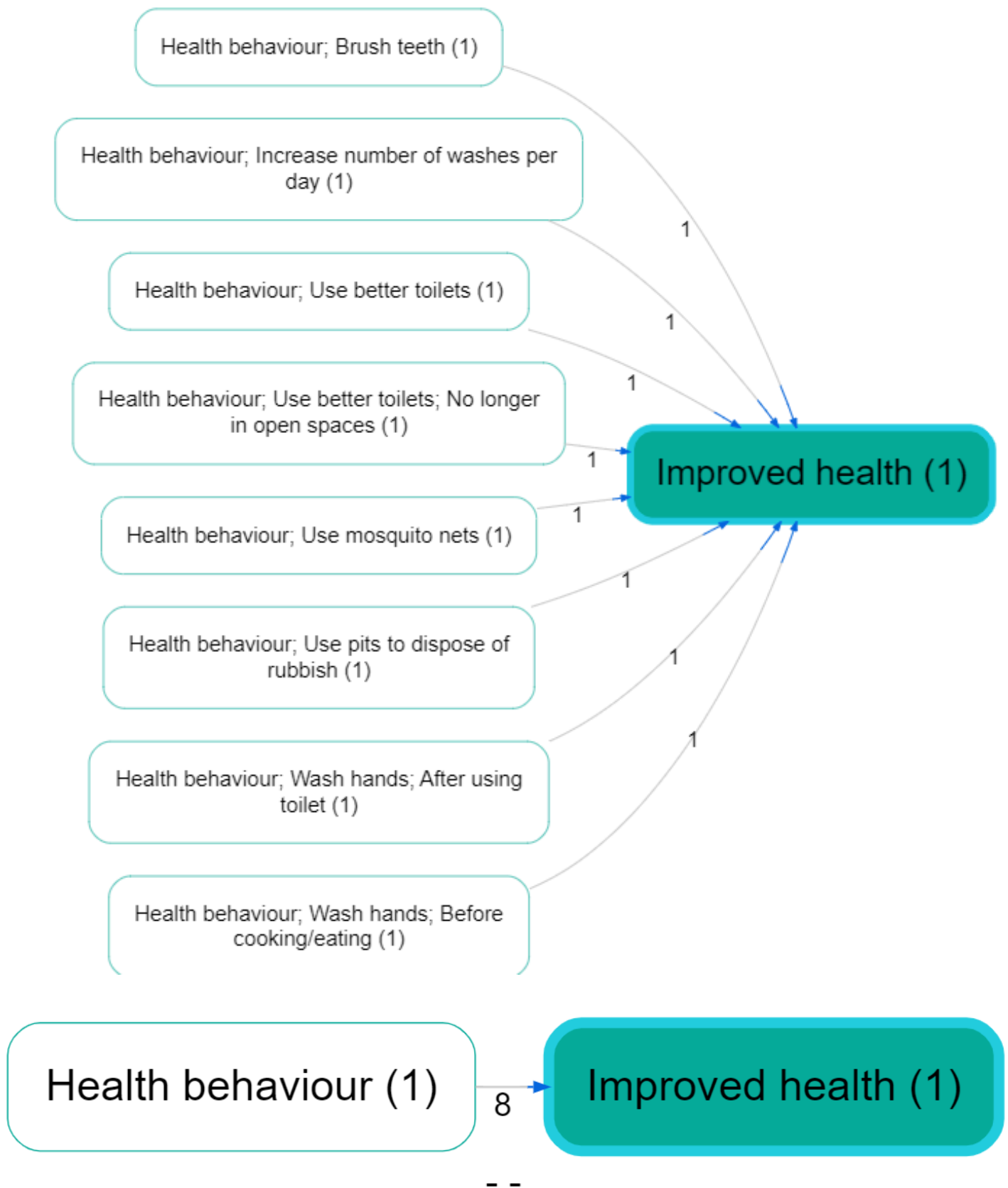
Zoom Filter

Ideas Garden: **Hierarchical coding (Zoom filter)**

🔍 **What this does:** Simplify complex factor labels by zooming to higher levels of a hierarchy. For example, turn "Health; Mental Health; Depression" into just "Health" (level 1) or "Health; Mental Health" (level 2). Perfect for getting a big-picture view of your data.

- 🖱️ **Hierarchy convention:** Use semicolons ; to express levels, e.g. **Health; Mental health; Depression**. (A space after ; is optional.)
- 🖱️ **Zoom level** (radio buttons): None, 1–9.
- **Examples:**
 - **Level 1:** "foo; bar; baz" becomes "foo"
 - **Level 2:** "foo; bar; baz" becomes "foo; bar"
 - **If the label has $\leq N$ levels:** it stays unchanged at zoom level N (and higher)
- **Tip (hierarchy reuse):** Once you have *any* hierarchical factor (i.e. a label containing ;), the **Cause** and **Effect** dropdown menus also include implied *parent* labels (e.g. **Health behaviour**) to make it easier to keep higher-level naming consistent.
- **Tip (find hierarchical factors):** Search for ; in factor labels, then zoom out to level 1 for a quick “top-level only” view.

Example (hierarchy before/after zooming):



Collapse Filter 🗑️

Ideas Garden: [Collapsing factor labels](#)

🔖 **What this does:** Merge similar factors under one common label. Type or select multiple similar terms like "money", "income", "salary" and they'll all be replaced with the first term. Great for cleaning up

data where the same concept is described in different ways.

Widgets:

-  **Search terms** (multi-select dropdown + free text): select existing labels or type parts of labels.
-  **Matching** (radio buttons): Start / Anywhere / Exact.
-  **Separate** (toggle): off = collapse all matches into the first search term; on = collapse into separate buckets (each match becomes the term it matched).

How it works:

- **Case-insensitive:** matching ignores case.
- **Multiple search terms** are treated as **OR** (any match triggers a collapse).
- **Separate = off:** any matching factor becomes the **first** search term.
Example (Any/Anywhere): search terms **Food**, **Diet** → **Diet**; **healthy** becomes **Food** (because **Food** is first).
- **Separate = on:** each matching factor becomes the **search term it matched** (so you can collapse into multiple buckets).
Example (Any/Anywhere): search terms **Food**, **Diet** → **Diet**; **healthy** becomes **Diet**, and **Food insecurity** becomes **Food**.

Combine Opposites filter ⇄

Ideas Garden: **Opposites**

 **What this does:** Unify opposite factor labels by matching tags. The simple version: `~good health` is the opposite of `good health`. It's another way to code `Poor health`. - This filter rewrites `~good health` as `good health` to combine them under one label but adds `flipped\_cause` and/or `flipped\_effect` columns to the links table to track which causes and effects were flipped in this way so that this information can be used in maps and tables. There's an alternative, more flexible version: - If you have pairs like `Foo [99]` and `Bar [~99]`, this filter rewrites `Bar [~99]` as `Foo [99]` to combine them under one label. That way you can even have multiple opposites: `[~27] Low Income` and `[~27] No Income` might be the opposites of `[27] High Income`.

Motivation:

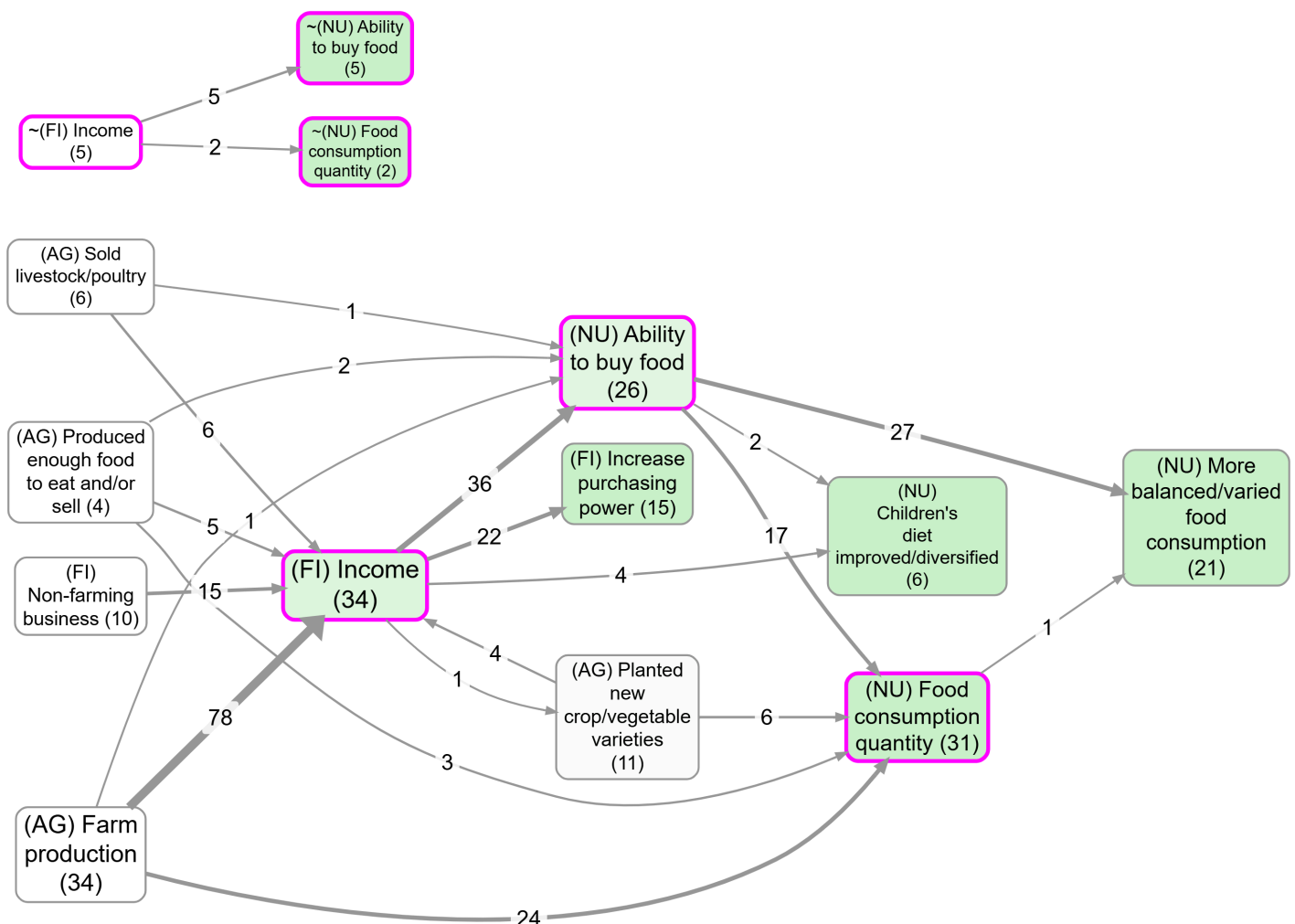
- In “barebones” causal coding, you often end up with pairs like **fit** and **~fit** (or **employment** and **~employment**). If the app doesn't know they're opposites, it's hard to compare the “positive” story against the “negative” story, and simple searching/filtering around **fit** can miss the **~fit** evidence.
- Combine Opposites is a lightweight alternative to “signed edges”: you keep factors as plain text labels, but you can still analyze them as a pair **without losing information** about which pole was originally coded (tracked via **flipped_cause** / **flipped_effect**).

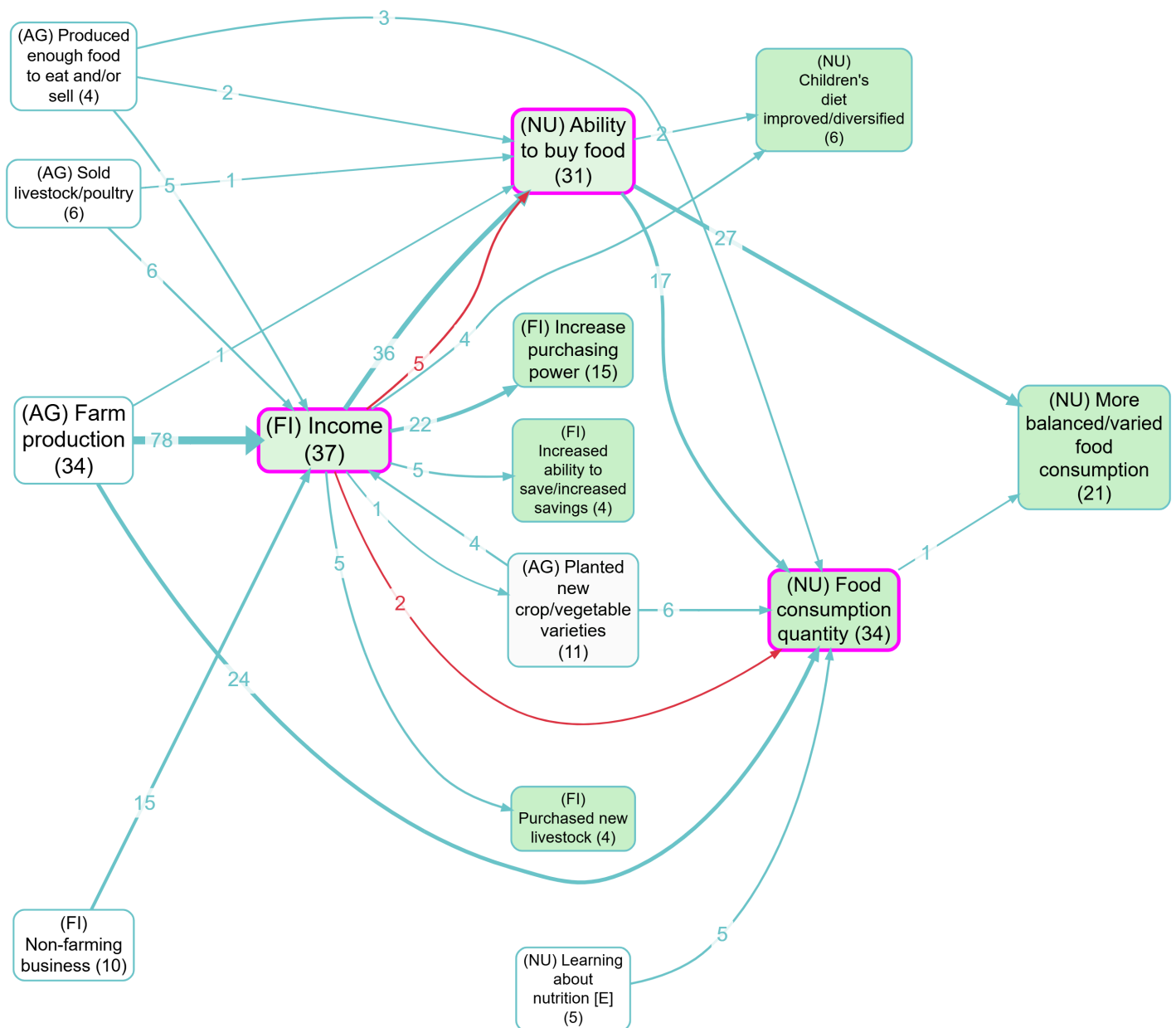
Notes:

- **Opposites, not sentiment:** ~ marks the opposite pole (e.g. **smoking** vs **~smoking**), not “bad”. Use it even when there’s no valence.
- **Order matters, and this filter belongs above the ones that replace labels (rule 3):** it reads the **[~3]/[3]** tags and the leading ~ to find each pair, so Remove Brackets (square), Collapse, Soft Relabel, Soft Recode Plus, Cluster or Auto Recode above it leaves nothing to pair, silently. As a transform filter it also changes what the filters below it work on, and which labels appear in their dropdowns.
- **Map colouring override:** when active, arrowhead colouring switches to Combine-Opposites colouring (flipped status / flipped share), so the Map Formatting **Link Colour** setting does not apply in the usual way.

Example bookmarks (contrast):

- Without combined opposites: **#985**
- With combined opposites: **#986**





Toggle – Turn the filter on/off.

Opposites mode toggles (you can use either or both):

- **~ prefix (e.g. ~ foo)** – Treat **foo** and **~ foo** as opposites (no brackets or numbers).
 - Works with hierarchies too: we flip **every** ;-separated component when comparing opposites. Example: **~Healthy habits**; **smoking** is the opposite of **Healthy habits**; **~smoking**. **Important:** this “flip every hierarchy component” behavior applies **only** to the **~ prefix** mode (not to the numeric tag mode below).
 - **~foo** is always rewritten to **foo** (even if **foo** does not appear elsewhere in the current factor list).
- **[~N] tags** – Treat numeric pairs like **Foo [99]** and **Bar [~99]** as opposites. (Square brackets are optional: **Foo 99 / Bar ~99**.)

More details on [~N] tags

Labels can be written in pairs like:

- Foo [99]
- Bar [~99]

where **Bar** represents the opposite of **Foo**. The square brackets are optional - you can use **Foo 99** and **Bar ~99** - but brackets make it easier to remove tags later using the [Remove brackets filter](#).

If there are any such pairs, with matching integers, and the filter is switched on:

rewrite any Bar [~99] filters as Foo [99] and add new columns...

- **flipped_cause** column tracks which causes were flipped
- **flipped_effect** column tracks which effects were flipped

... to the current augmented links table, so that if the label has been flipped, the value is True and otherwise False.

Strip tags from labels (default: on) – When enabled, removes [N] and [~N] tag patterns from labels after combining opposites. This keeps labels clean while preserving the tracking information in the **flipped_cause** and **flipped_effect** columns.

Hide colours (default: off) - When enabled, the map stops colouring links and factor borders by flipped status and goes back to its normal colouring, which is sentiment unless you have set a custom link colour. Bundling, link labels and the >F-> bundle notation are unaffected, so you keep the variant breakdown without the red and blue. Use it when the flipped colours fight with the colouring you want to show.

Bundling strategy (Separate vs Together):

- **Separate** (default) – Treat each flipped/unflipped **variant as its own bundle** (so counts are plain numbers per bundle). This means that you may often see two or even more links between two factors.
- **Together** – Put all variants into a **single bundle** (one link between two factors). Optionally show an **embellished per-variant breakdown** inside the link label (see “Map legend” below).

Link labels (Simple vs Detailed):

- **Simple** – Show only the total source/citation count (Together mode only).
- **Detailed** – Show the per-variant breakdown using the unicode markers (Together mode only).

Map legend (what you’ll see on the map) — 4 variants

Notes:

- The “Link labels” setting only matters when Bundling is **Together** (it is disabled/ignored in **Separate**).
- “Count” below means whatever your map is currently showing for link labels (e.g. **Sources** or **Citations**).
- **Colours:** Separate = per-end flipped status (tail=cause, head=effect; blue=no, red=yes). Together = **average** flipped share (tail=cause, head=effect; blue→red). Switch **Hide colours** on to drop both and keep the map's normal colouring.
- **Separate mode tip:** you can set **Link colour** (Map Formatting) to grey so the default (unflipped) links match maps without combined opposites better.

1) Separate + Simple

- In the map, you'll often see **multiple parallel links** between the same two factors (one per variant).
- Each link label is a **plain count** for that variant.
- Variants correspond to **flipped_cause** / **flipped_effect**:
 - `foo >--> bar` (unflipped/unflipped)
 - `foo >-F> bar` (unflipped/flipped)
 - `foo >F-> bar` (flipped/unflipped)
 - `foo >FF> bar` (flipped/flipped)

You'll see this special `>F->` notation in the Bundle column of the Links Table

2) Separate + Detailed

- Same as **Separate + Simple** (Link labels detail does not apply in Separate mode).

3) Together + Simple

- You'll see a **single link** between the two factors.
- The link label is the **total count across all variants** (no breakdown).

4) Together + Detailed

- You'll see a **single link** between the two factors.
- The link label shows a **per-variant breakdown** using these unicode markers:
 - `⚭n` = neither flipped (`--`)
 - `\n` = effect flipped (`-F`)
 - `/n` = cause flipped (`F-`)
 - `⚭n` = both flipped (`FF`)
- Example: `⚭5 \2 /1` means 5 `--`, 2 `-F`, 1 `F-` (zero variants are omitted).

Label by Group 📌

🔗 **What this does:** Configure how link labels appear on your map based on any field in your data. Choose what information to display (like tribe memberships or custom attributes) and how to show it (counts, percentages, or statistical significance).

Controls:

- **Field** - Dropdown of available fields from your filtered data (typically shows custom fields like tribe ID)
- **Counts** - Choose whether to count **Sources** (unique participants/documents) or **Citations** (links)
- **Display mode** - Choose how to show the data:
 - **Tally** - Show counts for each value (e.g., "T1:4 T2:3")
 - **Percentage** - Show what % of each value's total links appear in this bundle (e.g., "T1:34% T2:22%")
 - **Chi-square** - Show bundle size, then which values are significantly over-represented (↑) or under-represented (↓) (e.g., "45 (T1↑ T3↓)")
 - **Chi-square (with counts)** - Also show the observed count for each significant value (e.g., "45 (T1 4↑, T3 3↓)")
 - **Chi-square (with counts/totals)** - Also show observed/total for each significant value (e.g., "45 (T1 4/5↑, T3 3/6↓)")
- **Ordinal correction (numeric groups)** - When ON (and group values are numeric-like, including strings like "12 foobar"), Chi-square modes use an ordinal trend test and show only the overall total plus (↑) or (↓) (no per-group totals).
- **Sig level** - Significance threshold for Chi-square / Ordinal trend (default $p < .05$).

↑ / ↓ meaning (only in Chi-square modes):

- **Normal Chi-square:** ↑ over-represented vs expected; ↓ under-represented.
- **Ordinal correction ON:** ↑ increasing over numeric groups; ↓ decreasing.

To use:

1. Add the Label by Group to your pipeline
2. Select a field (e.g., `s_Tribes_5` after running Tribes and saving to Sources)
3. Choose a display mode
4. In Map Formatting, set Link Labels to "Label by Group"

Example use cases:

- **After Tribes:** Show which tribes contribute to each connection (T1:5 T2:2 T3:1)
- **Significance testing:** Identify connections where certain tribes are surprisingly over/under-represented (T1↑ T3↓)
- **Custom attributes:** Display any custom field you've added to your data

Example bookmark:

- [Differences between groups shown on map](#)

Map Custom Columns Filter

 **What this does:** Takes one of your link or source custom columns and turns it into map display data. This filter creates values for **Custom label**, **Custom width**, and **Custom colour**, which are then automatically applied in Map Formatting.

Use this when you want one custom link or source column to drive the map, for example:

- show **confidence** or **policy_area** as a label on each bundled edge
- show source metadata like age group, gender, location, or interview type on each bundled edge
- make edge width reflect a numeric custom score
- colour edges from a custom colour field or other grouped coding

Controls:

- **Label field** - choose which link or source custom column should feed **Custom label**
- **Label aggregation** - how bundled links should be combined:
 - **Unique** - distinct values only
 - **Tally** - counts by value
 - **All** - list all values
 - **Average** - numeric columns only
 - **Sum** - numeric columns only
- **Width field** - choose which link or source custom column should feed **Custom width**
- **Width aggregation** - numeric bundle summary, typically **Average**, **Sum**, or **Max**
- **Colour field** - choose which link or source custom column should feed **Custom colour**
- **Colour aggregation** - for numeric columns, choose **Mean**, **Max**, or **Mode**

Source custom columns appear with a **Source:** prefix in the field dropdowns. These are the **s_*** fields from the enriched Links data, so they can be aggregated across all links in the visible bundle just like link custom columns.

To use:

1. Add the Map Custom Columns filter to your pipeline.
2. Pick one or more link or source custom columns for label / width / colour output.
3. Choose the aggregation rule for each output.
4. In [Map Formatting](#), set Link Labels / Link Width / Link Colour to **Custom label**, **Custom width**, or **Custom colour** as needed.

Notes:

- This is a **map-output** filter. It does not create a permanent new data column for general analysis; it prepares reserved map fields for the current pipeline result.
- Bundle aggregation matters because one visible map edge can represent several underlying links.
- Custom columns stay **typeless**. If a link column already looks numeric and you enter a non-numeric value in the Link Editor, the app warns but does not block you.
- For **Custom colour**, the selected column must be numeric. The aggregated value is mapped onto a muted scale: high positive values = muted green, values near zero = whitish, and negative values = muted blue.
- Non-numeric values are ignored for **Custom colour**.

Remove Brackets Filter

 **What this does:** Clean up your factor labels by removing text in brackets. For example, "Education (primary school)" becomes just "Education". You can remove round brackets () and/or square brackets [].

- **Two switches:** Round () and Square [] (both off by default)
- Removes all text within the selected bracket type(s)

Soft Relabel Filter

 **What this does:** Temporarily relabel factors.

- **Old factor labels** listed on the left
- **New factor labels** editable, listed on the right
- **Load labels button** when pressed, adds into the Old labels list any current factor labels (in links as currently filtered) which are not yet listed in the Old labels list and adds the same Old label to the New column as default.
- **Clear button** to clear the New fields
- **Clear ALL button** to clear all rows

Effect: all factors exactly matching any of the labels in the Old list are relabelled with the corresponding labels from the New list. factors not listed are not relabelled but preserved.

Many use cases:

- temporarily merge multiple factors into one
- you are using magnets and you can't really use the formulation you want because you want to maximise similarity with existing labels
 - eg you are using "floods" as a magnet but you really want it as a hierarchical factor like "environmental problems; floods" but you can't use that as a magnet.

Keyboard shortcuts (Win/Linux ⇔ macOS):

- Tab / Shift+Tab: move focus down/up between NEW cells
- Arrow Up/Down: move focus up/down between NEW cells

- Alt+Arrow Up/Down (mac: Option+Arrow): move the current row up/down
- Ctrl+Arrow Up/Down (mac: Cmd+Arrow): move the current row up/down
- Delete current row:
 - Shift+Delete (mac: Shift+Fn+Backspace) or
 - Ctrl+Shift+K (mac: Cmd+Shift+K)

Potentially, one NEW label might have multiple OLD labels.

Temporary Factor Labels Filter 🔄

🔧 **What this does:** Temporarily map other link fields into `cause` and/or `effect` for this pipeline run.

- Pick a field for **Map to cause** and/or **Map to effect**.
- Leave either dropdown on **Keep current ...** to keep the original **cause/effect**.
- Typical use: map **cause_temp** / **effect_temp** created by AI recoding so you can compare alternative label sets without overwriting original labels.
- Downstream calculations (for example bundles, source counts, citation counts, factor frequencies) use the temporary mapped **cause/effect** values.
- This changes only the live pipeline output; your stored source links are unchanged.
- Related: [AI Answers](#) → [Sources, Links, and Factors](#) subtabs.

Soft Recode Plus filter ✎

Requires AI subscription

🧲 **What this does:** Group messy factor labels under clearer names you choose (called "magnets"). Example magnets: `Improved health`, `Education programs`, `Income changes`. The filter finds the closest magnet for each label and replaces it.

Example bookmarks:

- [Soft recode using all top-level labels](#)
- [Soft recode \(recycling losers\)](#)

Controls:

Create Suggestions for Magnets

(collapsed by default): Optional. Ask AI to propose clear names from your current labels. Insert adds them to your magnets box to review/edit.

- **Number of clusters** – Choose how many groups to find for AI suggestions.
- **Representatives per cluster** – How many example labels the AI sees for each cluster (8–20; default 8). Consider choosing more than 8 if you want to split clusters into positive/negative variants.
- **Labelling prompt** - With the usual buttons to save and recall previous prompts

- **Insert**

Main panel

- **NEW: Only unmatched** – A new toggle which appears right at the top, before the Create Suggestions subpanel. default off.
- **NEW: Apply to link tags** – When enabled, Soft Recode Plus processes **link tags** (instead of factor labels). Tags are split on commas, trimmed, and then matched to magnets just like labels. Reserved tags like **#plain_coding** are never recoded.
- **Magnets** – One magnet per line. Saved per project. Use Prev/Next to browse recent sets.
- **Similarity slider** – The raw labels are dropped if they are not at least this similar to at least one cluster.
- **Drop unmatched** – If on, remove links whose labels don't match any magnet. If off, keep them as they are. (When **Apply to link tags** is on, this drops **unmatched tags**, not links.)
- **Save** – Save magnets and apply the recode.
- **Remove hierarchy** – Strip any text before the final semicolon
- **Clear / Prev / Next** – Manage saved sets.
- **Recycle weakest magnets:** – A slider starting at 0, default is 0. If the slider is $n > 0$, then we look at the cluster assignments which would have been returned and find the n clusters which we are going to recycle. Reassign them to their nearest cluster, providing the similarity is still better than the similarity cutoff. This way we don't lose factors / links which are otherwise assigned to smaller clusters which may get excluded later on in the filter pipeline. When it is on zero, it makes no difference and we just use the solution based only on the magnets, similarity, and remove_hierarchy. The maximum value changes to match the total number of magnets.

If you have say 50 magnets but later apply a filter which only shows 5 factors, in that kind of case it can be nice to apply this slider, to reassign the links for the 45 factors which do not show, to the ones that do show, if they fit. It is not automatic because it depends on you looking to see how many factors are left in the FINAL output e.g. maps. It should work whether you drop unmatched or keep them.

Recoded columns

When you use Soft Recode Plus, the Links and Factors tables show special columns that track which labels have been recoded:

- **Links table:** Shows **_recoded_cause** and **_recoded_effect** columns (✓ for recoded, ✗ for not recoded)
- **Links table (tags mode):** Shows **_recoded_tags** column (✓ if any tag was recoded at current similarity cutoff)
- **Factors table:** Shows **_recoded** column (✓ if the factor appears at least once as recoded, ✗ otherwise)
- These columns only appear when Soft Recode Plus is active in your filter pipeline
- You can filter by these columns using the True/False dropdowns in the table headers

These columns track recoding from any filter that transforms labels: Soft Recode Plus, Zoom, Collapse, Remove Brackets, Soft Relabel, Cluster, Hierarchical Cluster, and Combine Opposites.

Process only unmatched NEW

the point of this is: what if I apply some (maybe standard) magnetisation and matches plenty of factors but there might be some important material left unmatched, not just noise. so i can use a PAIR of these filters. in the first one, I leave OFF its Discard Unmatched toggle and in the second filter switch ON its Only Unmatched filter. (if there is no preceding SRP filter with Discard Unmatched=OFF, this second SRP filter does nothing).

So now,

- the Create Suggestions (if used) optionally processes ONLY the UNMATCHED factor labels
- the magnetisation (if labels are non-empty) works only on the unmatched factor labels.
- the actual output of the second filter is now the union of both soft-recode processes, i.e. the original matches from the first and the new matches of the previously discarded material from the second.
- the Discard Unmatched on this second filter works as usual: if it is OFF, then we also return all the still-unmatched labels.

Meaning Space (2-D projection of embeddings)

Go to the [map formatting](#) and select Layout → Meaning Space to see a 2-D scatter plot of your factor-label embeddings (projected down to 2 dimensions).

- Magnets are shown with labels; raw factor labels are dots.
- Colour indicates the magnet group; magnet dot size represents group size.
- Meaning Space uses the **Similarity** and **Drop unmatched** settings from your most recent Soft Recode Plus filter:
 - If **Drop unmatched** = **ON**: unmatched raw labels are not shown (dot count shrinks as you increase Similarity).
 - If **Drop unmatched** = **OFF**: unmatched raw labels are still shown, but they render in **grey**.
- You can pan (drag) and zoom (mouse wheel and [zoom controls](#)).
- Double-click on an empty part of the map to zoom in at that point.
- Tooltips on dots show the original (raw) labels and the magnet label.

Motivation for Remove Hierarchy

"Remove hierarchy", default off. if on, strip any text before a final semi-colon, if no semi-colon, do not change the text.

```
something; another thing
```

is treated same as

another thing

.... but it continues to be treated as "something; another thing" in the rest of the filter pipeline.

Quick workflow:

1. (Optional) Open Create Suggestions for Magnets panel → set Number of clusters and use Insert to get AI suggestions.
2. Use these suggestions and/or edit them, paste or type your own magnets (one per line).
3. Click Save.
4. Clusters your current labels (factors as currently filtered), ranks typical examples, and asks AI to suggest clear names.
5. Returns suggested names into the magnets box; you can edit them before Save.

See [tips on using the history](#) to reuse both your labelling prompt and magnet sets.

Motivation for "recycle weakest magnets": suppose you create 20 magnets, and then apply more filters like say a [link frequency filter](#) so that you end up with say only 5 factors. If you then *remove* those factors from the magnets list which are *not* included in the final output, you will usually increase the coverage of your map (re-assigning raw labels which fit best with one of the "lost" labels but still fit well with one of the "surviving" labels). This is what the Recycle slider does: it recycles the specified number of smaller magnets and reassigns them to the larger magnets. So in the example, if you start off with 20 magnets but your final map only shows 5, try recycling say 10 or even 15 of the missing factors.

Note that Recycle Weakest Magnets is applied BEFORE Drop Unmatched.

Cluster filter

Requires AI subscription

 **What this does:** Automatically discover groups of similar factors in your data using machine learning. The system finds natural clusters of related concepts and labels them with cluster numbers. Great for exploratory analysis when you're not sure what causal themes exist.

- **Enable toggle** (starts disabled)
- **Number of clusters** (1-9)
- **Server-side processing** using [cluster_factors_pgvector](#) database function
- Uses k-means clustering on factor embeddings
- Labels clusters with numeric IDs

Auto Recode filter

📖 What this does: Quickly turns your current set of labels (after any previous filters like Zoom) into a simple tree you can "roll up" or "open out". Pick a small number of clusters for a big-picture view, then nudge the Balance and Similarity to tidy results. Designed for fast, practical exploration on real projects.

Motivation

Making sense of hundreds or thousands of factor labels is hard.

You might use something like soft Recode Plus, but often you'll ask for 20 clusters to cover a wide range of meanings. Then after filtering out insignificant data, you end up with only 7 clusters — losing coverage. Ideally you'd go back and recreate just 7 clusters, but that gives different results. Frustrating!

The point of this Auto Recode filter: have your cake and eat it. Ask for an foldable/unfoldable hierarchical solution. When you move the slider to 15, you get the best solution for 15 clusters. Slide it to 3, you instantly get the best solution for 3 clusters.

Controls:

- **Enable toggle** (starts disabled)
- **Balance (0..1):** 0 = prefer more distinct clusters; 1 = prefer more even sizes. Changing this can be slow because the tree has to be rebuilt
- **Number of clusters (K):** 2–50. Unfolds the returned tree locally to K. This is fast unless you increase beyond 20.
- **Similarity $\geq$:** prune locally by similarity to the centre of each cluster.

NEW: **AI labelling prompt** with history controls. Use this to suggest clearer names for each cluster:

- Saved in the prompts table as type `hierarchical_label` (shared across projects; history shows current first then others).
- A Save button stores your prompt; it also auto-saves on blur and after the first tree build.
- When you raise K (unfold deeper), we call AI in parallel only for the two new child clusters introduced by each applied split, using up to 8 representative labels per child as context. For K clusters this is K–1 requests. Folding to fewer clusters does not call AI; existing AI labels or medoid representatives are used.
- If the prompt is blank, we show the medoid representatives for each cluster.
- If earlier splits already have AI labels ($K > 1$), we include a reference list of those labels so new labels avoid overlapping meanings.

NEW: **Seed labels (optional)** with history and strength:

- Provide up to K seed labels (one per line). Seeds softly influence split formation but are not included in the final tree (not nodes, not representatives).

- Saved in the prompts table as type `hierarchical_seeds` with standard history controls (Prev/Next/Dropdown/Save).
- Seed strength (0..1) adjusts influence; 0 is a no-op (identical to no seeds). Changing strength or seeds triggers a single backend rebuild (like Balance). Changing K or Similarity does not re-call the backend.

How to use (quick):

- Add the filter and enable it. We build a quick draft tree from the labels you see now (respecting any filters above, like Zoom).
- Set **Balance** if you want more equal-sized groups; the first build may take a moment on large projects (one server call).
- Use **K** to choose how many clusters to show. Changing K is instant (no extra server calls).
- Use **Similarity** $\geq$ to drop weak matches. If either side of a link isn't matched, that link is hidden.

Notes:

- On very large projects, we automatically sample a representative set to build the tree, then assign the rest to the nearest cluster. This keeps things responsive while preserving the overall picture.
-  Tip: changing the number of factors should be instant if they are less than 20. Setting more than 20 can be slow. If you are going to want more than 20, set this number initially to the maximum number you are likely to want. You can then easily reduce it. Gradually decreasing the number is fine, but *gradually increasing* it will be very slow.

A good prompt looks something like this:

This is a list of many raw labels grouped into two different clusters, with their cluster IDs, together with a reference list of other labels. Return a list of two new labels, one for each cluster ID. Each label should capture the meaning of the whole cluster, using similar language to the original raw labels, but in such a way that the labels you create are distinct from one another in meaning. Try not to be too generic, try to be as concrete as you can. Do NOT provide labels which include causal ideas, like "X through Y" or "X leading to Y" or "X results in Y" or "X improves Y" etc. Equally, don't include conjunctions in the title like "X and Y". The meaning of the labels you give me should ideally not overlap in meaning with one another or with the labels in the reference list.

Optimized Cluster filter

 **DEPRECATED**

Requires AI subscription

 **This filter is deprecated.** Its functionality has been merged into [Soft Recode Plus](#). The filter still works for backward compatibility with existing bookmarks/URLs, but new instances cannot be

created. Use Soft Recode Plus instead for optimal clustering and recoding.

 **What this does:** Automatically finds the most optimal factor labels to use as centroids through genuine optimization. Unlike regular clustering that just groups similar items, this finds the best possible $n \leq N$ labels that maximize coverage with similarity $\geq S$. Perfect for discovering the most representative concepts in your data.

Controls:

- **Max Centroids (n)** - Maximum number of optimal centroids to find (2-50)
- **Similarity $\geq$** - Minimum similarity threshold for grouping labels (0-1)
- **Timeout (s)** - Optimization time limit in seconds (5-60)
- **Drop unmatched** - Remove labels that don't meet similarity threshold
- **Real-time status** - Shows optimization progress and results

How it works:

1. Extracts all unique labels from your current data (1K-30K labels supported)
2. Runs iterative optimization with multiple strategies (random, frequency-based, diverse selection)
3. Uses hill-climbing optimization to find the best possible centroids
4. Shows coverage percentage and timing information
5. Returns recoded links table with optimal centroid labels

Optimization Strategies:

- **Random selection** - Tests random starting points
- **Frequency-based** - Prioritizes most connected labels
- **Diverse selection** - Maximizes distance between centroids
- **Hybrid approach** - Combines best-so-far with random exploration

Performance Features:

- **Sampling strategy** for datasets >1000 labels (uses representative subset)
- **Early termination** when excellent coverage ($\geq 95\%$) is achieved
- **Configurable timeout** prevents infinite optimization loops
- **Multiple iterations** with different starting strategies for robustness
- **Smart caching** - Embeddings cached separately from algorithm parameters for fast parameter changes
- **Quote-safe processing** - Handles labels with quotes, apostrophes, and special characters

Technical Implementation:

- Client-side optimization using cosine similarity on embeddings
- Hill-climbing algorithm with local search improvements

- Genuine optimization problem solving (not just k-means clustering)
- Real-time UI feedback showing progress and final results
- Handles massive datasets efficiently through smart sampling
- **Original label preservation** - Stores original labels in `_recoded` metadata for map display
- **Chain compatibility** - Works seamlessly with zoom filter and other transformations

Soft Recode Integration:

- Optimized cluster results available as magnet source in Soft Recode filter
- AI can generate meaningful labels for optimal centroids
- Seamless workflow from optimization to AI-powered naming

This filter implements the optimization challenge described in the technical documentation: finding optimal centroids that maximize label coverage within similarity constraints.

Tribes filter

Requires Pro subscription

 **What this does:** Group your sources (participants/documents) by how similarly they describe causal relationships. This reveals different "tribes" or perspectives in your data - for example, optimists vs. pessimists, or urban vs. rural viewpoints.

Tribes button in the **Links tab** (next to Create/Filter links) opens a modal. Operates on the **currently filtered links** and saves results to a custom source column. Column names are coordinated with cluster count: 5 tribes → `Tribes_5`.

Modal controls:

- **Extra fields** - Optional multi-select (*s_ source columns first, then others*). Source-level (*s_*) count 1 per source; link-level per link. $\Delta s = s_with_all - s_without_field$; negative = removing that field improves silhouette
- **Preview** - Table of sources per tribe, p-value (chi-square), and silhouette for k=2–6 (Refresh preview to update)
- **Clusters** - Select 2, 3, 4, 5, or 6
- **Save to Sources** - Runs clustering for selected k and writes tribe ID (T1, T2, ...) to column `Tribes_<k>`

How it works: Builds sentiment-aware cause×effect matrices per source, TF-IDF weights them, clusters with k-means, and writes to `sources.metadata.custom_columns[col]`. Enriched links get `s_<col>` (e.g. `s_Tribes_4`) for use in [Source Groups](#), [Label by Group](#), and [Pivot Tables](#).

- The report's **Counts (Report)** toggle controls whether those tables use **Sources** (unique `source_id`) or **Citations** (links).



PROJECTS PANEL

📁 **What you can do here:** Organize and manage all your research projects in one place. Create new projects, share them with collaborators, add descriptive tags, and control who can view or edit your work. You can also merge multiple projects together or archive old ones to keep your workspace tidy. See also the [File menu](../file-menu/) for more project management options.

Project Management

- **New project** ➕ - Create with name and description
- **Load project** ▶ - Open selected project
- **Edit project** - Click row to modify name, description, tags, sharing (or use [Edit project Modal](#))
- **Archive/Unarchive** 🗄️ - Hide/show projects (archived projects are read-only and hidden from public view)
- **Read-only toggle** - Restrict editing even for owners/editors (independent of archive status)
- **Archive toggle** - Show/hide archived projects in the table

Bulk Operations

Select projects with checkboxes, then:

- **Delete** 🗑️ - Remove projects and all data
- **Apply Tags** 🏷️ - Add tags to selected projects
- **Remove Tags** ➖ - Remove tags
- **Toggle Archive** 🗄️ - Archive/unarchive
- **Merge** 🔗 - Combine multiple projects into one

Sharing and Permissions

- **Email-based collaboration** : add and remove colleagues' email addresses
- **Locked / Read-only permissions** for viewing without editing
- **Global sharing** 🌐 for public read-only access (free to turn on; keeping a project private instead needs an active paid plan, see [Private projects and subscription expiry](#))
- **Permission badges** next to project names
- **Admin only: admin panel** 👤 for user management

Note: In the Edit Project modal, the informational notice "Upgrade to a Private subscription to create or share private projects. New projects on the Free plan are public by default." is shown only to users who can edit the project (owner, editor, or admin). Viewers do not see this notice.

Versioning

The app automatically backs up your project, so you can restore earlier snapshots if you want.

- An automatic backup is made every 10 minutes if you have made changes.
- You can make a manual backup from the Project Info screen.
- You can use the Version dropdown menu in the Project Info screen to see all available backups with details.
- From here you can restore a backup, with a confirmation step before applying changes.
- After restoring an earlier version, you can always go back to the latest version if you want, using the same dropdown menu.

This panel shows a dropdown list of times when you made changes to the mapfile in UTC/GMT. Along with the size of your file which can help you identify which timepoint you want to revert to. It can be easy to forget what time you made alterations to your file, so if you're likely to want to restore a previous map it is best to note the time so that you can easily return to it.

Named versions. When you make a manual backup you can give it a name, for example **actor-focused** or **normal**. This is useful when you keep more than one style of coding in the same project.

- Named versions are kept forever; unnamed backups are pruned over time (see below).
- Names can be reused: the version list shows each entry as date and time plus its name, so the timestamp tells versions in the same line apart.
- The name box is prefilled with the name of your newest named version, so saving again into the same line is one click. Clear the box to make an unnamed backup, or type a new name to start a new line.
- Once a project has at least one named version, a shortcut button  appears in the Projects Bar, just left of the pencil, so you can open Versions with one click instead of going through the File menu.
- Restoring a named version puts a fresh copy of it at the top of the version list under the same name, so the newest named version always shows which coding line you are working in.
- Every bookmark records the exact backup that was current when you saved it (making a fresh backup first if needed). Slides in the Reports list and slideshow show the version name as a badge, so a deck can mix slides from different coding styles and you can see which is which. Backups referenced by a bookmark are never pruned while the bookmark exists.



SOURCES PANEL

 **What you can do here:** Upload your research documents (PDFs, Word docs, RTF, TXT, MD) or XLSX with a statements tab, and organize them with custom metadata like participant demographics or interview dates by adding and editing **custom columns**. This is your document library and metadata manager.

Upload & Setup

Upload Source Texts

- **New Project or Upload Sources** (green + in the Projects bar): besides files, you can **paste plain text** into **Or paste text**; it is imported like a TXT file as `pasted-text.txt`. You can combine paste with file picks in one step.
- **Click to select** one or more PDFs, DOCX, RTF, TXT, MD, or XLSX files
 - **XLSX:** Requires a `statements` tab with `source_id` and `text` columns; other columns become custom columns. Optional `sources` tab for metadata. Adds sources to the current project (no new project created).
- **Optionally split large documents into multiple sources** using separator patterns
- **Optionally set custom columns on upload** (applies to all uploaded sources)
 - Type an existing column name (or a new one) and optionally set a fixed value, e.g. `Country = Zambia`
 - Leave the value blank to just add the column (no value set during upload)
- **Confirm Upload modal** (shown when "Show Advanced" is on in [New Project or Upload Sources](#)) lets you adjust filenames→ID mapping, add custom columns, and split sources. When Show Advanced is off (default), uploads use defaults (keep filenames as IDs, no custom columns, no split) and skip this screen.
 - On completion, the app automatically:
 - selects the sources via [Sources Dropdown](#)
 - switches to the [Create Links](#) sub-tab and loads the first source's text into it
 - on the right, switches to Sources panel with and the View & Edit subpanel

Splitting documents into multiple sources with source separators

This feature helps you split individual documents you upload into multiple sources. These separators are *hard*: you use them just once, on uploading one or more documents, to produce multiple sources.

- In the Confirm Upload screen, there is a Sources Separator text box
- Text lines matching special "regex" patterns separate into multiple sources. So if you have sections marked with "Source Number 12", "Source Number 13" etc, just put "Source Number.*" in the box.

- This will produce multiple sources with "Source Number 12" etc as source_id
- Live preview of new source IDs with count
- User can leave blank for normal upload

Section Separators

This feature helps you split existing source texts into sections. These separators are *soft*: they don't permanently change the file and you can add or remove one or more separators on the fly.

- Into the box, type one **regex pattern per line**. The pattern must match the *whole separator line* in your source text, so use `.*` to match the rest of the line. For example, if your text contains lines like `statement_id: 42` or `Question 7: ...`, enter `statement_id:.*` or `Question.*` respectively.
- Matching lines are styled as **headings** in the text viewer.
- Each link is assigned to the section it falls under. The matched text (e.g. `statement_id: 42`) becomes the section value, available as a column (e.g. `section_statement_id`) in:
 - The [Everything Filter](#) — so you can show only links from specific sections
 - The [Sections Filter](#)
 - Pivot tables and statistics
- Links before the first separator in a source get the value `Initial Text`.

Sample Check

You want a table showing your sample: gender * region?

Use this simple customisable table to check your sample according to any [custom columns](#) you have defined.

- See also: [Pivot Tables](#) and [Analysis Filters](#).

Custom Column Analysis

- **Aggregate by** multi-select for cross-tabulation
- **1 column**: Simple count table
- **2 columns**: Cross-tabulation table
- **Maximum 10 values** per column for analysis
- **URL state preserved** for bookmarking

View & Edit Your Sources

- See also: [Tips for all tables](#) and [Custom columns](#).

Sources Table

- NEW column **Source Prompt** - this new column shows the first few characters of any text in this field. It can be edited as usual with the existing pencil icon/ edit button in each row.
- **Edit the source text:** the pencil/edit button opens the Edit Source modal, where the **Content** field is now editable along with the title, source prompt and custom fields. Changing the existing text will probably break any links coded beyond the point of the change, because each link remembers where in the text it was coded and those positions shift. Where possible, only add new text at the end. Saving after a text change asks you to confirm first.
- **Checkbox selection** for analysis pipeline
- **Row editing** with keyboard navigation
- **Custom columns** for metadata
- **Column visibility control:**
 - Click the small **:** in any header to hide that column (or hide selected columns) and quickly show hidden columns again.
 - **Manage Columns**  is the main source-table column manager. It includes a checkbox list for Sources table columns, with **Select all / Select none / All source / No source**, and drag-to-reorder.
- **Uncoded column** - Shows true/false for sources with no links; filterable to find uncoded sources
- **Fullscreen mode** available

Table Editing Features

- **Range selection** - Click and drag
- **Copy/paste** - Ctrl+C, Ctrl+V
- **Arrow key navigation**
- **Delete/Backspace** to clear cells
- **Column/row selection** - Click headers
- **Double-click editing**

Custom Columns

Custom columns are where you store extra data about each of your sources, e.g. gender, location, type; whatever you want. You can create, edit and remove custom columns. Then you can use them to filter your data.

- You can also use them for source-level memo columns; see [Link custom columns](#) for the same idea on individual causal links.
- **Manage Columns**  opens a modal to add/remove custom columns, choose which Sources table columns are visible, and reorder them
- After you add a custom column and reopen the modal, the new column appears as an existing project column; the add box starts empty again
- Column **names** are tidied automatically when you create or rename them: only letters, numbers, spaces, hyphens and underscores are kept, so other symbols (such as **\$** or **/**) are dropped. This stops

odd characters from breaking the column.

- **Double-click a cell in a custom column to edit it** or via source edit modal
- **Copy and paste** selections with ctrl-C, ctrl-V.
- **These columns are available elsewhere in the app, e.g.**
 - In the [Source Groups filter](#)
 - In the [Everything filter](#)
 - In the [Sample Check table](#)
- Link custom columns work the same way in the [Links Table](#), but apply to individual causal links rather than sources.



MAP PANEL

 **What you can do here:** See your causal relationships as an interactive network map. Drag nodes around, click on links to edit them, and use the controls to customize how the map looks. You can even drag one factor onto another to quickly create new links. This is where your data comes to life visually.

Map Controls

-  **Jump to factor** (type-to-search dropdown): quickly find and select factors on the map (supports multiple selections).
-  **C Refresh layout** (button): reset the map layout after zooming/moving.
-  **Copy image to clipboard** (button): copy a high-quality map image for reports/slides.
-  **Copy legend** (button): copy the map legend text.
-  **Zoom in/out** (controls): zoom the map view.
-  **Full screen** (button): expand the map to fill the whole window; click again or press Escape to return. You can still click factors and links to edit them while full screen. The same button appears on the Links, Factors, Projects and Sources tables.
-  **Double-click** (gesture): zoom in to that point on the background.

Map Legend

Discrete text legend showing:

- projectname and included sources
- Citation coverage percentage
- Visual encoding explanations (link sizes, colors, numbers)
- Applied filters summary
 -  Tip: Click **Copy legend** to copy this text to clipboard.
 - You can drag the legend box to reposition it on the map.

Evidence (shortcut to Links Print View)

-  **Evidence** (button, bottom-right on the map, opposite the legend): switches to the **Links** tab and opens **Print View** for the **same filtered links** you see on the map. It applies a preset—**Group by** Bundle and Source, **sort** by citation count (highest first), **Show details** and **Context** on, **page size** 50—and updates the URL so you can bookmark or share that layout. A toast links to Print View help; a short hint may point to the **Print view** toggle.

Map Formatting

Customisable formatting (Things you can tweak)

Layout and interaction

-  **Layout** (dropdown): choose how the map is laid out.
 - Interactive and most other layouts are good while you are conducting your research (fast + supports the [interactive features](#)).
 - Print/Graphviz is best for static images (reports/journal articles). In Graphviz SVG you can still pan/zoom (mouse wheel, double-click, Shift+double-click).
-  **Groups** (dropdown): layout maps with top-level factors as boxes which group together their "children". Choose how group titles are extracted from factor text: No groups / Level 1 (;) / First colon (:) / Square brackets [] / Round brackets ().
-  **Direction** (radio group): LR, RL, TB, or BT (for Interactive and Print/Graphviz layouts).
-  **Link direction** (dropdown): Normal (directed arrows) vs Undirected (dots at both ends).
 - In Undirected mode, dots use the same colours as arrowheads (including sentiment colouring). When sentiment is neutral (o), they use **Link Colour**.
 - Note: when the [Combine Opposites filter](#) is active, tail/head can still have different colours.

Factors

-  **Factor labels** (dropdown): what to show next to each factor (same data as the [Factors Panel](#)).
 - Citation count (default) / Source count / Sentiment (mean incoming) / None
-  **Factor colours** (dropdown): Outcomeness (default) / Influence / **Sentiment discrepancy** (orange scale: how much the factor's incoming links disagree in sentiment) / Citation count (total, in, or out) / Source count (total, in, or out) / **Label segment** (full label or the same segment patterns as **Groups**) / None
-  **Factor sizes** (dropdown): Citation count (default) / Source count / None
-  **Label Width** (slider): how wide a factor label may run before it wraps onto another line. The width is measured in **pixels**, not characters, and runs from 30px at the far left to 800px at the far right on a logarithmic scale, so equal steps make bigger jumps at the top end. The number under the slider is the width it currently means. This is the main control for the shape of the whole map: a **narrow** width wraps labels onto more lines, making factors tall and the map tall and thin; a **wide** width keeps labels on one line, making the map short and wide. So to get a taller map, move this slider **down**, not up. **Tidy** (below) sets this slider for you unless you turn it off.

Spacing (Print / Graphviz layout) {#map-spacing}

-  **H-Gap** and **V-Gap** (sliders): one sets the gap between the levels of the map, the other the gap between factors on the same level. Which is which depends on **Direction**, so the two labels swap when you change it: the slider marked with the horizontal arrows always changes horizontal spacing. In LR and RL the levels run left to right; in TB and BT they stack top to bottom.

- 🖱️ **Tidy** (tick box, on by default): sets **Label Width**, **H-Gap** and **V-Gap** for you. It aims to make the factor labels as large as they can be on screen while the arrows stay easy to follow, which usually also leaves the map with roughly the same proportions as the pane it sits in.
 - To change the proportions it fits to, drag the map pane's separator bar so the pane is wider or taller. The map tidies itself again.
 - To set the spacing yourself, drag any of those three sliders. That unticks Tidy and your settings stay put. Tick it again to hand control back.
 - Where a map's arrows leave their factors very steeply and bunch into a bundle, Tidy pushes the levels further apart, and it will accept smaller labels to do so. Legibility of the arrows comes first.
 - Some maps cannot be improved much. A long single chain, or a set of factor pairs with nothing joining them, will not sit well in a pane whatever the spacing.

Links

- 🖱️ **Link labels** (dropdown): what to show on each link.
 - Citation count (default) / Source count / Sentiment / Label by Group / Unique Sources / All Sources / Unique Tags / Unique Tags (Tally) / All Tags / None
- 🖱️ **Link widths** (dropdown): Citation count (default) / Source count / None
- 🖱️ **Link label font size** (control): change link label font sizes.
- 🖱️ **Arrowhead size** (control): scale arrowhead size (Interactive + Print/Graphviz). Default 100% keeps current appearance.
- 🖱️ **Link colour** (colour picker): sets the default link line colour (Interactive + Print/Graphviz). When sentiment is neutral (o), this colour is also used for arrowheads and node borders.
- 🖱️ **Links highlight** (dropdown): optional extra "halo" highlighting without changing the base colour scheme.
 - Off (default)
 - Reverse (backwards/same-rank in current layout direction)
 - Significant (when Label by Group shows ↑/↓)
 - Feedback loop (2 / ≤3 / ≤4 factors)
 - Feedback loop + reverse (combine the above)

Other

- 🖱️ **Show self-loops** (toggle, default on): show/hide A→A links on the map.

Why a factor's number can look bigger than its arrows add up to. A self-loop is a link from a factor back to itself (for example a "plain coding" that just records "this theme is present"). When both factor numbers and link numbers are set to citation count, a self-loop is counted on its factor **twice**: once as a citation going out of it, once as a citation coming into it. But on the map that self-loop is drawn as a single looping arrow with a single number. So if you add up the numbers on a factor's arrows by eye, any factor that has a self-loop comes out higher than you expect. This is normal and correct: it is the standard way graphs count connections (a self-loop adds one to both the factor's "in" tally and its "out" tally). If you do not want self-loops counted this way, turn off **Show self-loops**, or use the [Exclude self-loops filter](#).

Fixed visual appearance (things you can't tweak)

Some parts of the map's appearance are automatic (i.e. they are not controlled by the Map Formatting widgets above):

Leading emoji (Print / Graphviz only):

- If a factor label starts with an emoji, the **Print (Graphviz)** SVG post-process moves that emoji to a larger symbol offset near the top of the node and removes it from the inline label text (trivial matches like lone digits are left inline). **Interactive** layout keeps the full label, including any leading emoji, in the normal label position.

Link geometry (bundling):

- Links are bundled and drawn as curved edges for readability.

Automatic colouring overlays:

- Arrowhead colours reflect mean **sentiment** for that link bundle (neutral uses your chosen **Link colour**).
- When the [Combine Opposites filter](#) is active, arrowhead colours instead reflect **flipped share** (tail=cause, head=effect).

Automatic highlighting:

- Factors that match filters like [Factor Label](#) or [Path Tracing](#) show dashed coloured borders.
- Factor border colour reflects mean incoming edge sentiment (but when Combine Opposites is active: average flipped share, blue→red).

Interactive Features

These work for all layouts except Print/Graphviz layout (which is mostly for static export, but does support clicking nodes/links now).

- **Drag factors** to temporarily reposition them
- **Drag factor to factor** to create new links
- **Shift+drag** for box selection of multiple factors (opens edit modal)
- **Ctrl+drag** for box selection of multiple factors (direct selection, no modal)
- **Click a link** to edit.
- **Click a factor** to edit; shift-click or ctrl-click to add to selection without opening modal.

Editing and deleting (multiple) factors

- Select factor(s) by clicking a factor, shift-click or ctrl-click to add more, or shift+drag/ctrl+drag a box around multiple factors, then:
 - Move selected factors together
 - Delete matching factors everywhere or in current view only
 - Rename matching factors everywhere or in current view only

What does "everywhere or in current view only" mean?

- **everywhere:** all links containing factors with exactly the selected labels will be deleted
- **in current view only:** all links containing factors with exactly the selected labels (and matching the current filters, i.e. those you can see in the current map) will be deleted

💡 Tip: By control-clicking or shift-clicking multiple factors you can easily rename several at once, e.g. you can merge multiple factors as a single factor.

Grid layout

What this is for: Sometimes you want factors to appear in a rough **grid**—like “this cause is clearly to the left of that one”—instead of leaving the layout algorithm to guess. You do that by putting a small **coordinate tag** in the factor label (anywhere in the text). The app reads the tag to place the box, and (by default) hides the tag so respondents don’t see a string of numbers.

This is especially useful when:

- your factors fit logically into some kind of positional story like a theory of change and you want to keep them there
- you are showing many different versions of a larger map (e.g. districts within a country) and you want them to be more easily comparable.

How to write a tag: Use two whole numbers: a **column along the main flow** of the map and a **row** across the flow. For example (2,1) or (2.1) in the label means “second step along the story, first slot across.” You can use square brackets instead of parentheses, and you can leave one side blank if you only care about one direction—for example (3,) fixes the step along the flow but not the slot across.

In Map Formatting

- **Grid layout** — Turn this **off** if you want to ignore tags and let the map place everything as usual. It only appears when at least one factor in your current links actually has a tag. Default is **on**.
- **Strip grid tags** — Turn this **off** if you want to **see** the (...) or [...] text on the map. Default is **on** (tags still affect layout when **Grid layout** is on, but the numbers are hidden on the label). These two switches work independently.

Interactive map (the usual live map): tagged factors snap to a grid; anything without a tag is still laid out automatically, but is kept **near** the grid so the picture stays readable.

Print / Graphviz (static SVG): the app can't pin exact pixel positions like the live map, but it still respects **left-to-right order of the first number** along the direction you chose, and does its best to respect the second number **within** each step. If you only give a second number and not the first, Print layout can't place that factor on the "step" axis—use both numbers when you care about order.

Heads-up: The grid helpers only apply to the **default interactive** layout and to **Print**. If you switch the live map to layouts like **Cola**, **Circle**, or **CiSE**, coordinate tags are **not** used for placement.

Vignettes

 **What you can do here:** Generate AI-powered narrative summaries of your causal maps. Choose between a **whole-map** summary that covers the relationships in your current filtered view, or a **typical-source** story that focuses on one representative respondent. Useful for reports, annexes, and explaining your analysis in plain language.

For practitioners (getting started)

What are vignettes and why use them?

Vignettes turn your coded causal map into readable prose. After you have links (and usually quotes on those links), the AI can draft a narrative that names themes, tensions, and patterns in your material. You stay in control: you choose the prompt (tone, audience, structure), and you can edit the text afterwards. They are a fast way to move from “many links on a map” to “something I can paste into a report or share with stakeholders”—not a substitute for your judgement, but a structured first draft.

Whole-map vignette

Use this when you want a bird's-eye story of **everything that is currently in the map** (respecting your filters and source selection). The app sends a compact summary of factors and bundled links (with sentiment), **evidence snippets** (quotes and source IDs from highlighted bundles where available), and up to **30 “typical” sources** scored by how much they represent common bundles—each with bundles, link-level quotes, and a small **metadata preview** (title, filename, and simple custom fields). If you want the narrative to emphasise particular edges, set **Map Formatting → Links highlight** to **Significant** or **Feedback loop** first; that snapshot is included so the model can focus there. **AI region** (where Gemini runs) follows **Project Details → AI region**, not a separate control on this card.

Typical-source vignette

Use this when you want a **single-respondent case study**: the app picks the most “representative” source for the **current** map view (using link counts and coverage of bundles, with a bias so very long documents do not always win). It sends that source’s **full text** plus its links with quotes and sentiment, so the model can write as if telling **one** story. Do not ask it to generalise across the whole project in this mode—the prompt and data are scoped to that one source.

How to use (UI):

1. Open the **Vignettes** card on the Map side panel and expand it if needed.
2. Choose **model** (and optional **thinking** controls if your model supports them). **Region** for AI is set per project under **Edit project** (AI region), not on this card.
3. (Optional) Leave **Enable checking (second AI pass)** on so a checker can review the draft; notes appear in a collapsed panel when relevant.
4. Edit the **Whole-map prompt** and/or **Typical-source prompt** (prompt history: dropdown and prev/next; see [tips on using prompt history](#)).
5. Click **Write Whole-map Vignette** or **Write Typical-source Vignette** (two separate buttons).

Tip (optional): tell the AI which links matter most (whole-map mode):

- Go to **Map Formatting** → **Links highlight** and choose **Significant** or **Feedback loop**.
- When you run the **whole-map** vignette, the app includes a small snapshot of those highlighted edges so the model can focus on them.
- If **Links highlight** is **Off** (or **Reverse** only, with no edges flagged), or nothing qualifies, **no extra highlight list is sent**.

What each mode sends (summary):

- **Whole map**: Aggregated nodes/edges for the current view, optional **links highlight** snapshot, **bundle evidence** (quotes/source IDs sampled from top bundles so large projects stay within limits), and up to **30 typical sources** with bundles, per-link quotes, and compact source metadata—not the entire raw links table.
- **Typical source**: One chosen source’s **full text**, its links with quotes and sentiment, simple node frequencies, and optional links-highlight context for that source.

Output format: Markdown (headers, bold, lists, blockquotes, code blocks). You can copy from the result area.

Bookmarking & restore:

- Each run saves a **bookmark** for the current view (description: **Vignette (whole|typical): <your prompt>**) and appends a link at the bottom of the vignette output.
- The bookmark footer includes the **model name** used.

- Vignette-related settings are stored in **URL state** where applicable (model, thinking, checking, prompts); project **AI region** lives in project metadata.



FACTORS PANEL

Ideas Garden: [Factors Panel](#)



What you can do here: See all the factors (causes and effects) in your data, ranked by how often they appear. Select multiple factors to rename them, merge similar ones, or delete unwanted entries. If you've added demographic data to your sources, you can also see statistical breakdowns showing which groups mention certain factors more often.

above both links and factors tables add a toggle "Use filters". If on (default) the table is filtered by the links filters. If off, we bypass this part of the pipeline and filter only by project and sources.

The Factors panel displays all unique labels from the current filter pipeline.

Table Features:

- Columns include:
 - **Citation Count** – total number of citations of this factor (as cause or effect)
 - **Source Count** – number of different sources mentioning this factor
 - **Citation Count: In** – number of citations of this factor as an effect of something
 - **Citation Count: Out** – number of citations of this factor as a cause of something
 - **Outcomeness** – $\frac{\text{in}}{\text{in}+\text{out}}$ (how much a factor is an outcome vs a cause)
 - **Influence** – “influence as a cause” score (Katz-style centrality in the causal direction, cause → effect). We shift the numbers so the smallest factor is 0 (makes differences easier to see), but we **don't** scale to 0..1.
 - **Source Count: In** – number of sources mentioning this factor as an effect of something
 - **Source Count: Out** – number of sources mentioning this factor as a cause of something

Why “Importance” can be better than Citation Count: Out:

- **Citation Count: Out** just counts how often a factor is used as a cause. If you point to *anything* a lot, you score high.
- **Influence** also looks at *what you point to*: causing a factor that goes on to cause lots of other things gives a factor more influence than causing a dead-end.
- Sorted by citation count (descending order)
- Click-to-select (no checkbox column)
- Server-side pagination consistent with other tables
- Actions column with edit button to open factor edit modal

Action Buttons:

-  Delete: Remove selected factors
-  Relabel: Rename selected factors
-  Search/Replace: Find and replace text in factor names
-  Merge: Combine multiple factors into one
- Buttons disabled until factors are selected
 -  Tip: Use [Search/replace](#) for quick, scoped relabeling.

Find out more about bulk delete and relabel of factors [here](#).

Bulk factor labels editor

The search/replace functions in the factors and links tables are useful, but what if you have thousands of factors to look at? You might prefer this bulk editor.

Toggle the **Bulk Edit** switch to edit multiple factor labels at once. The table header remains visible for sorting and filtering, while the table body is replaced with a line-by-line editor.

Features:

- **Multi-cursor editing:** Use **Alt+Click** or **Ctrl+Alt+Up/Down** to add multiple cursors
- **Find occurrences:** Use **Ctrl+Alt+Right** to add next occurrence, **Ctrl+Alt+Left** for previous
- **Select all matches:** Use **Ctrl+Shift+L** to select all occurrences of selected text
- **Line-by-line editing:** You can only edit existing labels - you cannot add, remove, or reorder lines
- **Recoded labels:** Labels that have been recoded (shown with yellow background) are read-only and cannot be edited
- **Sort and filter:** The editor automatically updates when you sort or filter the table (any unsaved edits are discarded)
- **Important:** If any **label-transforming filters** are active in the filter pipeline (e.g. Soft Relabel / Zoom / Collapse / Remove brackets / Combine opposites / Soft Recode), the Factors table is showing **transformed labels**. Bulk Edit saves by **exact label text match**, so editing a displayed label like **foo** will only rename factors whose actual stored label is exactly **foo** (it will not rename all original labels currently shown as **foo**). Consider turning off label-transforming filters before bulk editing labels.

 **TIP to recode several factors into one:** Simply overwrite all the old factor labels with your desired label.

 **TIP to translate your factors:** Select all the labels, copy them, paste them into any AI chat (ChatGPT, Claude, etc.) and ask it to translate them, then copy the result back over the labels here. Keep the same number of lines. It's the same idea as the built-in AI answers, just quick and rough. Works for tidying up wording too, not only translation.

How it works:

1. Toggle **Bulk Edit** on
2. Edit factor labels directly in the editor
3. Press **Save Changes** to apply your edits
4. A confirmation dialog shows which labels will change and how many links will be affected
5. After saving, the editor refreshes to show the updated labels

Remember, what it shows depends on:

- any filters you set (sources, most frequent links, etc) — just like the map does
- any filters or sorting you set in the table itself. The bulk editor is really just a special version of the normal factors table, so it respects pagination etc. If you want to see more factors, set the "Page Size" (default is 10) to a larger number.

The editor has a second column on the right which provides some live info about the selected factor: source and citation counts, and a list of sources mentioning it.

Search/replace

Near the top is a row containing a search box. If you type something into it,

- a replace box and a Replace button also appear.
- the table is filtered to show only matching rows

The search is **case sensitive**.

You can then alter what you see in the Replace box:

- in the factor label column in the table, you see a preview of what the affected rows would look like;
- if you delete all the text so the replace is empty, the preview shows the effect of deleting the search text from each label.

Then when you are satisfied, check all the checkboxes where you want to update the labels as shown. If you want, select all rows using the checkbox at the top of the column. Note, if there are more hits than fit on this page of your table, you'll want to either treat each page separately or increase the page size with the Page Size selector.

Finally, hit the Replace button to actually update the labels as shown in the rows you selected. What actually happens is that the Cause and Effect labels in all the currently selected links are changed. As you'd expect, this search/replace only affects the factors for the currently selected links: for example if you have only selected the first three sources, this update will not affect the links in the other sources.

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Demographic Breakdowns

Ideas Garden: [The factors table \(group comparisons + tests\)](#)

- **Breakdown selector** - Choose custom columns to analyze by demographics
- **Count type** - Source count (default) or citation count
- **Display mode** - Counts (default) or % of baseline (cell as a percent of that breakdown group's total across all factors)
- **Statistical testing** - Chi-squared analysis to identify significant patterns
 - See also [Statistical Significance Testing](#)

Statistical Significance Testing

When you select exactly **one custom column** for breakdown, the factors table includes powerful chi-squared significance testing to identify factors that are preferentially mentioned by different groups.

Example bookmark:

- [Factors: significant differences between age groups](#)

Show Differences dropdown appears with threshold options:

- **Off** (default)
- **p < .1** (marginally significant)
- **p < .05** (significant)
- **p < .01** (highly significant)
- **p < .005** (very highly significant)

Visual indicators:

- **Significant column** - Shows "Yes" (red highlight) or "No"
- **Cell coloring** - Blue = mentioned more than expected, Orange = mentioned less than expected

Ordinal testing (numeric breakdowns):

- If the chosen breakdown is numeric-like ($\geq 95\%$ of non-missing values parse as numbers), an extra column **Ordinal Sig.** appears.
- It uses the Mantel linear-by-linear association (Cochran–Armitage trend) with ranks $1..k$ and the same $2 \times k$ totals as chi-squared.
- The existing **Significant** column (chi-squared) remains; you can compare both.
- The threshold from Show Differences applies to both tests.

in the factors table when factor-show-differences is on, we calculate chi-sq. but if over 95% of non-missing values in the column selected in #factor-custom-column-input can be interpreted as numeric, we should use an ordinal test instead, or apply an ordinal correction to make the chisq test more powerful

Developer note: Percent mode divides each factor's cell by the group total for that breakdown column.

State keys: `factorDisplayMode`, `significanceThreshold`.



LINKS PANEL

Ideas Garden: [Links Panel](#)

 **What you can do here:** View and manage all your causal links in a spreadsheet-like table. You can sort, filter, and edit individual links, view the quotes like a printed page, or export your data to Excel. Each row shows one causal relationship with its source quote and any additional details you've added. Great for detailed review and bulk editing of your causal map data.

Links Table

Links Table Features:

- Standard column filters, sorting , and pagination
- Sentiment column with numeric values (-1 to 1) and blue/white/red conditional formatting (blue = higher, red = lower, white = mid-range relative to the current view)
- **Citation Count** – total number of links in each bundle (cause >> effect pair), with muted green → white conditional formatting (darker green = more links in that bundle relative to the current view)
- **Source Count** – number of different sources contributing links to each bundle, with the same muted green → white conditional formatting
- **AI run** – short id of the [ai_runs](#) row for links created by [AI coding](#) (hover for full UUID); empty for manual links or older imports
- **Package** and **Package type** – standard columns for [coding packages](#): links claimed to work together (e.g. "you need both oxygen and a spark to make fire") share a code in Package, and Package type names the connective ([AND](#), [OR](#), [DESPITE](#)). Both are editable inline.
- Checkbox selection for bulk operations
- Edit functionality opens causal overlay for link modification
- Action button to open coding in the Sources pane and scroll to the corresponding highlight
-  Clear Table Filters option
 -  Tip: For label changes, prefer [Factors Search/replace](#) when working on labels across bundles.

Link Editing:

- Single link click opens editor popup
- Multiple link selection opens chooser interface
- Consistent with coding panel behavior
- **AI run id after manual save:** the link editor uses a split update path:
 - **1 cause × 1 effect:** updates that existing row in place (keeps row id; [ai_run_id](#) remains unless changed elsewhere).

- **multi cause/effect (cross-product):** deletes the original row and inserts new rows; those new rows do not carry the original `ai_run_id`. The original AI call is still traceable in **Responses**.

Link Custom Columns

Alongside **tags** and **sentiment**, each link can now carry its own named custom fields. Use these when you want more structured information on each causal claim, for example `confidence`, `policy_area`, `mechanism`, `evidence_strength`, `time_horizon`, or a numeric score.

This is also the main place to add QDA-style memos while you code. Create one or more memo columns, for example `link_memo`, `evidence_note`, `interpretive_note`, or `coder_reflection`, and use them to record why you coded a link in a particular way, doubts about the evidence, or analytic notes you want to revisit later.

- Create/remove column names in **Manage Link Table Columns**.
- Show/hide Links table columns in the same modal (checkbox list), or from the header : menu.
- Invert a text column's header filter to **does not contain** from the header : menu (**Invert filter**); the column header turns pink while inverted. The choice persists in the URL. The same toggle is on the Sources table (Title, Filename) and the Projects table (Name, Owner). Implemented once in `js/negatable-header-filter.js` (`NegatableHeaderFilters`), shared by the links, sources and projects tables.
- Create new fields directly from the **Link Editor** custom-fields picker.
- Edit values in the **Link Editor** or directly in the **Links Table**; this works the same way as editing `source custom columns`, but on link rows instead of source rows.
- Use them elsewhere in the app, for example in the `Everything filter`, links-table grouping/breakdowns, the `Link Editor screen`, and map display via the `Map Custom Columns filter`.

Links Utilities

- Download as Excel
- Take a screenshot and copy it to clipboard
- Clear any filters at the top of the table columns
- Bulk delete any selected rows in the table
- **Package** 📦 the selected rows: give them a fresh shared package code. The dialog lists every selected link with its own package type field (`AND`, `OR`, `DESPITE`, or your own), so a package can mix types where that is what the source said. `AND` and `OR` buttons at the top fill every row in one click, which covers most packages. **Unpackage** 📦 clears the codes and types. See `coding packages`.

Coding packages

Sometimes a source presents several influences as working *together*: "you need both oxygen and a spark to make fire". Two separate links would drop that; one combined factor label ("oxygen and a spark") would not be parsable. Instead, give the links that belong together a shared code in the **Package** column,

and name the connective in **Package type**: **AND** (a claimed conjunction), **OR** (presented as alternatives), or **DESPITE** (a **despite link** packaged with the influence that overcame it, e.g. "the fan failed to cool him down because the sun was too strong"). The code itself is arbitrary; the app mints codes for you, either from the **Link Editor's Package toggle** while coding, or from the bulk Package button here, whose dialog also takes the package type. Both columns stay editable inline afterwards. Filter with the **Everything filter** on **package** (one package, or any code) or **package_type**. Full rationale: the Garden working papers "Minimalist coding" and "Coding packages".

Row Grouping and Print View

- **Group by selector** - Choose one or more columns to group rows by values. This applies both the the links table and Print View.

Useful Columns:

- **Bundle** - Shows "cause >> effect" pairs
- **Original Bundle** - If you have used filters which transform the links, like Zoom or Soft Recode, use this to also view the original causes and effects

Bulk Tags editor

Toggle **Bulk Tags** to switch the Links table into a line-by-line editor for **unique link tags** (one tag per line), sorted alphabetically.

- The list contains **all tags** found in the **current filtered links** (the pipeline output: current Project + Sources selection + Analysis Filters). It does **not** use Links table header filters or pagination.
- While Bulk Tags is on, the Links table (including header filters and pagination controls) is hidden.
- You **cannot** add/remove/reorder lines.
- Edit a line to **rename** that tag; make a line **blank** to **remove** that tag.
- Clicking **Save Changes** applies the rename/remove across **all currently filtered links**.
- Tag matching is **case-insensitive exact match** (after trimming).

Print View

The purpose Print View is to make it easy to explore and read actual quotes from the currently filtered links. What it does is show, instead of the contents of the Tabulator table, a printed version of the same information, leaving the table headers and filters in place. The toggle switches between table contents view and print view.

This view prints out the quotes from each row in the table, grouped by the Group By columns formatted as nested headings, and we suppress repeated headings until they change.

We also reveal two more toggles:

- Show Details: Print the values of all the extra columns such as tags and any **Custom Columns**

- Context: for each quote we add an additional three sentences at each side, highlighting the actual quotes.
- **PDF**: prints the **full** list (not just the current on-screen page) in a new tab. Use **Print** → **Save as PDF**. The export uses slightly **smaller** body type. Each **group** from **Group by** is a separate table so the **group title repeats at the top of each printed sheet** when one group spans many pages. The new tab's **title** (and the usual default **Save as PDF** name in **Chrome** / **Edge**) is **CausalMap-Links-<project>-<YYYY-MM-DD-HHmm>**. In **Chrome 131+** and current **Edge**, the **bottom page margin** lists the **current project name, date/time, Causal Map | causalmap.app**, and **page i / n**. If the system print dialog also adds its own header/footer, turn that off there to avoid duplicates.

You can manually sort the texts using to the sorting widgets in the tabulator headers, as far as allowed by the nested headers.

Search/replace

This works exactly the same as [search/replace in the factors table](#), except that it works on the Cause label and/or the Effect label.

Near the top is a row containing a search box. If you type something into it,

- a replace box and a Replace button also appear.
- the table is filtered to show only matching rows.

The search is **case sensitive**.

You can then alter what you see in the Replace box:

- in the label columns in the table, you see a preview of what the affected rows text so the would look like;
- if you delete all the replace text so it is empty, the preview shows the effect of deleting the search text from each label.

Then when you are satisfied, check all the checkboxes where you want to update the labels as shown. If you want, select all rows using the checkbox at the top of the column. Note, if there are more hits than fit on this page of your table, you'll want to either treat each page separately or increase the page size with the Page Size selector.

Finally, hit the Replace button to actually update the labels as shown in the rows you selected. As you'd expect, this search/replace only affects the factors for the currently selected links: for example if you have only selected the first three sources, this update will not affect the links in the other sources.

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Assessing link evidence quality

Use this when you want one assessed summary row per causal bundle (**cause >> effect**) while keeping all original unassessed citation rows unchanged.

The **Assessment** control sits **below the Project bar**, to the **right** of the Label Set widget. The button label is just **Assessed** or **Unassessed**. Click it to toggle modes; its tooltip explains the two modes. **Assessed** is only available once the project has at least one assessed row.

Bundle assessment UI lives on the Links pane, sub-tab **Assess links** (next to Create links / Filter links). Open that sub-tab to edit bundle-level assessed rows when the global mode is **Unassessed**; in **Assessed** mode the editor is read-only and the app switches you away from **Assess links** if needed.

Because assessed rows can use link custom fields, you can add a narrative judgement field as well as, or instead of, a simple score. For example: *Solid evidence overall, but more sketchy testimony from younger respondents.*

The card's header × sets **Unassessed**, switches to the **Filter links** sub-tab, and briefly highlights the Assessment control.

Ideas Garden: [Assessing quality or robustness of evidence for a causal link](#)

Main controls

1. 🖱️ (**Assessed / Unassessed**) (below Project bar): click to toggle between assessed and unassessed links.
2. 🖱️ (**Assess links**) (Links sub-tab): opens the bundle assessment card (when the global mode allows editing).
3. 🖱️ (×) (Card header): switches to **Unassessed**, opens **Filter links**, and briefly highlights the Assessment control.
4. 🖱️ (**Bundle selector**) (Dropdown): chooses which bundle to edit. Picking a bundle narrows the links table to just that bundle's unassessed rows so you can focus on it. The default **All bundles** shows every unassessed link; choose it again at any time to return to the full list. Opening **Assess links** starts on **All bundles**, so nothing is hidden until you pick one.
5. 🖱️ (◀ / ▶) (Buttons): move to previous/next bundle.
6. 🖱️ (**Help**) (Button): opens this section in the help drawer.
7. 🖱️ (**Tags**) (Input): sets bundle-level assessed tags.
8. 🖱️ (**Sentiment**) (Buttons + number input): sets assessed sentiment (**-1**, **0**, **1**, or custom numeric value).
9. 🖱️ (**Favorites**) (Buttons): toggles heart / exclamation / star on the assessed row.
10. 🖱️ (**Add**) (Button): adds a visible custom field to the assessed-row editor.
11. 🖱️ (× **on custom field row**) (Button): hides that field from the editor (does not delete saved values).

AI-assisted bundle assessment

1. 🖱️ **(Run scope)** (Dropdown): run on **Current bundle** or **All filtered bundles**.
2. 🖱️ **(Evidence fields)** (Dropdown): use **All row fields** or a selected subset.
3. 🖱️ **(Evidence fields list)** (Multi-select): picks fields passed to AI when using selected mode.
4. 🖱️ **(Prompt history controls)** (Buttons + dropdown): reuse previous prompts.
5. 🖱️ **(Save assessed link)** (Button): runs AI and writes results to the bundle's assessed row.
6. 🖱️ **(Status text)**: shows progress and completion counts (saved / skipped / failed).

When you save an assessed link, CausalMap stores the prompt text used for that assessment on the assessed row so you can see what instruction produced it.



PIVOT TABLES

Use this panel to build pivot tables and charts from your project data with a simple drag-and-drop interface. Three subtabs: **Pivot table**, **Summary** (one-dimensional heatmaps for categorical vars with <10 categories, mean/median for continuous), and **All-by-all checks** (pairwise significance tests).

Example bookmark:

- [Comparing groups \(gender\) — heatmap pivot table](#)

Quick start

This panel is powered by PivotTable.js, which generates most of the UI dynamically. These are the **actual widgets** you can see/use:

1. 🖱️ **(Links / Factors / Sources)** (Dropdown): **Which table to analyse?** — chooses which dataset to pivot.
2. 🖱️ **(After pipeline / Before pipeline)** (Radio buttons): **Which stage to analyse?** — chooses post-pipeline (matches other panels) vs raw data for the chosen dataset.
3. 🖱️ **(Refresh)** (Button): reloads + re-renders the pivot with latest data.
4. 🖱️ **(Sig level)** (Dropdown): significance threshold for 2-way table tests (0.10, 0.05, 0.01, 0.001).
5. 🖱️ **(Shading)** (Dropdown): conditional formatting for every table in the panel — see [Conditional formatting](#).
6. 🖱️ **(Copy to Clipboard)** (Button): copies the current pivot table/chart as an image.
7. 🖱️ **(Copy as Table)** (Button): copies the pivot as tab-separated text — paste into Excel/Sheets to get a properly aligned table (**Table renderer only**).
8. 🖱️ **(Download XLSX)** (Button): downloads the current pivot as **.xlsx** (**Table renderer only**). Merged cells (rowspan/colspan) are expanded so columns align correctly.
9. 🖱️ **(Help)** (Button): opens help for Pivot Tables.
10. 🖱️ **(Search fields)** (Text field): searches the available field/column names in the drag-and-drop list.
11. 🖱️ **Drag-and-drop fields** (field chip list): shows available fields (columns). Drag a field chip into **Rows**, **Cols**, or **Vals**.
12. 🖱️ **(Rows / Cols / Vals)** (Drag-and-drop drop zones): defines how the pivot is laid out and what values are summarised.
13. 🖱️ **(Aggregator)** (Dropdown): chooses the aggregation function (e.g. **Count**, **Sum**, **Average**).
14. 🖱️ **(Vals)** (Dropdown): chooses the numeric field to aggregate (only appears for aggregators that need it).
15. 🖱️ **(Renderer)** (Dropdown): chooses the output type (table, heatmaps, Plotly charts).

16. 🖱️ (**Row Order / Col Order**) (Dropdowns): chooses how row/column keys are sorted.
17. 🖱️ (**Filter popup on each field chip**) (Popup): include/exclude values for that field.
 - 🖱️ (**Search**) (Text field): searches within the field's value list.
 - 🖱️ (**Checkbox list**): ticks/unticks specific values.
18. 🖱️ (**✕ on a field chip**) (Button): removes that field from **Rows/Cols/Vals**.

Arrange fields (drag and drop)

- 🖱️ (**Search fields**): type part of a column name to narrow the available field list.
- 🖱️ **Drag-and-drop list**: the “pool” of fields you can use.
- 🖱️ (**Rows**) (Drop zone): fields listed down the left side of the output.
- 🖱️ (**Cols**) (Drop zone): fields listed across the top of the output.
- 🖱️ (**Vals**) (Drop zone): numeric field(s) to summarise (when needed by the chosen aggregator).
- 🖱️ (**Drag within a zone**): reorders fields.
- 🖱️ (**✕**) (Button): removes a field from a zone.

Choose the calculation ("Aggregator")

- 🖱️ (**Aggregator**) (Dropdown): chooses how each cell is calculated. **Optional** — you can leave this at the default **Count** (no additional variable needed).
 - 🖱️ (**Count**): how many rows fall into each cell.
 - 🖱️ (**Sum / Average / Min / Max**): summarises a numeric field.
 - 🖱️ (**Unique Count**): counts distinct values of a field.
- 🖱️ (**Vals**) (Dropdown): choose which numeric field to summarise (**optional**, only shown when needed).

Filter or exclude values

- 🖱️ (**Filter popup on a field chip**) (Popup): include/exclude values for that field.
- 🖱️ (**Search**) (Text field): narrows the value list.
- 🖱️ (**Checkbox list**): include/exclude specific values (includes a “select all” control).

Sorting

- 🖱️ (**Row Order**) (Dropdown): sorts row keys (e.g. by key or by value, depending on the option).
- 🖱️ (**Col Order**) (Dropdown): sorts column keys.

Heatmaps and charts

- 🖱️ (**Renderer**) (Dropdown): switches between:
 - 🖱️ **Heatmaps** (e.g. **Heatmap**, **Row Heatmap**, **Col Heatmap**)
 - 🖱️ **Plotly charts** (e.g. **Bar**, **Line**, **Scatter**, **Stacked Bar**, **Area**, **Multiple Pie**)

Conditional formatting

One **Shading** dropdown in the toolbar colours the tables in all three subtabs the same way, so the pivot table, the Summary heatmaps and the All-by-all tables always match.

- 🖱️ (**Off**): no colouring.
- 🖱️ (**Rows**) (default): shades each cell against the other cells in its own row, so you read across a row to compare.
- 🖱️ (**Columns**): shades each cell against the other cells in its column, so you read down instead.
- 🖱️ (**Cells**): shades how far each cell sits from the count you would expect if the two variables were unrelated. Blue means more cases than expected, red means fewer, and white means about as expected. Hover a cell to see its observed and expected counts. This is the view that shows you where an association actually comes from.

Row and column totals always keep their own blue shading. The pivot table's own **Heatmap** renderers colour their cells themselves, so Shading leaves those alone. The setting is saved to the URL, so a bookmark reopens with the same formatting.

Export and sharing

- 🖱️ (**Copy to Clipboard**) (Button): copies the current pivot output as an image.
- 🖱️ (**Copy as Table**) (Button): copies the pivot as TSV — pastes as a correctly aligned table in Excel or Google Sheets (**Table renderer only**).
- 🖱️ (**Download XLSX**) (Button): exports the pivot table to **.xlsx** with merged cells expanded so columns align (**Table renderer only**).
- 🖱️ (**URL state**): the pivot configuration is saved to the URL automatically, so you can bookmark/share it. That includes which subtab you were on, the shading mode, and the field tick boxes on Summary and All-by-all, so a bookmark reopens on the same view.

Significance testing

When you build a **2-way count table** (one field in Rows, one in Cols, Count aggregator), the app runs a statistical test and shows the result below the table.

- **Sig level** (dropdown): choose your significance threshold (0.10, 0.05, 0.01, or 0.001). The test result shows whether the association is significant at that level.
- **Which test?** The app picks the right test for your data:
 - **Chi-squared** when both variables are nominal (e.g. categories with no natural order).
 - **Mantel** (linear-by-linear) when one or both variables are ordinal (e.g. Likert scales, age bands). This test is more sensitive to ordered trends.
- **All-by-all checks**: the third subtab, which runs as soon as you open it. The app tests every pair of categorical variables with 2–10 values, lists them by p-value, and shows tables for the significant

ones. Useful for exploratory analysis when you have several group or rating variables. The **Rows** and **Cols** tick boxes above the results narrow which pairs are shown; leave them empty for all of them.

Notes on the datasets

- **Links:** every causal link plus metadata; includes AI fields (e.g. confidence) and reserved columns like `original_cause`, `original_effect`.
- **Factors:** unique factors with frequency, source count, citations, and `original_label` (ALL underlying original labels for the displayed factor, concatenated with line breaks, derived from the current stage's links like `original_cause/effect`).
- **Sources:** document metadata and flattened custom fields (`custom_*`).

💡 Tip: For results that match other panels, use **After Analysis Pipeline**.



LOGS PANEL

The Logs tab shows a concise history of what has happened in your projects – for example, when links or sources were edited or deleted, when projects were created, or when AI processing ran.

- Use the **scope buttons** (This project / All projects) to switch between seeing activity only for the current project or across all projects you have access to.
- Use the **filters above the table** to narrow by user, project name, action type, date, or text in the details column.
- Non-admin users see only activity for projects they own or can edit; admins can see activity across all projects.
- Navigation and other UI clicks are logged only very briefly and are hidden from non-admin users.



BOOKMARKS AND REPORTS

The [yellow button in the navbar](#) is the fastest way to save useful views of your current project: maps or tables. From that one entry point you can quickly save views, update existing bookmarks, and copy links, images, legends, or a combined HTML block (link + image + legend) for reporting.

Report views (admin)

Admin only, in testing

The **Report views** button in the Reports toolbar builds a standard set of maps for the current project in one press: an **Overview** (top 12 factors, top 30 links by citations), **What most people say** (counted by how many people mentioned each factor and link, so one talkative respondent cannot carry them), and a **Headline** map of the few factors and links that dominate. Each is saved as a bookmark with a screenshot, and the slideshow opens on the three of them.

No AI is involved. The views come from a fixed list, so the same project gives the same three maps every time, and each one can be explained by the filters that produced it. Pressing the button again replaces the previous set rather than adding to it.

The bookmarks table itself is not shown for users below Pro level, i.e. to users only on Free or Private plans.

AI report deck (MapCat)

Requires AI plan, Edit my data with AI on

Ask [MapCat](#) to build a report, slides or a deck and it reviews your project data, asks a few preference questions in chat (tables, groups, sentiment, disagreement, quotes, and so on), then builds a slide deck of the strongest views, each with short written commentary. It can open the deck in a slideshow over the chat.

- The first deck defaults to about five slides so it lands fast. Read it, then ask MapCat to extend it or build a fuller version.
- MapCat keeps the stronger of any two views that would read the same (for example a top-factors map and a top-links map over the same filter), so you do not see near-duplicate slides.
- Each view is saved as a bookmark, so you can reorder, edit or export the deck from the [Bookmarks table](#). MapCat can also retile or delete a slide and export the deck as a PDF on request.

Bookmarks Panel

In this section you can learn about bookmarking and how to manage bookmarks.

Requires Pro subscription

🚀 **What you can do here:** Save and organize your favorite views of your data. Bookmark specific filter combinations, map layouts, or analysis states so you can quickly return to important insights later, share clean links, and build reports from saved views.

To create a bookmark from anywhere, use the [Navbar bookmark button](#). This adds a bookmark to the [Bookmarks table](#).

After saving, the navbar bookmark button briefly shows a green tick to confirm the save.

When you save a bookmark in the session,

- a small popover opens near the button with a Description textarea and a **Save** button (description is optional, but useful).
- inside **Existing bookmarks**, you can pick a bookmark and **Overwrite** it with the current screen state (keeping the same bookmark number/link).
- each bookmark can then be used to copy:
 - a short plain link (`?bookmark=ID`)
 - a formatted link (HTML)
 - the bookmark image
 - combined HTML that includes the bookmark link, image, and legend

The Bookmarks Table

Use the full table / report workflow when:

- you want to work across different projects
- you need more detailed tools for managing many bookmarks
- you want to build a full report from some or all bookmarks
- Each row is one bookmark.
- Click the checkbox to select rows.
- Click Load to recall that bookmark, restoring tabs and outputs.
- Quick load: at the very top of the Bookmarks / Reporting tab, you can type a bookmark number and press **Load** (or Enter) to load it directly.
- Click Copy Formatted Link to copy the bookmark's URL to the clipboard.
- Double-click on the Description field to edit it.
- Click Delete to remove the bookmark.
- Click **Show slideshow** to present the loaded bookmarks in their table order. The slideshow opens in a large modal with previous/next controls, left/right keyboard navigation, a title slide, bookmark

actions, and a **Load bookmark** button.

- Use **Share** next to **Show slideshow** to copy a link that opens the current project directly in slideshow mode. If the link includes `bookmark=ID`, that bookmark is opened first.
- Turn on **Hide from slideshow** for any bookmark that should remain in reports but not appear in the slideshow. This is saved with the bookmark.
- Click the badge at top right when viewing a map or table to add a new bookmark to it.
- When a new current project is loaded from a URL or by changing the project dropdown, the "Project" filter in the table is pre-filled with the name of the current project.

Main Features:

- **Bookmark badge** ★ - Save/remove current URL state from navbar
- **Bookmarks table** - Manage saved views with sorting and filtering
- **Bulk operations** - Select multiple bookmarks for deletion
- **Editable descriptions** - Double-click to edit bookmark names
- **Auto-normalization** - URLs cleaned and standardized for consistency

Bookmarks Table:

- **Actions** - Load URL, Copy link (plain), Copy formatted link (HTML), Edit URL, Delete individual bookmarks
- **Copy buttons** - Both create short `?bookmark=ID` URLs instead of full parameter strings
- **Columns** - Project, Description (editable), User, URL, Created, ID
- **Bulk delete** - Select multiple with checkboxes, delete with "Bulk:" trash button
- **Badge integration** - Shows bookmark ID when current view is bookmarked

URL Editor:

Click the Edit button  next to any bookmark to open the URL Editor. You can edit the bookmark description (Markdown supported). The editor also provides a user-friendly interface to understand complex bookmark parameters. Instead of viewing raw query strings, you see a structured breakdown with proper labels and grouping - showing your selected project, sources, active filters (displayed as individual cards), table settings, and map configurations. The editor categorizes parameters into logical groups (Navigation, Data Selection, Filters & Processing, etc.) and displays filter pipeline details with sequential numbering and status indicators. This makes it easy to understand exactly what state each bookmark preserves without needing technical knowledge of URL parameters.

Building reports from the Bookmarks table

The Bookmarks table lets you create professional reports by combining multiple bookmarked views with custom descriptions and variant filtering.

Requires Pro subscription

 **What you can do here:** Build multi-slide reports from your bookmarks. Add markdown descriptions, choose variant fields/values from your current filtered data, reorder slides, and export as formatted HTML or PDF. This is useful for producing stakeholder-ready outputs from saved app views.

Key Features:

- **Drag-and-drop reordering** - Arrange slides in any order
- **Markdown descriptions** - Add rich text titles and explanations with heading styles, lists, and formatting
- **Variant generation** - Create multiple versions of the same bookmark view filtered by selected field values
- **Include/Exclude toggles** - Selectively include slides in your export
- **HTML/PDF export** - Copy formatted HTML or export PDF with clickable bookmark links and legends
- **Persistent settings** - Your slide order, descriptions, and include/exclude states are saved automatically

Legends:

- For maps we use the full legend
- For tables we use the relevant parts of the legend, omitting material about link and factor colours, sizes and annotation.

Variants:

- Variants simply add an extra filter on top of the bookmark's existing filters (sources + filter pipeline). We do not change any other filters. For example, if the bookmark has 4 selected sources and you choose village=X and Y, then the X variant uses only those of the 4 sources with village=X, and the Y variant uses only those with village=Y. If a value has no data after the existing filters, no variant is produced.

Where to find the variants: when the accordion section is open, the variants appear as thumbnails above the main bookmark image, each preceded by which variable/value it represents (e.g., village: Y). On PDF export and copy to clipboard, first the original bookmark image and its legend are shown, and then each variant with its fullsize image and legend.

How to use:

1. Create bookmarks of your maps and tables
2. Switch to the Report tab (yellow bookmark icon)
3. Click on slide descriptions to edit them (supports markdown: **# Heading**, **## Subheading**, **- List items**)
4. Drag slides to reorder them
5. Use the variant controls to generate multiple versions filtered by selected field values

6. Toggle "Include/Exclude" to control which slides appear in your export. **Hidden from slideshow** bookmarks are also skipped by Copy HTML and Export PDF.
7. Click "Copy HTML" (or "Export PDF") to export all included slides
8. Paste into Word/Google Docs - headings and links will be preserved



AI ANSWERS PANEL

Requires an AI subscription

You can open this panel by clicking the corresponding tab lower down on the right-hand side of the app.



What you can do here: Ask questions about your data in plain English and get AI-powered answers. Type questions like "What are the main barriers to education?" and the AI will search through your currently selected sources to provide relevant answers with supporting quotes. Perfect for exploring themes and getting quick insights from large amounts of text.

Main Features:

- **Query input** - Type your questions in plain English
- **Context (optional)** - Briefly describe what your sources are about (used to guide expansions and answers)
- **Expand Question (optional)** - Generate and edit expanded search phrases before asking
- **Automatic chunking** - Sources split into searchable pieces when needed
- **Similarity slider** - Control search precision (0.1-0.9)
- **Max Chunks slider** - Maximum number of the most relevant chunks to send to the AI
- **Enable checking (second AI pass)** - Optional checker reviews the draft answer, reports issues in a collapsed panel, and returns a corrected final answer (toggle is on by default).
- **Prompt history** - Navigate previous questions with prev/next buttons. See these [tips](#)

Bookmarking & restore:

- Each time you click **Ask**, the app automatically saves a **bookmark** for the current view (description: **AI Answers: <your question>**), and appends the bookmark link at the bottom of the answer.
- The bookmark footer also prints the **model name** and the **similarity/max-chunks settings** used for that answer.
- AI Answers settings are saved into the **URL state**, so bookmarks restore them (scope, sliders, model, thinking budget, checking toggle, and text fields).

Search Modes

AI Answers offers two search modes, automatically optimized based on your data size:

Full Sources Mode

Searches the complete text of your sources (documents/interviews).

How it works:

1. Type a question about the text of the currently selected sources
2. System automatically chunks sources into searchable segments (if not already done)
3. Searches through document chunks using AI embeddings and semantic similarity
4. Most relevant chunks are sent to AI for analysis
5. AI generates answers with supporting quotes from your sources

Question expansion and HyDE (Hypothetical Document Embeddings)

This is now **optional**:

- If you fill **Context**, it is included in both the expansion prompt and the final answer prompt (to reduce generic expansions/answers).
- If you click **Expand Question**, the app calls genAI to generate expansion phrases and shows them in an editable textbox (one per line). You can edit them, or use **Clear expansions** to hide/remove them.
- If you click **Ask** directly, the app **skips expansion** and searches using only your original question.

When expansions are provided, we match each phrase against chunks and sum scores per chunk, then select the top n by the max_chunks slider.

So for example if the user asks what is the connection between money and happiness, the AI produces question variants like:

- having money, being joyful
- being wealthy
- being happy
- connection between money and happiness

And answer variants like:

- financial security enables emotional wellbeing
- wealth contributes to life satisfaction
- economic resources support positive mental health outcomes

Best for: Exploratory questions about raw text, finding themes not yet coded, discovering new patterns.

Link Contexts Mode

Searches only through your coded causal links and their surrounding context (the quote + 3 sentences before/after).

How it works:

1. Gets filtered links from your current filter pipeline (respects Sources dropdown and all Source Groups filters)

2. For each link, extracts the selected quote plus surrounding context
3. Organizes contexts by source, with source metadata (title, custom columns)
4. For ≤ 500 links: Sends all contexts directly to AI
5. For > 500 links: Same pattern as SRP — RPC + client-triggered embedding creation, no long-running

Edge Function

- Client gets query embedding (one call), then calls RPC
`find_relevant_link_contexts(project_name, link_ids, query_embedding, top_k, threshold)`
 - RPC fetches links from DB, builds a **normalized key** per link (see Normalization note below), checks `embeddings`; if any missing (or `embedding IS NULL`) returns `needs_embeddings` + `missing_texts`
 - Client calls `DataService.createEmbeddingsBatch(missing_texts, ..., { useProvidedKeys: true })` (server-side generation + DB upsert), then retries RPC
 - RPC runs similarity in Postgres (pgvector), returns `relevant_link_ids` + `similarities` (and may include a `debug` block); no embedding vectors to frontend
 - Selection is **source-diversified** (round-robin: best link per source first), and avoids identical contexts within a source (exact de-dupe on the normalized key)
 - Prompt to AI uses full `chunk_text` (quote + surrounding 3 sentences) built on the client
6. AI analyzes contexts showing cause → effect relationships
 7. AI is instructed to use the coded factor labels as the vocabulary (wrap labels in backticks like `this`) and to be sceptical about evidence coverage and alternative narratives.

Context format sent to AI:

```
'## Source: Interview with Participant 001
ID: ABC-123
custom_Country: Kenya | custom_Gender: Female | custom_Age: 34

Links from this source:

[ABC-123-1] Lack of resources → Poor school performance
Context: "We don't have enough books or supplies. The children struggle because..."

[ABC-123-2] Teacher training → Better outcomes
Context: "When teachers receive proper training, we see improvements in..."
```

Best for: Questions about causal relationships you've already coded, comparing patterns across sources, analyzing specific demographic groups using Source Groups filters.

Key advantages of Link Contexts mode:

- Uses your coded causal structure, not just raw text
- Respects all your filters (Sources dropdown + Source Groups)
- Includes source metadata in AI context (country, demographics, etc.)

- More focused and structured than full text search
- Automatically scales to large datasets (>500 links) using backend semantic search

etc etc

Sources, Links, and Factors subtabs

Use **Sources** for source rows, **Links** for link rows, and **Factors** for factor labels.

Why this exists: it lets you do fast, repeatable coding/scoring in bulk instead of editing one row at a time.

Each subtab now has two modes:

- **Answer mode:** asks a question over the currently selected payload fields and current filtered rows, then shows a text answer in the subtab (no row writes).
- **Create/Modify columns mode:** runs row-by-row writes.

How Create/Modify mode works:

1. Type an instruction in **Your Question** (for example, “score each row 0–3 for frustration”).
2. Choose a subtab and click **Run**.
3. The app only processes rows currently included by filters.
4. AI returns one value per requested target column per row.
5. The app writes those values back.

Per-subtab write behavior:

- **Sources:** writes to source fields/custom columns for filtered sources.
- **Links:** writes to link fields/custom columns for filtered links.
- **Factors:** does not create factor columns (there is no persisted factors table); it relabels matching values on currently filtered links, writing to either **cause/effect** (blank suffix) or **cause_<suffix>/effect_<suffix>** (suffix set).
- Related: [Temporary Factor Labels filter](#).

Links + Factors (Create/Modify mode): target columns follow the **Label set** in the toolbar (there is no separate “target set” in Answers). Writes may overwrite existing non-blank values where the run applies; when targets are suffixed **cause_*** / **effect_***, filtered-out rows may be backfilled from base **cause** / **effect** in the same run.

Choose payload columns (optional):

- All three subtabs use checkbox-pill choosers.
- Chosen fields are prepended to each row payload as **Field = value**.
- Default payloads:
 - **Sources:** **content**

- **Links:** *cause, effect*
- **Factors:** *label*

Safety behavior (Create/Modify mode):

- If a run would overwrite a standard (non-custom) column, the app shows one confirmation modal before processing.
- It is shown once per run (not once per batch).
- Protected/system columns are never overwritten.
- Non-filtered rows are unchanged.

Label-transform filters note:

- In **Answer mode**, transformed-label warnings/blocks are not shown.
- In **Factors → Create/Modify mode**, relabeling is blocked when label-transforming filters are active.



SETTINGS PANEL

⚙️ **What you can do here:** Enable live collaboration. Other settings coming soon.

Live collaboration

Watch as your colleagues make changes to the causal map (and links and factors tables) in real time!



ACCOUNT PANEL

👤 What you can do here: View and manage your personal account settings. Change your password, update your project information, and control your privacy and security settings. This is also where you can export your data or delete your account if needed.

Making projects private requires a Private subscription

User account management and project settings.

Onboarding (first sign-up): When you sign up, we ask a few questions including whether you want AI options switched on and active right at the start, and what you want to do first in the app. If you choose an AI-first workflow, AI is switched on automatically. If AI is off, there are no AI services at all (except basic MapCat help); you can change this anytime in Account settings. If AI is on, you get 100 free AI credits per month, and the AI switch is turned on by default; you can turn it off later in Account settings.

AI coding toggle ("AI options switched on and active"): Turn on to use AI; turn off and there are no AI services at all (except basic MapCat help). You can change this anytime in Account settings. When you turn it **on**, a warning modal appears: your data is sent to AI providers when you use AI; it is **not** used to train models; OpenAI (GPT) may retain data for up to 30 days for abuse monitoring; Vertex AI and DashScope do not retain data; see our [privacy policy](#). Credits when on depend on your plan; see [AI credits](#). **Plans without AI** (e.g. Private Manual) can turn on to try with the free allowance (free credits do not stack with paid AI plans).

Where AI runs is a project setting, not a user setting. Each project has an **AI Processing Region** (Project → Edit project → AI Processing Region) with three options: **EU (Belgium, europe-west1)**, **UK (London, europe-west2)**, or **US (Virginia, us-east5)**. New projects default to EU.

When the region is **EU or UK**, the app guarantees every AI call stays in Europe:

- The coding model picker offers the European-resident Geminis only: **Gemini 3.5 Flash** (the coding default), **Gemini 3.6 Flash**, **Gemini 3.5 Flash-Lite**, **Gemini 3.1 Flash-Lite**, **Gemini 3.1 Pro Preview** and **Gemini 2.5 Flash**. Gemini 2.5 runs in a single European region and every Gemini 3 model runs in Google's EU multiregion, so processing stays inside the EU. Models with no European route (GPT-5, Qwen) are hidden.
- Any AI request that would process data outside Europe is blocked client-side before it leaves the browser. A notification explains which model or region caused the block and how to fix it.
- Document embeddings (used by RAG and similarity search) automatically route to **europe-west1** (or **europe-west2** for UK), instead of the default **us-central1**.

When the region is **US**, the model picker shows the full list, including GPT-5, Qwen and others. These process your prompts and data outside Europe.

Switching the region for an existing project takes effect immediately — the dropdowns re-render and the gate updates without a reload.

You can change this toggle anytime.

Account Features:

-  project information and settings
-  Password and security management
- Account deletion and data export
- Subscription and billing information

2FA (beginner guide)

If your organisation asks you to use 2FA, do this in **Account → Two-factor authentication**:

1. Turn on **Require two-factor authentication on sign-in**.
2. Sign out, then sign in again.
3. If asked, scan the QR code in your authenticator app.
4. Enter the 6-digit code to finish setup.

If codes do not work:

- Try a different Causal Map entry in your authenticator (you may have old entries).
- Use **Delete all 2FA factors** in the same card, then sign out/in and set up again.

For strict org policy:

- Click **Require 2FA permanently**.
- This is one-way in the app UI and keeps 2FA locked on for that account.

Image export resolution

Controls resolution for all image capture (map copy/download, bookmarks, pivot/table screenshots, PDF export). Options: 1× (default), 2×, 3×. Resets to 1× on page load. **1× is already very good** for most uses; higher values produce larger files and may hit clipboard limits on large maps.

Subscriptions

Subscriptions List

Users without a subscription are either:

- anonymous (not logged in) (this is disabled at present)
- free (logged in)

Subscriptions (via LemonSqueezy) are available in the Account panel. Monthly and Annual plans are recurring subscriptions: a Monthly plan renews and is charged each month until the subscription manager cancels it.

To manage billing, update payment details, or cancel a subscription, open **Account → Your subscriptions** and click **Manage in LS**. This opens the Lemon Squeezy customer portal. Only the subscription manager/purchaser sees this button; users who only have a seat on someone else's subscription should ask the manager to make billing changes.

Admins can also create **manual subscriptions** (for testing/support). These are stored in the same `subscriptions_purchased` table but do **not** come from Lemon Squeezy.

The subscriptions list uses one row per type (private, pro, team) with seat-count, square radio buttons for monthly/annual and Manual/AI, and a live-updating price (from a JSON price file). Each row includes a text description.

There are three dimensions to the subscriptions,

Manual vs AI,

Type:

- private
- pro
- team

Monthly vs Annual.

AI credits (when you have AI): Free users who opt in get 100 credits/month. Private AI: 1000. Pro AI: 2000. Team, Academic and Corporate: 4000. Credits renew at the start of each month and do not roll over. See [Responses Panel](#) for usage. Plans without AI (e.g. Private Manual) can turn on the AI toggle in Account to try with the free 100 credits/month. These figures are the user-facing copy of `subscription-prices.json`, which is the source of truth; update them together.

User can purchase multiples of one or more subs to distribute to colleagues.

Subscriptions Table

The Subscriptions card shows a simple table of the subscriptions where you are the manager or a subscriber, with the manager and subscriber emails. If you manage a subscription, an **Edit** button lets you change who has a seat.

Renewals: your renewal date updates automatically whenever you open or refresh the app while signed in.

Reminder emails: we email you before a subscription ends, 7 days before, 1 day before, and when it expires.

Private projects and subscription expiry

The paid value is privacy: a **private** project is one that only you, and the people you invite, can see. Keeping a project private needs an active paid plan.

On the free plan, projects are **public**: anyone with the link can view them. You cannot make a project private without a paid plan.

On a paid plan that includes private projects (Private, Pro, Team and similar), you can make projects **private** and work on them privately.

If your paid plan expires, your private projects cannot be opened, viewed or edited, until you do one of these:

- **Renew** your plan: the projects stay private and open normally again, or
- **Make a project public**: you can do this without a plan, behind a confirmation. After that anyone can view it, and you can open and edit it again.

Your private projects are never published for you. Turning a project public is always your own explicit choice.

When you allocate seats, the email box pre-fills your own email in the first slot and tells you how many seats are left, or if you have added too many emails.



RESPONSES PANEL

Requires an AI subscription

 **What you can do here:** Review all your AI interactions and usage. See a complete log of AI requests, responses, costs, and performance metrics. Useful for tracking your AI usage, reviewing past queries, and understanding costs. Great for administrators and power users. This panel shows a searchable table of AI calls (rows from the `ai_logs` table). You must be logged in to see your logs.

What kinds of AI work get logged (Type column):

- **ai-coding**: “AI coding” runs that read source text (often in chunks) and generate things like coded links / structured outputs.
- **vignette:whole** / **vignette:typical**: Vignette generation (AI-written narrative summaries).
- **answers:question_expansion**: AI Answers expanding your question into multiple search phrases.
- **answers:answer**: AI Answers generating an answer from retrieved evidence.
- **filter:soft-recode:generate_magnets**: Filter Pipeline (Soft Recode / SRP) using AI to propose magnet labels.
- **filter:soft-recode:label_cluster_node**: Filter Pipeline (Soft Recode / SRP) using AI to label individual cluster nodes.
- **filter:cluster:label_clusters**: Filter Pipeline (cluster filter) using AI to label clusters.
- **docs:chat**: MapCat documentation chat. Logged but **not counted** toward your monthly credits.
- **– (blank)**: Older log rows (before we added `request_type`) or any legacy code path that inserts into `ai_logs` without setting a type. You can still click the eye to see the full prompt/response.

How to use the table:

- **Scope toggle (top-right):**
 - **This project**: shows only AI calls from the currently selected project (exact match).
 - **All projects**: shows AI calls across all projects you have access to.
- **Sort**: click a column header (e.g. **Timestamp**, **Cost**) to sort by that column; click again to reverse the sort.
- **Filter row (the grey row under the headers)**: type/select a value and the table refreshes automatically.
 - **Date**: pick a date. If you pick a past date, it filters **from that date up to today**. If you pick today, it filters **today only**.
 - **Project**: free-text filter (substring match). Most useful when Scope = **All projects**.
 - **Model**: dropdown.

- **Status: Success or Error.**
- **Source:** filter by `source_id`.
- **Prompt:** filter by text inside the prompt (substring match).
- **Clear filters (X button):** appears only when any filters are active; click it to reset the filters.
- **Pagination:** use the pager at the bottom to move through pages; use the **page size** dropdown to change rows per page.

What the columns mean (quick guide):

- **Timestamp:** when the AI call was made.
- **Project:** which project it was for.
- **Model / Region:** which model ran and which region it used (where applicable).
- **Status:** success / error (some rows may show “pending”).
- **Time (s):** response time.
- **Chunk / Iter / Batch / Source:** metadata used mainly for chunked/batched runs.
- **Prompt:** a short preview (hover to see more).
- **P Tok / C Tok / Cost:** prompt tokens, completion tokens. **Admins** see cost in USD (\$). **Non-admins** see **Credits** (users without paid AI plans get 100 free credits per month; credits renew monthly and do not roll over; see [AI credits](#)).
- **User:** the user email associated with the call (when available).
- **Type:** what feature created the call (see “What kinds of AI work...” above).

Credits and limits: Non-admins see **Credits this month: X / Y** in the summary (usage / limit).

Credits renew at the start of each month and do not roll over. MapCat ([docs:chat](#)) is logged but excluded from the usage count. If you exceed your monthly limit, a warning appears and AI features are disabled until next month.

Click the eye (Details column) to see the full record:

- Shows **full prompt**, the **full response** (and sometimes renders it as a table if it looks like JSON/tabular text), plus **error messages** and the **raw JSON response** for debugging.



HELP SYSTEM

? What you can do here: Get instant help and guidance while using CausalMap. Search for specific topics, browse documentation by section, or click the question mark icons throughout the app for context-sensitive help. The help system adapts to what you're currently doing. Also view the entire help contents as a separate Guide.

Main Features:

- **Help drawer:** Right-side drawer with collapsible sections
- **Smart search box** at the top of the help drawer. If you enter multiple words, either word matches. To match exact phrases, use quotes.
- **Context-sensitive help:**  Click the blue buttons within the app next to open the help drawer.
- **App hints:** When you click a section in the help drawer which refers to part of the app, that element flashes briefly.
- **Standalone help:** The same help information is available as a [section](#) within our [Guide to causal mapping](#).



MapCat is the small chat helper at the bottom-right. **MapCat is available to all users**, even if you do not have an AI subscription or if AI is disabled in your account settings. It is the only AI feature that does not require a subscription or for AI to be enabled, because it does not see your project data.

What MapCat is for

MapCat answers questions about the app and about causal mapping while you work. With **Edit my data with AI** on and a project open (AI plan), it also works alongside you: you can ask it to do things and it carries them out, showing a confirm step before any change.

Without that switch on (the default), you can:

- Ask how a control, panel or workflow works.
- Ask it to open help, highlight a control, or switch tab.
- Read its nudge when the cat wiggles: it has a suggested next step.

With **Edit my data with AI** on and a project loaded, you can also:

- Ask questions about your own data: your factors, sources and links, or what a filtered view holds.
- Set what you want to find out and let MapCat hold that aim across the chat.
- Run AI coding by chat. MapCat asks a couple of plain questions about how you want the sources coded, then starts the run for you, so you do not need the settings screen. It shows the conventions it chose in a coding-decisions summary before the run.
- Build a report. MapCat asks what it should cover, builds a slide deck, and can open it in a slideshow over the chat (see [AI report deck](#)).
- Ask about your findings, and MapCat answers by running a report view and telling you what it shows.
- Change map, filter and pivot settings, and edit links and sources, by asking.
- Have MapCat build a view and save it as a bookmark for you, so you can ask for a set of maps in one go and come back to each one later.
- Ask it to reopen a saved bookmark, change something about it and save it back, so a set of views can be corrected without rebuilding each one by hand.

Every change shows a confirm step first, so nothing happens to your project until you approve it.

Header controls (top of MapCat panel)

- **Window controls** (group of 3 icons):

- **Maximize icon:** open MapCat in large centered mode.
- **Restore icon:** dock MapCat in the normal bottom-right position.
- **Minus icon:** minimize back to the cat chat-head button.

Footer controls (bottom of MapCat panel)

- **Contact support:** sends the latest question/answer context to support.
- **Edit my data with AI** (AI plan only). The switch was called **Superpowers** until August 2026; the name changed because ticking it is your consent to send this project's data to the AI, and the old name said nothing about that.
 - Default is **OFF** (safe mode).
 - When OFF, MapCat can still answer documentation questions and use non-data UI helpers (for example help/hints/tab switching).
 - When ON, MapCat can run data-aware actions (for example sources selection, filter pipeline edits, pivot config), which means project state data is sent to the AI. It can also change your project, and every change asks you to confirm first.
- **Clear:** clears MapCat chat history.
- **Send:** sends the current prompt.

Privacy and data-sharing behavior

- With **Edit my data with AI** OFF (default), MapCat runs in safe mode intended for users who do not want project data sent for AI actions.
- With **Edit my data with AI** ON, MapCat may send project-derived context needed to execute data-aware requests (such as source IDs, filtered link/factor context, and related view state) to the AI service.

Safety rules

- UI actions are whitelist-only.
- Project deletion actions are explicitly blocked (**delete/remove/destroy/drop project**) even if requested by a tool directive.



HINTS SYSTEM

Hints are small popovers that help you discover features. They never appear on their own timer: you see one only when you ask (the lightbulb button in the navbar), after certain first-time actions, or when MapCat points you at a relevant control.

- Hints are **tied to specific UI elements** (buttons, inputs, panels).
- Hints are **capability gated** (you won't see hints for features your plan doesn't include).
- When the **Link Editor (causal overlay)** is open, you only see **overlay-specific hints**.
- Repeated lightbulb clicks walk the current panel's hints in a **deterministic sequence** (so you don't keep seeing the same ones at random).
- **Don't show** turns hints off permanently; the lightbulb still shows them on demand.



MOBILE VIEW

- A mobile-only two-button toggle (Left side / Right side) sits next to the brand.
- Tapping Right side opens the offcanvas that contains the right pane (tabs and content).
- Tapping Left side closes it. The active side shows a tick.
- The offcanvas shows a "Back to left side" button at the top to close it.
- On desktop ($\geq$ lg), the right pane behaves exactly as before; offcanvas visuals are disabled via CSS.
- A mobile-only Menu consolidates: About, Guide, Help, Bookmark, and Account/Login/Logout.



FAQ - FREQUENTLY ASKED QUESTIONS

How do I subscribe or unsubscribe?

Go to **Account** → **Subscriptions**, choose the plan, seats, Monthly or Annual billing, and whether you want the AI option, then click the purchase button. Monthly and Annual plans are recurring Lemon Squeezy subscriptions, so they renew automatically until cancelled.

To unsubscribe or change billing details, go to **Account** → **Your subscriptions** and click **Manage in LS**. This opens the Lemon Squeezy customer portal, where the subscription manager can cancel the subscription, update payment details, or manage billing. If you only have a seat on someone else's subscription, ask the subscription manager to do this.

Tips on coding

First of all, there's nothing to worry about, it's fun!

The versioning / backups feature means you can always go back to any version of your file at any earlier time point.

Also, [bulk editing of factor labels](#) makes it easy to rapidly change one or many links or factors. And you can do it either globally, i.e. changing one factor everywhere in the file, or you can do it for particular sources or specific kinds of links by using filters.

Usually, don't bother coding the same link more than once for the same source, unless they bring up distinctively different evidence each time.

- It's okay not to code a source at all. If there's nothing in it, or if people are just making vague and general sources.
- You'll find you're constantly shifting between sometimes creating new factors, and then going back and reviewing them and merging them and organising them using the [bulk editing of factor labels](#).
- Don't forget you can combine two or more factors into one using the [bulk editing of factor labels](#).
- Don't forget when you want to search rapidly through already coded links through all of the sources, you can click on the rows in [the bulk editing of factor labels](#) to go back to the relevant sources directly.
- Occasionally, a source will make a comment about something which is worth coding, even though there isn't actually a causal link. For example, they might make general comments about some outcome without saying what causes it. In this case just use plain coding. (But if you find you are doing this a lot, you might need to rethink your research design.)

- If you are using [hierarchical/nested coding](#) (and you probably should) don't forget you can see the whole map zoomed out to the top level: just press the appropriate button in the Filters panel.

Help, I have too many factor labels!!

That's a classic problem with causal mapping (or any free text coding). We have several ways to help you regain control when you have dozens or even thousands of factor labels with overlapping meaning.

1. Use the [Bulk editor](#). It's a really powerful way to inspect and change all or part of your factor labels. You can sort alphabetically, by citation count ... and you can highlight and edit multiple matching labels at once!
2. Use the quick [Search/Replace box](#) at the top of the Factors Table tab. It's simple but also powerful. If you want to rename your factors in the context of thinking about causes and effects, there is a similar widget in the [links table](#).
3. To look at similar labels in the map, type one or more fragments (e.g. "income", "Money") into a [Factor label filter](#) so you can see where the corresponding factors appear in your map
4. If you have more than a hundred labels and you have an AI subscription, the [Soft Recode](#) filter is amazing!

Do I have the latest version?

See [Navbar → About menu](#) to view the version number. The app should silently update itself when a new version is available.

How can I adjust my links or sources data in bulk (round-tripping)?

As an alternative to editing your [links](#) and [sources](#) tables manually in the app, you can do what we call round-tripping: [download](#) your file, tweak this Excel file manually (e.g. by adding additional columns to the sources tab) and [upload](#) it again into a new Causal Map project. Like this you always have to create a new project, which helps you not get mixed up with which version is which.

How can I deal with closed questions like in QuIP?

In Causal Map 4 there is no special treatment of QuIP-style closed questions. What you can do is this:

1. if you want to be able to see the closed question answers while coding, include the answers to the closed questions simply as part of the text of the interview with an appropriate question number. None of this has any meaning to the app, but it might be useful to have for coding.
2. If you want to also use the closed question scores for further analysis, e.g. to make a map of all the interviews which answered an average of better than 0 to some question, then just add a custom column for each question and add the average scores for each question into each column. Then you can apply these values as filters, see [here](#).

What does the 'recycle weakest magnets' slider do?

The slider temporarily removes the N weakest magnets from your list and reassigns their raw labels to the stronger magnets.

For example, if you created 50 magnets but after filtering you only have 5 factors showing with 9% coverage, those 45 weak magnets might be taking evidence away from your main ones. Try moving the slider to 40 to recycle those weakest magnets - this gives their evidence a chance to match with the stronger magnets instead (using the same similarity criterion).

This is most useful when you have lots of fiddly magnets that nibble away at your main ones but then disappear without trace. Note that the slider is a bit unpredictable if you have intervening filters.

My pasted images are failing after a while in my markdown editor

If you paste not just an image but the html including image, legend etc from a bookmark, it works ok if you paste it into an app like Word, but with a markdown editor like Obsidian, the image remains as a URL, but our URLs are only temporary (we don't save images forever at Causal Map), so after a while the image will fail. So, you need to install a plugin for your markdown editor, like the Obsidian plugin called Local Images Plus.